Held, Rocking, Locked: The Islands of a Light Star Around an Eccentric Binary

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Abstract

We follow a planet-mass body around an eccentric pair of equal stars, near the 7:1 and 8:1 period ratios, by direct three-body integration for up to 60,000 orbits, to see up close why three bodies are hard. The star is held where two clocks keep time together: one angle, φ_N, comparing the pair’s orbit with the star’s lap, stays put while every other angle turns. Escape is one-sided: with the star’s eccentricity at 0.20, 163 of 166 recorded escapes, from light and heavy stars alike, leave from the island’s side nearest the pair, a door the published stability rules smooth over, and one that 41 stars started on circular orbits, as stability maps start theirs, did not find in 10^5 of the pair’s periods. An island read over 600 orbits can be slow chaos: at the pair’s eccentricity 0.25, 22 of 28 inner starts lose their hold within 6,000, and the room that keeps its stars, births 7.465 to 7.505, lies below the centre the first 600 orbits point to. Mardling’s (2013) resonance pendulum passes a direct test where its assumptions hold, its rate–width relation within 1.1% and its widths within about 30%, and is off by measured amounts outside them: the island’s centre doubles once, at the star’s eccentricity 0.348 to 0.350 rather than 0.41, and then every nearby start escapes. Inside the island the rocking locks to simple fractions of a cycle per orbit, at a fifth to within a few billionths, and odd fractions fill about three times as often as even ones. The hold drags on the star’s ellipse: Mardling’s coefficient, with nothing fitted, gives the drag’s slope over six hundred orbits within about 6% at the star’s eccentricities 0.20 to 0.26, and near 0.28 the centre’s ellipse stands still, then turns backward. A heavy third star moves the islands just below the whole numbers, and the pair’s short-period swing is about 1.12 times Georgakarakos’s (2003) formula. At a tolerance that looks tight, the integrator’s arithmetic drags every unlocked rocking rate down and fakes a slow slide of locks, and with the heavy star it shortens a slow beat by a third over 30,000 orbits; a hundredfold tighter tolerance removes both, though a far smaller drift remains, and at a second birth a test finds the slow beat keeping pace with the pair’s falling eccentricity. The method was first checked on maps with known answers, reproducing the frictionless and damped period-doubling constants and the dial between them, the area an orbit keeps per trip. Many tests began as marks made by eye on plain pictures, each written down before any number was read: a star polygon drawn on one picture is the inscribed-angle theorem, and a scatter of readings that looked chaotic is, zoomed out and read at relative tolerance 10^{-12}, a rotation of two angles, out to near returns 31,063 orbits apart in a run of 60,000, from which a twin nudged by 10^{-10} drifts off in a straight line, as a regular orbit’s twin does; sorted blind, reads naming a relation held more often than reads naming a shape, 19 of 30 against 11 of 28. Of 490 predictions written down before their runs, 236 came out as written; every miss is kept. A preprint, without peer review and with a limited literature pass.

What this paper found

1. **Held by coupled clocks.** Near seven (or eight) of the pair’s periods per orbit, one angle, φ_N, stays put while every other angle turns: the pair’s orbit and the star’s lap keep time together. (Section 5; Figure 7.)
2. **A one-sided door.** With the star’s eccentricity at 0.20, 163 of 166 recorded escapes, from light and heavy stars alike, leave from the island’s side nearest the pair (Figure 1); from 0.30 a second door opens on the far side. The published stability rules smooth the door over, and stars started circular do not find it. (Section 6.)
3. **Islands that are slow chaos.** At the pair’s eccentricity 0.25, what 600 orbits read as an island is, over 6,000, a row of rooms divided by chaotic threads, and the room that keeps its stars, births 7.465 to 7.505, lies below the centre the first 600 orbits point to, near 7.525. (Section 6; Figure 9.)
4. **Mardling’s pendulum, tested piece by piece.** Its rate–width relation holds within 1.1% where her assumptions hold, and her widths come within about 30% of ours. Once a chaotic layer eats the island, the relation holds instead, within about 10%, for the island and its layer together at the pair’s eccentricities 0.05 to 0.20, a reading made after the fact that held when tested forward at 0.225 and missed narrowly at 0.25. (Sections 5, 6 and 10; Appendices B and C.)
5. **Locks to a few billionths.** The rocking locks on simple fractions of a cycle per orbit, the core of the lock at a fifth to within 3.5×10^{-9} over 6,000 orbits, and odd fractions fill about three times as often as even ones. (Section 8; Figures 11 and 12.)

6. **The hold drags the ellipse, then turns it back.** At the island's centre the star's ellipse turns at two-thirds of its free rate. Across the island the drag follows the hold's grip, and Mardling's coefficient, taken at each line's mean eccentricity with nothing fitted, gives its slope over 600 orbits within about 6% at the star's eccentricities 0.20 to 0.26. Near 0.28 the centre's ellipse stands still, and above it turns backward. (Section 9; Figure 13.)
7. **One doubling, then escape.** The island's centre doubles once, between the star's eccentricities 0.348 and 0.350, where the pendulum would only at 0.41; then every nearby start escapes. (Section 10; Figure 14.)
8. **A heavy third star.** Its islands sit just below the whole numbers instead of above, and the pair's short-period swing is about 1.12 times Georgakarakos's (2003) formula, at half the period ratio he tested. (Section 11; Figures 15 and 16.)
9. **The arithmetic can fake resonance physics.** At relative tolerance 10^{-9} every unlocked rocking rate falls by about 2.8×10^{-8} cycles per orbit with each orbit, on the line measured most closely, enough to fake a slide of locks; a hundredfold tighter tolerance shrinks the fall to 0.0093 of itself. With the heavy star the same drift shortens a slow beat by a third over 30,000 orbits, and at a second birth a test finds the slow beat keeping pace with the pair's falling eccentricity. Ten times tighter again, over 60,000 orbits, the creep is about a hundred times smaller. At looser tolerances it moves the orbit's two rates in the proportion a nudge in birth does, leaving the eccentricity's second line almost still, though it is no nudge in birth: it drains the energy where a nudge outward adds to it, and it keeps its sign where the drifts of the energy and of the angular momentum both change theirs. Made again on a second machine, the long runs gave the same verdict in every test registered before they were sent there. (Section 3; Appendix A.5.)
10. **The bench and the guesses.** On maps with known answers the method reproduced the frictionless and damped doubling constants, 8.721082 and 4.669197, and found the dial between them (Section 4; Figure 6). Of 490 predictions written before their runs, 236 came out as written, 211 did not and 43 could not be read. (Section 3.)
11. **How the eye found it.** Several results began as the author's marks on plain pictures, each written down as a test before anything was measured: a clockwork star polygon that is the inscribed-angle theorem, dead zones that runs he never drew on avoid, and the beat hidden in the centre's rocking. Sorted blind, picture-reads that name a relation held 19 times in 30, those that name a shape 11 in 28. (Section 3; Figure 4.)

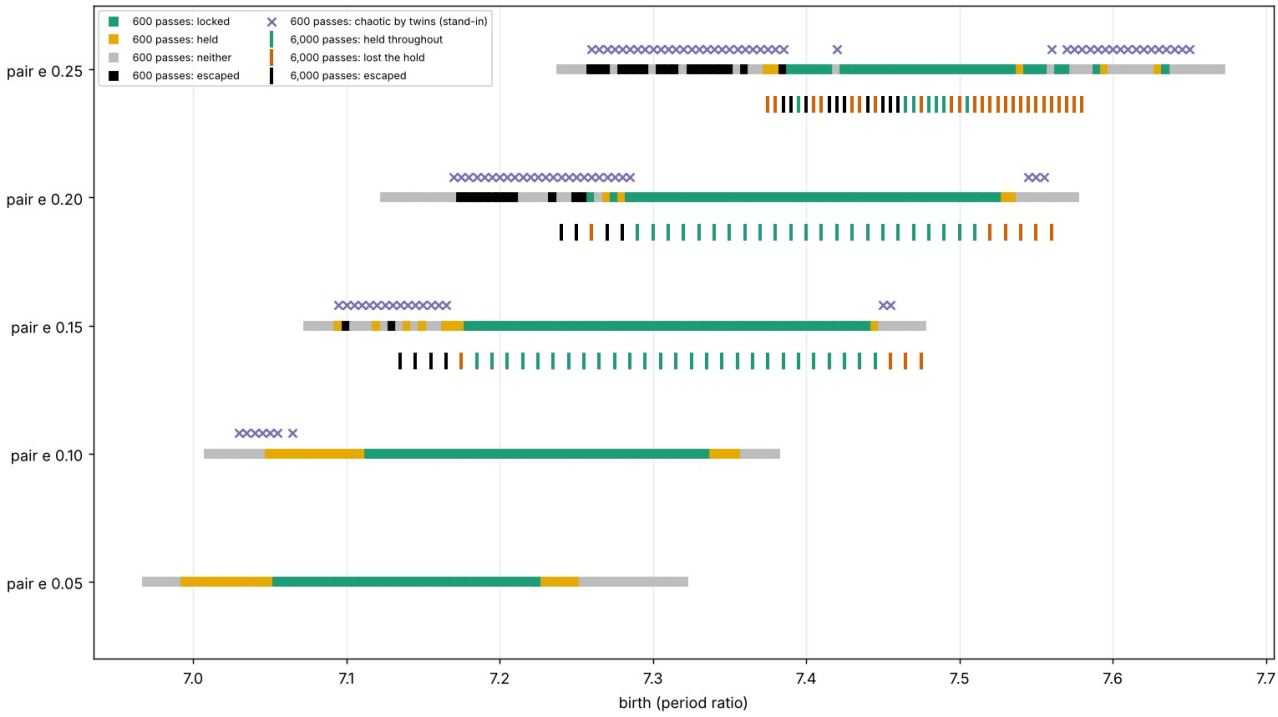


Figure 1. The door. Island 7 and its chaotic layer at five eccentricities of the pair, the star's at 0.20, side by side in birth, read over 600 passes: locked, held, neither or escaped, with a cross where a birth's twin went another way or ended 10^{-3} turn or more from it. That cross stands in for the twin test of A.4, which needs each run's series, not kept for every scan; on the two scans whose series were kept, the two agree at 144 of 146 births. The island moves up in birth as the pair grows more eccentric, and its chaotic layer grows on the low-birth side. Under the rows at 0.15, 0.20 and 0.25, how the same births ended when watched for 6,000 passes: at 0.15 and 0.20 the island holds nearly whole, giving up only its rims, and at 0.25 most of it loses its hold. From probe 34 (run 20261008_070206) and probe 39 (run 20261012_213903) over 600 passes, and probes 43 (run 20261013_233146) and 42 (run 20261013_152519) over 6,000.

1. Introduction

Where this paper came from

This paper began as a sandbox, not a plan. The three-body problem was opened to learn why it is so hard, beside toy models that are easier to hold: damped pendulums, and the maps with and without friction and the loop pushed by a slow beat of Section 4. It was never meant to be a paper; like many of the author's projects, it just happened. The islands grew from that play into a paper of their own. About then the author asked Claude what questions it still had and what it wondered about, and its answers started a second project, the Loop Atlas, which tests what loops leave behind. That one gave birth to a third, A Rider in Two Waves, on a rider on a tilted washboard passed by a ripple, the picture behind phase locking in driven circuits. The three were worked side by side, and the star and the rider turned out to share one frame, because they share one piece of physics: each is something that turns, read once per tick of a second clock, and each locks where the two clocks agree on a simple fraction. The one difference that matters most is friction. Without it, as here, every start keeps its own orbit and the islands sit in a chaotic sea; with it, the rider settles, and its doubling constant is the damped 4.669 rather than the frictionless 8.721 (Section 4). What travelled between the three was a way of working, reading a lock by the whole turns it makes and writing each guess down before its run, so that the results came back in a form that could be laid side by side.

Why three bodies are hard

Two bodies are simple: each traces the same ellipse for ever. Add a third, even one far too light to move the other two, and that simplicity goes. Two starts side by side can lead one to a steady orbit and the other to chaos; a pattern that holds for hundreds of orbits can let go over thousands; and the published rules for which orbits are safe smooth over part of what happens (Section 6). We set out to see why, up close, in one corner small enough to measure: a light star going round a pair of stars near where it goes round once for every seven or eight of the pair's turns.

Each part of the paper turned up one more piece of the answer. The light star is held only by timing, two clocks that must keep agreeing (Section 5). The hold's edge is lopsided, with a door on one side, and what looks steady over hundreds of orbits can be slow chaos over thousands (Section 6). Inside the hold the star rocks, and its rocking can lock onto exact fractions to within a few billionths, while most stretches that look locked are not (Sections 7 and 8). The hold reaches into the star's own orbit and drags on its slow turning, until the turning stops and runs backward (Section 9). Pushed harder, the hold's centre doubles once and then lets every nearby star go (Section 10). A heavier third star pulls back on the pair and trades eccentricity with it, so the pull that holds it changes as it goes (Section 11). And the tools that look at all this can fool the one looking: the integrator's own arithmetic can fake a slow drift, and the eye can see a shape that a finer grid dissolves, as well as shapes that are really there (Section 3).

That last lesson is also how much of this work was done. The author read many of the pictures by eye and said what he saw, his reads were written down as tests before anything was measured, and Section 3 shows his marked pictures and what became of them. Section 12 gathers the pieces.

The trapeze, before the physics

Picture a flying trapeze. The pair of stars is the catcher's bar, two equal stars going round each other on a slightly oval track. Far out, a light third star is the flyer, who comes round once for every seven turns of the bar. Each time the flyer swings closest, the bar gives it a small tug, and whether the tug hurries the flyer or holds it back depends on where the bar is in its turn at that moment. Near seven turns per lap the tugs keep correcting the flyer's timing, so the two keep meeting at the same point of the bar's turn: the flyer is held. That tug comes from the oval of the bar's track; on a perfectly round track this catch would vanish. Held, the flyer still arrives a little early, then a little late, then early again: it rocks about the catch, and sometimes the rocking itself falls into step, early and late repeating exactly every three or every five laps.

We find where the hold ends the way you would find the top of a swing: a flyer started just outside it slips out of time ever more slowly the nearer it starts, as a trapeze lingers at the top of its arc. That edge is lopsided: almost all the flyers that leave go from the side nearest the pair. Where the bar's track is most oval, the hold is only lent. A flyer keeps time for a thousand laps, drifts off, and sometimes falls back into step. Watch long enough and the hold turns out to be a row of rooms divided by thin walls, where flyers lose their timing soonest and from which some leave the act altogether; a little past the last wall, none has left in all the time we have watched. And the flyers that rocked least at first are not the ones that keep time longest: the calm that lasts sits a little to one side of where the first few hundred laps said it would. The picture is only a picture, since a real catcher grips and this bar only tugs, but it is the one this paper carries: the catch, the rocking and the locks, measured one run at a time.

What is known, and what this paper adds

Each kind of motion in this paper is known on its own. What is new, as far as we found, is measuring them together in one system, every number checkable by making the same run again, and what that turned up: a door out of the island on one side only, and slow chaos hiding in what short maps read as islands. The kinds of thing measured here are known: mean-motion resonance islands, their chaotic edges, secondary resonances, slices whose highest point resembles a vanishing twist, and doubling centres (Section 14). What this paper adds, as far as our literature passes found, is their measurement in one circumbinary setting by direct integration, over runs of up to 60,000 orbits, with every number tied to a run that reproduces on a laptop. It tests Mardling's pendulum in the test-particle limit, where it holds and where it misses. It finds a one-sided door at a resonance beyond the stability limit, which the published stability rules smooth over and which a star started circular, as the stability maps start theirs, does not find; and slow chaos that 600-orbit maps read as islands. It measures the rocking's lock at a fifth to a few billionths of a cycle per orbit, which fractions' locks fill most, the drag's size from Mardling's coefficient with nothing fitted and where, over longer records, it stops being one number; a single doubling followed by escape; and the heavy star's tide as Georgakarakos's short-period response. And it puts a size on the drift that a tolerance which looks tight fakes in resonance frequencies. Each section opens with what it found, with what is known beside it.

2. The setting, and the words we use

In the trapeze picture of Section 1, the pair is the bar, the light star the flyer, a pass the moment of closest approach when the tug comes, and φ_N where the bar is in its turn, seen from the flyer, at that moment. The words below say each of these exactly.

- **The pair (the binary).** Two stars of equal mass on an orbit of eccentricity 0.10, everything in one plane, in units where the gravitational constant is 1.
- **The light star.** A third body of mass 10^{-6} of either star, too light to move the pair, on a prograde orbit of eccentricity 0.20 around the pair's centre of mass. That is a planet's mass, a third of an Earth's beside Sun-like stars, so the setting is a circumbinary planet's, in one configuration: coplanar, around a pair of equal stars, near the 7:1 and 8:1 resonances.

- **The birth.** The star's starting period over the pair's period, the number we set (7.05 to 7.55 near island 7 in the atlas). The star then settles to its own average.
- **A pass.** One close approach of the star to the pair. We read everything once per pass.
- **The lap ratio.** The time between passes over the pair's period: the period the star actually keeps.
- **φ_N.** At each pass (Figure 2), φ_N = 2(θ − ϖ_bin) − M_bin. Here θ is the star's direction, ϖ_bin the direction of the pair's long axis, and M_bin the pair's mean anomaly (how far round its orbit it has gone, in equal-time steps). In words, it compares where the star passes with where the pair is in its dance, doubled because the pair's oval looks the same from both ends. It is the angle of the strongest pull near a whole period ratio, and that pull comes from the pair's eccentricity.
- **Held and locked.** A run is *held* when φ_N makes no net turn over the run. It is *locked* when φ_N also stays within half a turn.
- **Coupled clocks.** Two clocks that each keep time by their own events and are tied to each other: here the pair's orbit, counted in its own turns, and the star's lap, counted from one close approach to the next. A lock is where they agree.
- **Swing.** φ_N's back-and-forth over a run; its size is the range φ_N covers, also called its band.
- **Rocking rate.** How many back-and-forth swings φ_N makes per pass, read from its spectrum.
- **The apse turn.** How far the star's ellipse turns per pass.
- **The beat.** Twice the apse turn: the rate at which everything the main pull does not hold turns past it. It is the spacing between neighbouring harmonics of the pull, which beat against each other at that rate (Section 7).
- **The slow beat.** With the heavy star, a long swell in the pair's orbit-averaged eccentricity, made where three turns of the angle about its circle's centre nearly cancel two turns of ψ: 722 passes at birth 7.200 and 746 at 7.300 (Section 3). It is not the beat above.
- **The kit's tolerance.** The integrator's accuracy that every run uses unless it says otherwise: relative 10^{-9}, absolute 10^{-11} (Appendix A.1). The tight tolerance is a hundred times tighter.
- **Door.** The side of an island from which its stars escape: a side, not a lock and not a passage.
- **Threads, walls and rooms.** Watched for 6,000 passes at the pair's eccentricity 0.25, the island is divided by *threads*: thin stretches of birth, from under 0.002 to about 0.01 wide, where stars lose their hold soonest. A thread that divides the island is also called a *wall*, and a *room* is the stretch of births between two walls, whose stars keep their hold for a time typical of that room. A wall here is only a dividing line in birth, not a barrier of energy.
- **Island.** A stretch of births whose stars are held. Island 7 lies near seven of the pair's periods per orbit and island 8 near eight; an island's *chaotic layer* is the thin chaotic band along one of its rims.
- **Pass and orbit.** The star makes one pass per orbit, so a count of passes is a count of its orbits.
- **ψ.** The angle the circular part of the pair's pull would hold. In the island it goes round while φ_N stays put (Section 5).
- **Locks and chains.** The rocking can lock to a fraction p/q of a cycle per pass; the locks that share one denominator q make a chain.
- **The atlas, start lines and slices.** The atlas is a map of 5,656 runs of 600 passes: 101 births, seven starting angles within 0.05 turn of φ_N = 0, and eight of the star's eccentricities from 0.20 to 0.34, with the pair at 0.10. A start line is one starting angle's row of births across the island; a slice is such a row at one setting of the eccentricities.
- **Drag.** What the hold does to the slow turning of the star's ellipse. Free of the island the ellipse turns about 0.0128 of a turn per pass. Held, its turning falls in a straight line with the run's *grip*, the average of cos(2π φ_N) over its passes: slowest at the centre, two-thirds of free, and faster than free in the widest swings near the rim. The drag's size is the slope of that line (Section 9).
- **Kick.** With a heavy third star, the nudge each close approach gives the pair's eccentricity averaged over its own orbit: up or down by an amount that follows sin 2πφ_N at that pass. The kick's size is how large that nudge is (Appendix B; Section 13).

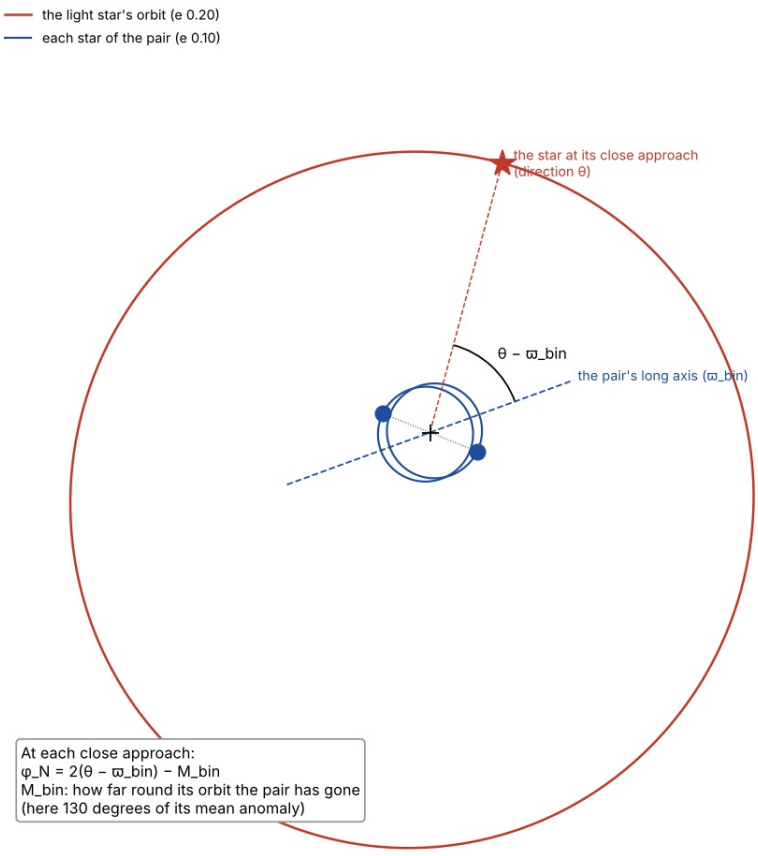


Figure 2. The setting. The system from above, to scale. The pair on its oval track, the light star's larger ellipse around it, the star's close approach marked, and the angles that make φ_N at that moment: the star's direction against the pair's long axis, and how far round its own orbit the pair has gone. A schematic, not a run (drawn 20261014_090804).

The pair's 0.10 and the star's 0.20 are where the runs start. Later sections turn both, and say so each time.

3. How we read it, and how we guessed

Timing a trapeze act means watching every catch, knowing how good the stopwatch is, and knowing which patterns the watcher only imagines. So this section sets out the readers and their floors, how the eye found patterns in the pictures, how each guess was written down before its run and then counted, and what the arithmetic fakes.

Each run integrates all three bodies directly, with no averaging and no expansion. It reads the five angles at each close approach. Angles are tracked by counting whole turns of the pair between passes from the lap ratio, so that an angle that slips by more than half a turn between readings is never misread. Every reader was first shown answers known by construction: exact Kepler orbits, synthetic rocking at known rates (Figure 3), synthetic locks. Every result below is from a reader that passed. Chaos is tested by twins: each start is run again with its birth larger by 10^{-10}, and the twins are called chaotic when they part by more than a thousandth of a turn.

Most of what follows is read from runs of 600 passes, and twins see chaos only when it is fast enough to part them within the run. That sets a floor for the twin test. Twins started 10^{-10} apart reach a thousandth of a turn only after about 16 e-foldings of their gap, so in 600 passes they see chaos only when the gap e-folds in fewer than about 37 passes; slower chaos reads as regular. The arithmetic has a floor of its own. Run again with every tolerance a hundred times tighter, a regular birth's net turns over 600 passes move by about 10^{-4} turn, far below the one turn that decides held and the half turn that decides locked. Our fidelity checks, which ask a tool to reproduce a run of record within 2×10^{-6} turn, therefore test reproducibility on one machine's arithmetic, not accuracy to that digit. The twin test is not blinded by that floor, because twins share their arithmetic: started 10^{-10} apart on a regular orbit at the pair's eccentricity 0.10, they end 600 passes within a median 1.5×10^{-6} turn of each other, far below the error either run carries alone. At the pair's eccentricity 0.10 the outermost held births at both rims of island 7 stayed held for 6,000 passes. At 0.25 the island read over 600 passes did not last: run for 6,000 passes with their twins, most of its births

lost their hold, and every twin pair, held or not, parted exponentially (Section 6). At 0.15 and 0.20 the islands last: run for 6,000 passes with their twins, every inner birth of both islands held, and at 0.15 every inner pair of twins drifted apart only in proportion to the pass count, as twins on a regular orbit do (Section 6; Appendix C).

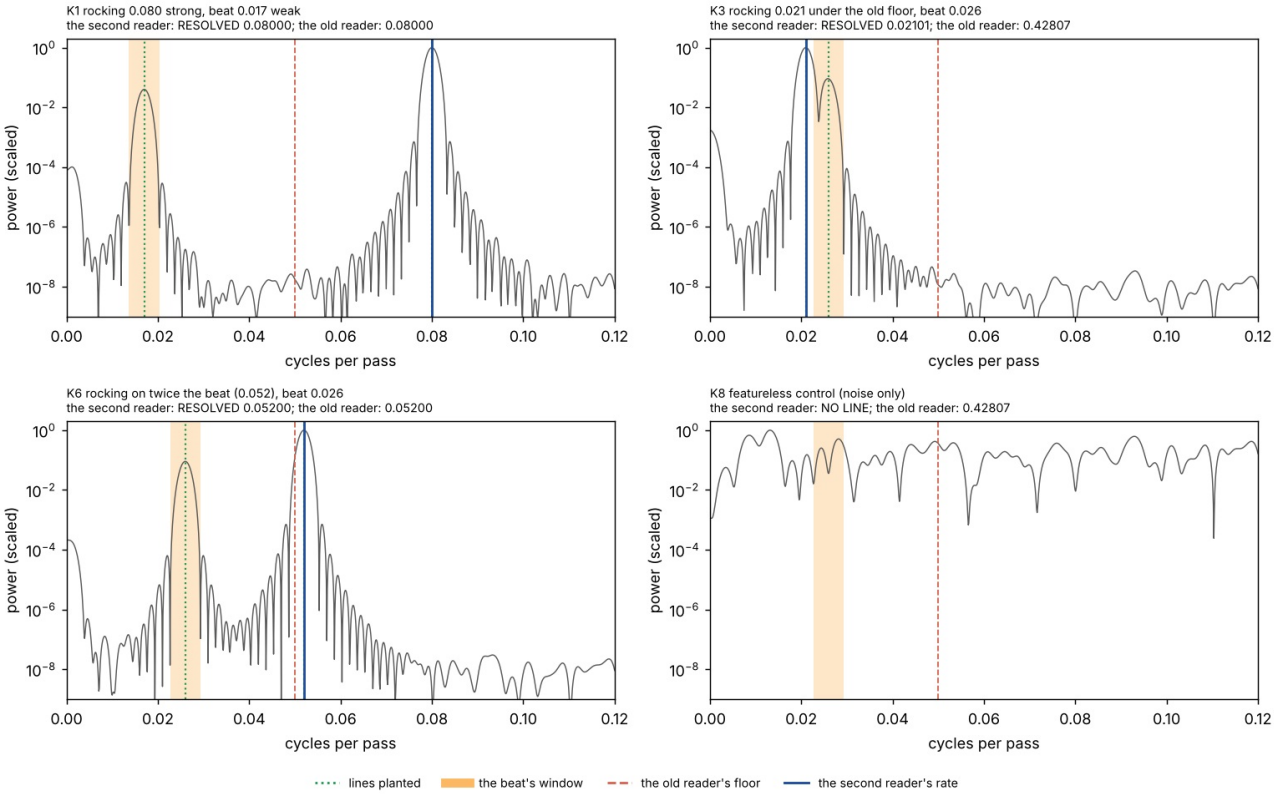


Figure 3. The reader's known-answer test. Four made-up series with lines planted by construction, their spectra, the beat's window, the old reader's floor at 0.05 cycles per pass and what each reader returned: a rocking beside the beat, a rocking below the old floor (which only the second reader finds), a rocking on twice the beat, and noise alone. The four are the second reader's own known answers K1, K3, K6 and K8 (drawn 20261014_091008).

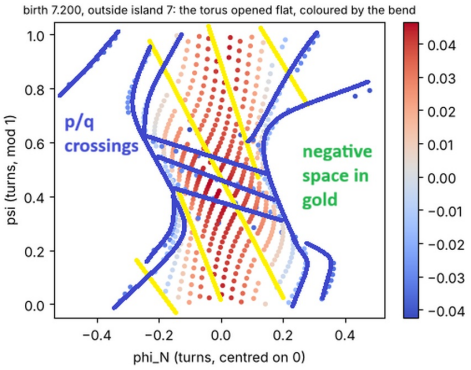
The way of working, reading a hold as a count of whole turns instead of a size, showing every reader answers known by construction before trusting it, and writing each test down before its run, is the one set out in the author's field guide (Schulz 2026a). An angle that lives on a circle, read by the whole turns it makes, is the circle-valued kind of function of Novikov's theory, whose turn counts the author's translation companion reads in its first chapter (Schulz 2026b); the link is one of kind, not a numerical bridge. Methods and tests are in Appendix A, and the code in Appendix F.

How the eye found it

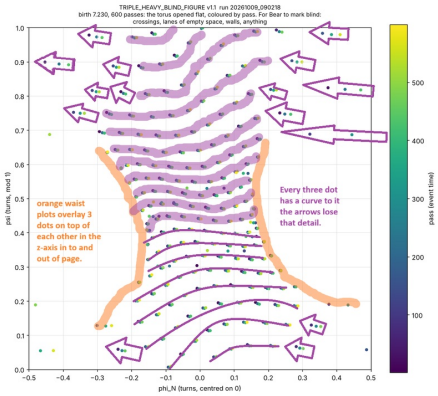
A note from the author. The next couple of pages are about how a human can help an AI interpret visual data. What I learned was that a human can see and notice more nuanced patterns in data. The best thing a human and an AI can work together on is how to interpret visual data. So each of our papers started learning how to see patterns in the soup. I don't stare at data plots regularly, but the human eye notices things (even when they aren't there). But between Claude and me we have a pretty good system figured out. It is more work on the human's side, but it produces results. As I shared with Claude, even a few newly observed elements or synergies can take a project in unexpected and wonderful directions.

Several of this paper's results began as a picture looked at by eye, and from 9 October on this became a habit of the work. Claude drew a run plainly, large and with no title across it, and sent it; the author looked at it, sometimes squeezed or tilted on his own screen, marked what he saw and said it in a line; Claude wrote the marks down in his words and, after an early miss in translation, read its translation back to him; the read was registered as a test before anything was measured; then the numbers decided. Claude's own reads by eye went through the same loop, and were withdrawn as plainly when they failed. Figure 4 shows six of his marked pictures in the order he marked them, Figure 5 the zoom-out that his first marks led to, and Figure 9 (right) his marked strobe picture. The aim was never to find every pattern in a picture. In the author's words, the important thing is not finding every one, but finding a few new observations that connect to other observations.

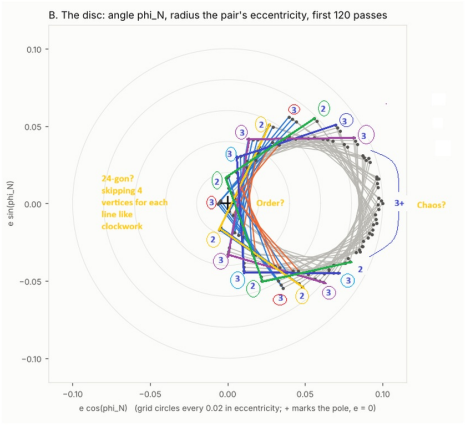
(a) 9 October: two angles opened flat (birth 7.200)



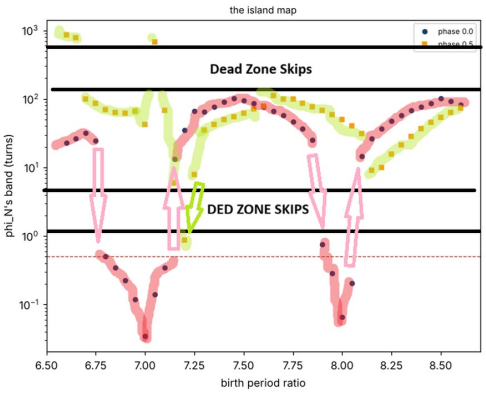
(b) 9 October: a picture drawn to be marked blind (birth 7.230)



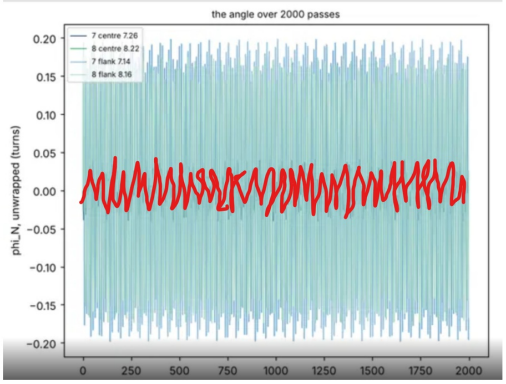
(c) 11 October: the maypole, 'like clockwork' (birth 7.200)



(d) 11 October: 'Dead Zone Skips' on an old island map



(e) 15 October: 'a very regular pattern' inside the centre



(f) 15 October: the ratio atlas, marked after its test

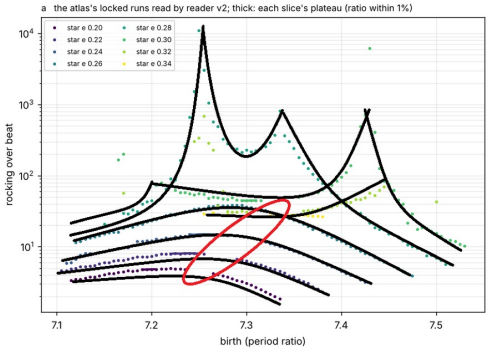
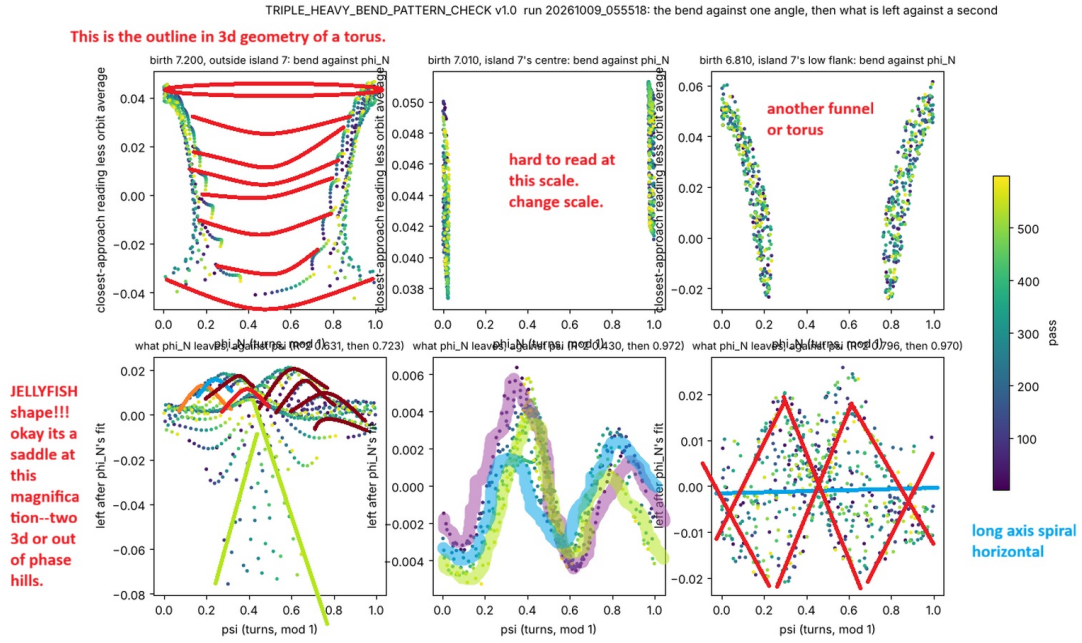


Figure 4. How the eye found it. Six of the author's working pictures in the order he marked them. Each was marked by eye and its marks written down as tests before anything was measured from them, except (f), whose marks were measured afterwards as a look and not scored. Each is cropped to one panel and scaled, with no mark moved, added or removed. (a) Two angles of one run opened flat, birth 7.200 (run 20261009_071003). (b) A picture drawn for him to mark blind, birth 7.230 (run 20261009_090218). (c) The maypole: one run at birth 7.200 drawn on a disc, angle φ_N and radius the pair's eccentricity (run 20261011_201030). (a) to (d) are runs with the heavy third star of Section 11. (d) Probe 25's island map, φ_N's band against birth (run 20261006_213610). (e) The centres' angle over 2,000 passes, from his screen (run 20261006_204230). (f) The ratio atlas, drawn to test his line 'the ratio stays fixed, the rates don't' (run 20261015_140219). The tests, told in the text and Table 1, are runs 20261009_085502, 20261010_010034, 20261010_012847, 20261012_033532, 20261012_062504 and 20261015_133950. Set side by side by triple_paper_figure_eye (run 20261025_033626).

His first marks, on 9 October, were on the first pictures of the bend: the pair's eccentricity read at each close approach less its average over the pair's own orbit, for 600 passes of three heavy-star runs (Figure 5a). He drew the outline of a torus, a surface of two angles, and wrote of the centre's run that it was hard to read at that scale: change scale. Then he bet that if we zoomed out we would see the weave more clearly. Written down in those words before it ran, the run at birth 7.200 was followed for 6,000 passes, ten times as long (Figure 5b), and zooming out made the difference he expected. Set on the two angles, each reading's bend differs from its five nearest neighbours' by 0.1126 of the bend's whole spread over 600 passes and by 0.0603 over 6,000, where scattered noise gives about 1.

(a) 9 October, 600 passes: his marks on the bend's first pictures (births 7.200, 7.010, 6.810)



(b) The zoom-out he asked for: birth 7.200 over 6,000 passes, unmarked

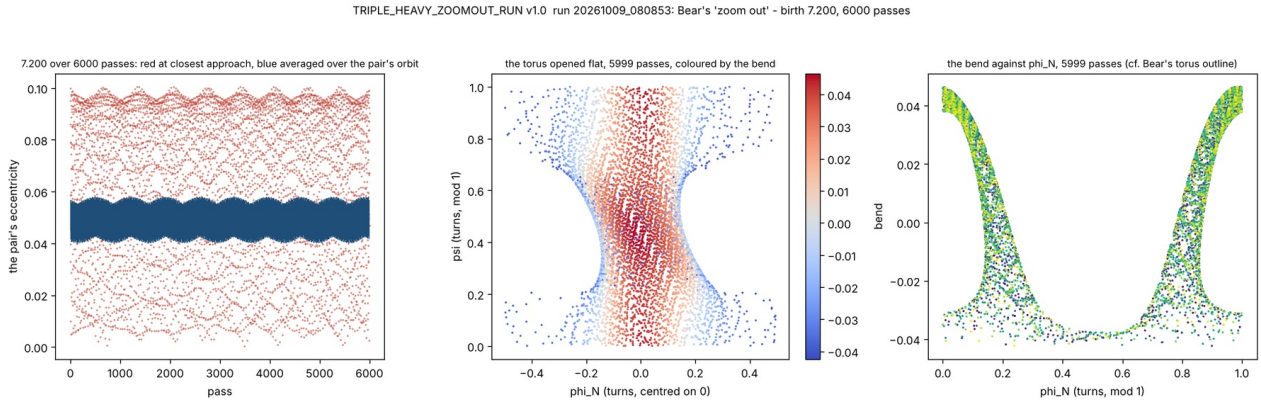


Figure 5. Zooming out. (a) The first pictures of the bend, 600 passes of three heavy-star runs at births 7.200, 7.010 and 6.810 (run 20261009_055518), as the author marked them on 9 October: the bend against φ_N (top) and what is left of it against ψ (bottom), with his 'outline in 3d geometry of a torus' and, at the centre's run, 'hard to read at this scale. change scale.' (b) The zoom-out he then asked for, birth 7.200 followed for 6,000 passes (run 20261009_080853), unmarked: left, the pair's eccentricity read at each close approach (red) and averaged over its orbit (blue); middle, the two angles opened flat, coloured by the bend; right, the bend against φ_N, his torus outline filled in. Both are as drawn, scaled only. Set one above the other by triple_paper_figure_zoom (run 20261025_180150).

On the left of Figure 5b the pair's eccentricity read at each close approach looks like a scatter of dots, and is layer upon layer of curves. It is not chaos: over 600 passes the run's twin, started 10^{-10} further out in birth, stays with it, and with every tolerance a hundred times tighter its readings over 6,000 passes are a rotation of two angles: the angle of the pair's eccentricity vector about the centre of its circle, the frame in which the author read his clockwork on the

maypole, and ψ. The readings come back near themselves on a ladder of gaps, 1, 3, 6, 9, 19, 28, 47 and 701 passes, and at every rung, measured in that frame, as close as the two-angle rotation gives, within 10%. The Rider in Two Waves session, reading the pixels of the 600-pass picture blind, found the dots linked 5 passes apart, and on the data there are 242 such links against 66.3 when the passes are shuffled. The slow swell along the band of the orbit average is the slow beat, 722 passes at the tight tolerance and a little shorter in this run at the kit's: three turns of the one angle against two of the other, nearly cancelling. Near returns, not exact ones, are what lay the dots down in layers, and run five times longer, for 30,000 passes, the surface grows smoother still (0.0251 by the measure above). Followed at the tight tolerance for 30,000 passes, it stays on its rotation out to the rotation's next near returns, 4,328 and 13,732 passes, at 0.94 and 1.57 times the rotation's own closeness. The 1.57 is the tight run's own slow drift: its rates still move, ψ's by 2.8 × 10⁻⁶ over 24,000 passes, and the pairs of readings centred on the run's middle, where a steady drift cancels, come back as close as the rotation does, 1.00. Away from the middle the loosening grows on both sides as a steady creep predicts, with nothing fitted, within 15% at every 2,000 passes of distance. At a tolerance ten times tighter still, followed for 60,000 passes, the loosening is gone from the whole run: at the rotation's near returns after 15,181, 15,882 and 31,063 passes, all pairs of readings come back as close as the rotation does, 0.98, 0.99 and 0.93 times. Its cleaner rates move the near returns beyond 701 passes: by them the rotation comes back at 5,076 and 10,105 passes, where the 30,000-pass run's own rates put it at 4,328 and 13,732; the near return at 701 holds in both. A twin started 10⁻¹⁰ further out in birth drifts away in a straight line, 6.4 × 10⁻⁶ turn after 60,000 passes, as two neighbouring rotations do; chaos would pull them apart ever faster. Two rules that would build each long rung from earlier ones were written down and tested; each missed at least one rung, and each fits about one random pair of rates in eight to ten. The long rungs are where the orbit's two rates line up, and no building rule tested so far adds to that. Nor is the 30,000-pass run's slow approach after 4,328 passes special: over the wait to 15,000, 79 of 200 random pairs of rates gain no more on their closest return than this orbit does. How alike the two halves of a stretch of passes fill the band rises and falls as the stretch is lengthened, in time with the slow beat: on a second birth, 7.300, whose tight-tolerance run no one had looked at before the test was written, it swings every 736 passes against that orbit's slow beat of 746. Not every test on the way held, and the misses are kept. A rotation read in φ_N, which slides and jumps where the angle about the circle's centre turns evenly, did not give the data's near returns; two redesigns asking for the same list of near returns each missed by one rung; over 30,000 passes at the kit's tolerance the data fell far from any rotation with fixed rates, until the tight run showed the arithmetic moving the rates; and asked first at exactly whole beats, the evenness test missed before its redesign held.

On 9 October, on the two angles of one run opened flat into a square (Figure 4a), he marked strings of dots in blue, three 'p/q crossings', and lanes of 'negative space in gold'. With them he taught the method: lay a grid over the picture, ask each dot its history (its slope, the change of its slope, and the change of that), and follow its neighbours out to five cells. Built literally as a tool, his change of change of slope found three lines across the band's waist where he had drawn his crossings, a description, since it was seen after the run. His gold lanes were not seen, because the sensors built for them looked only five cells each way. The same day a picture drawn for him to mark blind (Figure 4b) came back with rows, arrows and an orange waist where, he wrote, three dots lie on top of each other. They do: the dots are triplets, one place visited every 201 passes (575 of 599 dots). His arrows lie along φ_N. His rows arch below the waist and lift above it, 6 of 7 and 6 of 8, one row short of the bar we had set. And the visits pile up at his walls, their step 0.778 of its usual size, smaller than 2,000 random sets of dots ever gave (never below 0.856), but short of the half we had asked for.

On the maypole (Figure 4c), one run drawn as a disc, he saw a star polygon, 'skipping 4 vertices for each line like clockwork'. Measured over the first 120 passes, the median look steps 0.2154 turn about the circle's own centre, against his 5/24 = 0.2083, and 0.2579 with the order shuffled: the inscribed-angle theorem, which makes the step about the centre twice the step about the pole. He read the picture in the circle's frame and the tools read it in the pole's, and both were right. On an old island map (Figure 4d) he drew two bands that no run lands in, 'Dead Zone Skips'. On runs he had never drawn on, 3 of 484 land in the lower one, and above the upper one 1,462 of 1,475 rows agree that the only runs there are those whose pair was knocked into a new orbit.

On 14 October the author added ways of looking of his own. A run's readings, drawn once a pass with every pass a dot, were looked at whole, then squeezed along the passes, tilted to a grazing angle, and outlined by the density of their dots about an axis of symmetry, with the bands of dots and the empty rungs between them marked apart. What the eye found was written down as a guess before any number was read, and then measured. At the pair's eccentricity 0.25 (Figure 9, right) it found the island symmetric about φ_N = 0, pinched in the room that keeps its stars, and with its waist over the first 600 passes about 0.02 higher in birth than over 6,000; all three held when measured, and the last is the offset that Section 6 measures at 0.02 to 0.05. It also read a straight-sided narrowing, which the finer grid of Appendix C showed to be the outline of a speckle.

Under a thick band of rocking at the island's centre (Figure 4e) he traced a slow swell and wrote that there is a very regular pattern inside it. His trace has 37 highs, 53.5 passes apart, between the two centres' beats of 57.2 and 46.5 passes, and the band itself turned out to be two notes, the beat and the rocking (Section 7). And on the picture drawn to test his line 'the ratio stays fixed, the rates don't' (Figure 4f) he traced each slice and circled a diagonal of gaps.

Read afterwards, the spikes in the slices he traced are where the beat crosses zero, and his diagonal of gaps is the island’s centres, which, carried on, reach the spike at the star’s eccentricity 0.28, where the centre sits on the beat’s zero and its ellipse stands still (Section 9).

the read by eye	written down as	what the numbers said	where it went
The outline of a torus, and ‘change scale’; then ‘zoom out’ (the author, 9 October; Figure 5)	the bend a smooth surface of two angles at three births; then, over 6,000 passes, smoother than over 600	held at two births of three and missed at the third, 0.2225 against 0.2; zoomed out, held: 0.0603 against 0.1126	the bend read at closest approach (Section 11)
Lanes of empty space, in gold (the author, 9 October; Figure 4a)	lanes that the tool’s empty cells should see	not seen: the cells looked only five cells each way; the miss was in the translation	settle each translation with the one who marked, before the run
Three reads of our pictures from their pixels alone, the data held back (the Rider in Two Waves session, 9 October)	the same claims, judged on the numbers behind the pictures	three of three held, two with their size off	the first entries of a scorecard of eye against ruler
A star polygon ‘like clockwork’ (the author, 11 October; Figure 4c)	a step near 5/24 of a turn about the circle’s own centre	held: 0.2154 against 0.2083; shuffled, 0.2579	the inscribed-angle theorem
‘Dead Zone Skips’ (the author, 11 October; Figure 4d)	the two zones, on runs he never drew on	held: 3 of 484 runs in the lower zone; on the upper, 1,462 of 1,475 rows agree	the heavy star’s island maps, read on probes 35 and 36 (this section)
Symmetry, a pinch, and a waist that moves, after squeezing and tilting (the author, 14 October; Figure 9, right)	three tests, each read at two outline scales	all three held, the last exactly at its bar	the offset of Section 6
A straight-sided narrowing (the author, 14 October)	a V narrowing to a pinch	not borne out: on a grid five times finer, the outline of a speckle	Appendix C
A very regular pattern inside the centre (the author, 15 October; Figure 4e)	the beat, and the centre’s angle a chord of two notes	held: 37 highs, 53.5 passes apart; two notes at 44 of 51 runs	Section 7, ‘Two notes’
‘The ratio stays fixed, the rates don’t’ (the author, 15 October)	the ratio of rocking to beat steady across a plateau at six slices of eight	over 600 passes, steady at three; over 2,400 at the star’s 0.20 to 0.24, every birth within 1% of the median while the rates spread 2.7 to 3.2 times as much	the slide (Section 8); the beat’s zero near 0.28 (Section 9)
‘Fractals and fractals and fractals’ (the author, 15 October)	chains of smaller islands at simple fractions	the chains are there, 121 locked runs at the fractions against 34.5 at decoys, and odd chains fuller than even	the odd chains of Section 8
A shark’s fin on the drag (the author, 14 October; Figure 13, left)	the fin on nine lines	its tip at the island’s edge on 9 of 9; its outline on 4, 1 and 6 of 9	Appendix C
One curve of one quantity, crossing a tenth near 0.49 (Claude, 16 October)	one curve across fifteen dampings, and where it crosses	held: median crossing 0.4905	the dial of Section 4

Table 1. Reads by eye, what each was written down as before its run, and what the numbers said.

Counted, the eye did well but not always, and it matters which kind of read it makes. The 74 picture-reads then in the register, most of them the author’s and the rest Claude’s and the Rider in Two Waves session’s, were sorted blind into reads that name a relation (where one thing sits against another, which way something turns, which side something is on), reads that name a shape (a V, a floor, a ledge, a fin), and ten that name neither. The sorter saw only their wording, with every word that gave away a result taken out. Of those that could be scored, relation reads came out as written 19 times in 30 (0.633), shape reads 11 times in 28 (0.393): a difference of 0.240 against the 0.2 we had registered, narrowly (run 20261017_040300). The fin in Table 1 is the same lesson on one picture: where its tip sits is a relation, and held on all nine lines; its outline is a shape, and mostly did not. So the rule for reading with the eye is to trust where it puts things before the outline it draws.

The teaching went both ways. The author taught the project how he looks: zoom out and zoom in (Figure 5); lay a grid over the picture and ask each dot its history; mark the empty space as well as the dots; squeeze one axis until columns of dots become one outline; tilt the screen to sight along a slow trend, as a carpenter sights down a board; turn the picture a quarter turn, so that an up-and-down mirror becomes a left-and-right one, which the eye judges at a glance; and, in his words, say out loud what you see in a figure, because the AI may not have picked up the pattern you did. The project in turn built the habits that let a mark become a result: pictures drawn plain and large, their code kept, so that marks can be read back through the plot's own axes pixel for pixel; each read written in the marker's words and, after the gold lanes, agreed with him before the run; a shuffled or featureless control in the same tool; a grid several times finer before a shape gets a name; and the rule that the eye proposes and the ruler decides. Both directions learned from misses as much as from hits. The gold lanes failed in translation, and the empty rungs between the author's bands of dots on the strobe picture turned out to be the picture's own grid of births, a pattern made by the drawing and not by the stars.

How the guesses were counted

Writing each test down before its run also lets the record be counted. Every prediction registered for this paper, from its first session on, was read out of the register, in four parts on 18 October and one by one as later tests were read, one row for each reading, and counted by script (triple_ledger_count v1.1, run 20261028_210006, from TRIPLE_PREDICTION_LEDGER_2026-10-18, which has grown with the work). A prediction is what was written before its run: the registered statement or, where a guess about its outcome was written beside it, that guess. A prediction read more than once counts once, from its reading of record: restated in a later summary, read again by a mended tool after a slip, or scored on two machines. Of 490 predictions, 236 came out as written, 211 did not, and 43 could not be read, because a check failed or the test lacked what it needed; 92 were redesigns of earlier misses, and 35 of those came out as written; a prediction counts as a redesign when any of its readings names the miss it redesigns. Five tests of 20 October, on whether a run repeats bit for bit and how it answers a nudge in its last digit, were made for another question and are not counted. Forty predictions drawn at random were read again against the register by a separate reader; one outcome in the forty was wrong, and it and two more of its kind are corrected. The 46 predictions added from 26 to 28 October were each read again the same way, and all 46 stood; one of these checks also found two older predictions whose redesign was named on only one of their readings, which the script now counts (TRIPLE_PREDICTION_LEDGER_CHECK_2026-10-28). Where a section says that a registered guess was not borne out, it is among the misses here.

A note from the author. Now, there isn't a big scoreboard tracking all your guesses in your project. Or, at least, I hope not. But the method is the conversation. When you are working on a project, look at the data plots and share what you see or don't see: fragments of curves, clusters, dead zones, plots ordered by time or by event. And the cool part is that Claude often knows mathematically what the shapes mean and what should be happening when performing calculations and creating plots, so sharing your observations with Claude is the magic of the method. Claude isn't going to make fun of you for saying or seeing something that isn't there. And the process of looking at the plots is something any data scientist would and should be doing to support their AI research partner.

What the arithmetic fakes

The integrator's step-by-step errors are tiny, but they add up, and within a single run of 600 orbits they can make a frequency seem to creep. What it fakes, the slow slide of locks met in Section 8 and a turn of the still ellipse of Section 9, met in Appendix C, is set out here first, so that no later section is read through it.

One result of the method belongs here, before the physics. At the tolerance used for most runs (relative 10^{-9}, absolute 10^{-11}), the integrator's arithmetic lets the pair's eccentricity fall a few parts in 10^5 over 600 passes, and with it every unlocked rocking rate falls steadily as a run goes on: by about 2.8×10^{-8} cycles per pass in each pass on the line where the star's eccentricity 0.24 crosses an eighth, and 2.6×10^{-8} on the tenth's line. A rate locked on a fraction does not move with it. Any comparison of unlocked rates read over records of different lengths carries that fall, and a steady sweep can carry a rate into or out of a lock: between records of 2,400 and 6,000 passes it moves a rate by about 5×10^{-5}, the size of the slides first read in the shells around the locks (Section 8). With every tolerance a hundred times tighter the fall shrinks to 0.0093 of itself and the pair's eccentricity all but stops falling (5.1×10^{-11} a pass against 1.13×10^{-8}); the shells' slide disappears, and not every loss at a lock's edge that remains is the arithmetic's: those in the seventh's core are chaos, read by the twin test (Section 8; Appendix A.1). Frequency analysis reads chaos in a frequency that drifts from one stretch of a record to the next (Laskar, Froeschlé and Celletti 1992); a drift made by the arithmetic would read the same way, so long frequency records need the tighter tolerance. The fall is stronger in the longest records and at the star's higher eccentricities: over 24,000 passes at 0.28 the kit's tolerance lowers the rocking rate by 8.02×10^{-8} to 2.77×10^{-7} cycles per pass in each pass, three to ten times the eighth's line at 0.24, fastest beside the point where the centre's ellipse stands still, and makes that still ellipse seem to turn; at the tight tolerance the same runs change by at most 1.04×10^{-9} and the ellipse stays still (Section 9; Appendix C).

The same fingerprint appears with the heavy star. Over 30,000 passes at the kit's tolerance, at birth 7.200, the pair's mean eccentricity falls from 0.04899 to 0.04883 and the rates of the two angles that carry its readings drift steadily, and a near-cancellation between them magnifies the drift: the slow beat of Figure 5 shortens from 700 passes to 450

across the run. With every tolerance a hundred times tighter the rates change 200 to 300 times less and the slow beat holds at 722 passes (Section 3, ‘How the eye found it’). Loosen the tight tolerance threefold and the creep grows 5.4 times; tenfold, 25 times: it grows faster than the tolerance does, a little more steeply near the tight end. Over the tight run’s 30,000 passes it is a straight ramp: ψ’s rate rises at a steady pace, with no bend beyond its noise. Tighten the tolerance instead: three and a third times tighter than the tight run (relative 3×10^{-12}), over 30,000 passes, the creep is 6.7 times smaller, a power of 1.58; ten times tighter, over 60,000 passes, about a hundred times smaller (106 times), more than the tenfold to fiftyfold we had written down. The tighter the tolerance already is, the faster the creep falls with each further tightening, and the integrator’s own error does the same: over 6,000 passes the three bodies’ energy drift shrinks with powers of 1.46 and 1.92 over the two tenfold steps where ψ’s creep has 1.41 and about 2.0. At both looser tolerances measured, the creep moves the orbit’s two rates in the proportion a nudge in birth does, the angle about the centre’s rate by −0.49 of ψ’s for the twin against −0.54 for the creep. Yet the creep drains the energy where a nudge outward in birth adds to it, and at the tightest tolerance, over 60,000 passes, the energy’s drift turns to rising while ψ’s creep keeps its sign. Nor does the creep follow the three bodies’ angular momentum, the other quantity the equations conserve: its drift changes sign between relative 10^{-10} and 10^{-11}, where ψ’s creep keeps its sign, and at 10^{-11} it is about a six-hundredth of its size at 10^{-10}. Read from the saved states of the run at 3×10^{-12}, the energy still drains, by 2.67×10^{-13} a pass, as we had registered, so its turn to rising comes between that tolerance and 10^{-12}; the angular momentum rises there, as at 10^{-11} and 10^{-12}, by 7.0×10^{-14} a pass, more than at either. The energy alone shows that the creep is not a slide along the family of births; what it shares with a nudge in birth is that ψ plus twice the angle, the eccentricity’s second line, hardly moves under either. Twins at 7.300 give the same proportion, −0.492, and there that line moves by under a fiftieth of ψ’s change: a nudge in birth hardly touches it. The same holds above island 7: over the first 600 passes of the 41 births from 7.100 to 7.300 whose twins stay together, the angle’s rate changes by 0.211 of a turn a pass, falling 1.06 for each unit of birth as we had registered, while ψ plus twice the angle changes by only 0.011, about a twentieth as much. A second birth with its own near-cancellation tests the reading. At 7.300 it is three times the one rate less twice the other that comes close to a whole number of turns, and over 12,000 passes at the kit’s tolerance its slow beat shortens from 603 passes to 281, in steps that keep pace with the fall of the pair’s mean eccentricity (their ratio within 25% of its mean at each step) and 7.2 times faster than at 7.200, because with that sign the two rates’ drifts add, where at 7.200 they partly cancel. At the tight tolerance its slow beat holds at 746 passes, changing by under a five-hundredth of the kit’s step. The slow beat is the steadier gauge of the two: its steps spread by 0.18% of their size, the eccentricity’s by 26%. Most of the eccentricity’s spread is the leftover, at the blocks’ edges, of its own fast swing, once every 3.17 passes, and not of the slow beat, whose swing in it is only 2.6×10^{-6}; smoothed over the fast swing, its steps spread by 3.5%. That fast swing is the maypole’s own turning, seen in the eccentricity: at both births its period is that of the angle about the circle’s centre (3.17 passes at 7.300, 4.70 at 7.200), with a second line at ψ plus twice that angle, and it peaks as the hand points away from the pole, like a clock face whose pivot sits off its centre. At both births, then, the slow beat between two nearly cancelling rates is a sensitive alarm for a run’s arithmetic.

4. Before the stars: a bench of known maps

An act is rehearsed over a net before it goes up. Before any doubling was read in the stars, the method was tried on maps whose answers are known. It found them: the frictionless and damped period-doubling constants, 8.721 and 4.669, and the one dial that decides between them (Figure 6).

Hénon’s map, with and without friction

Hénon’s map is a short rule that hops a point around a page, step after step. As one knob is turned, its repeating loops double in length again and again, and the gaps between doublings shrink by a fixed ratio whose value is known: one value with friction, another without. Our method had to find both. On Hénon’s map (Hénon 1976) its doubling finder reproduced the map’s first two thresholds, worked out on paper, within 4.4×10^{-16}. Followed down the cascade, the ratio of successive gaps between doublings reached 8.721082 at the ninth doubling with the map frictionless, keeping area at every step, and 4.669197 at the tenth with Hénon’s own damping: the frictionless and damped constants (Greene, MacKay, Vivaldi and Feigenbaum 1981; Feigenbaum 1978). The first number belongs to systems like the stars, where every start keeps its own orbit; the second to systems with friction, which settle.

The dial

Real systems are seldom perfectly frictionless or strongly damped, so we asked what decides which of the two ratios a slightly damped system shows. With slight damping the ratios read frictionless at first and then fall to the damped value. Across fifteen dampings, keeping 0.998 to 0.80 of the area at each step, how far a damped cascade’s ratio departs from the frictionless one is a single curve of the area an orbit keeps over one trip round itself, whatever the damping, and it passes a tenth where that area falls to 0.49 (0.4889 to 0.4950 across the twelve dampings that cross within the doublings read). The area kept per trip is the dial that decides whether a cascade looks frictionless or damped. That single curve is the universal crossover curve van der Weele, Capel, Post and Calkoen (1986) found on the same map, a curve their abstract calls non-monotonic; by the mark of Appendix A.6, ours does not turn back.

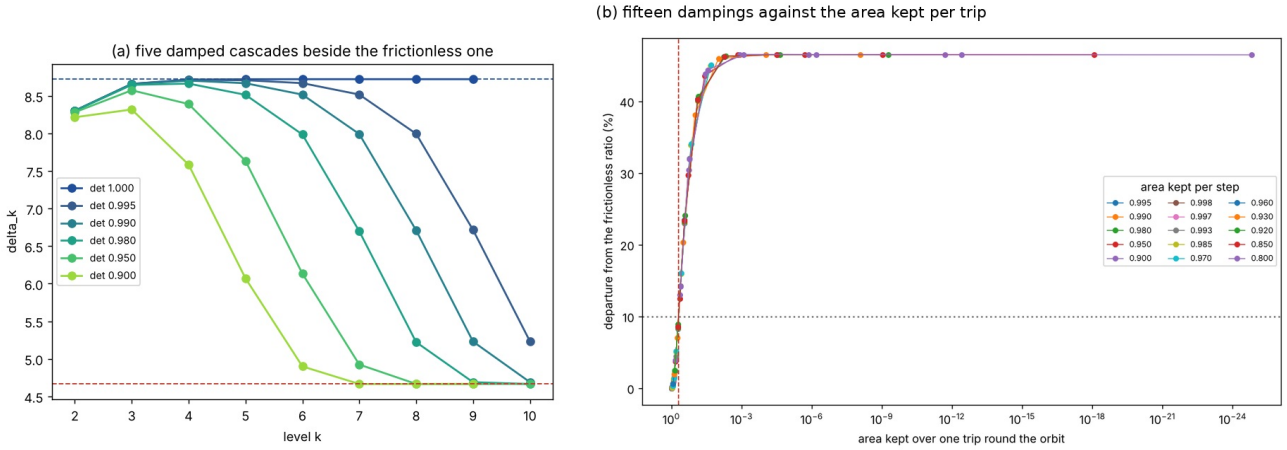


Figure 6. The friction dial. (a) The ratio of successive gaps between doublings, level by level, on the frictionless map, which keeps all its area at each step (det 1.000), and on five damped ones: each starts near the frictionless 8.721 and falls to the damped 4.669, sooner the stronger the damping (run 20261007_051639). (b) Fifteen dampings, keeping 0.998 to 0.80 of the area per step, against the area an orbit keeps over one trip round itself: one curve, whatever the damping, departing from the frictionless ratio by a tenth where that area falls to 0.49 (run 20261016_201627). Set side by side by triple_paper_bench_figure (run 20261019_164526).

A loop pushed by a slow beat

The stars' island is pushed by its beat (Section 7), a slow one, so the bench also asks what a slow push does to a doubling. Here a loop is an orbit of the map that repeats; it doubles when it gives way to a newborn loop that takes two trips to repeat.

Last, a loop pushed by a slowly turning angle, the bench's stand-in for the stars' beat, doubles once near the threshold of the same loop unpushed (3.0001470 against 3), and its newborn then locks to the push, rocking at half the push's rate, instead of doubling again; the cascade goes on only by hopping to a second loop beside it, which locks in turn. In the stars the newborn's neighbourhood empties instead of locking (Section 10).

5. Held: the island and its coupled clocks

This is the catch itself. The star is held because two clocks keep time together, the pair going round its orbit and the star going round the pair, and one angle that compares them, φ_N, stays put while every other angle turns.

Mardling's pendulum for this resonance passes a direct test in the test-particle limit, where its assumptions hold: the small-swing rocking rate is half the island's half-width within 1.1% at the pair's eccentricity 0.10, and the island sits off the whole number by twice the star's apse turn. Outside the conditions it was built for, it is off by amounts Sections 6 and 10 and Appendices B and C measure. We found the angle that holds before reading her paper, by working out which angle the strongest term of the pair's pull carries and counting its whole turns (Section 3); the angle is hers.

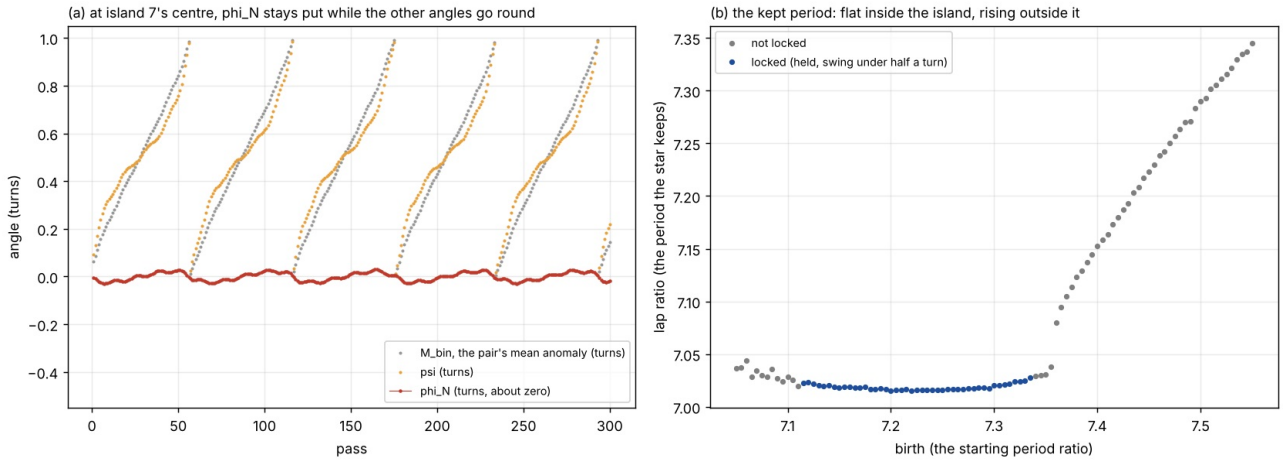


Figure 7. Held. Left, φ_N against pass for a star at island 7's centre, staying within 0.03 turn of zero, while ψ and M_bin go round; right, the lap ratio against birth, a flat step inside the island and a rising line outside it. Left, one 300-pass run at birth 7.250 in the atlas's setup (run 20261014_090911); right, the atlas at the star's eccentricity 0.20 (probe 30, run 20261007_004206).

Started at the right birth, the light star is held. On a coarse grid of births 0.02 apart, every run born between 7.12 and 7.32 locks φ_N, and so does every run born between 8.14 and 8.26; runs that hold without locking reach a little beyond. In those same runs the pair's mean anomaly goes round 5 to 8 times in 300 passes, and so does the angle the circular part of the pair's pull would hold. Neither clock is locked on its own; the lock is on the two of them together with the star's ellipse.

Inside the island the lap ratio is pinned. Births from 7.12 to 7.32 all run at lap ratios 7.015 to 7.024 (Figure 7). The pinned value is not seven. It is seven plus twice the star's apse turn per pass: 7.0185 measured against 7.0196 predicted, and 8.0219 against 8.0226 at island 8. That is the coupled clocks' agreement of the first finding (Section 12). Outside the island the star's slip goes as one over its distance from the island's centre (a log-log slope of −0.919): a free ride with no grip.

The birth is a label for the start, not the period kept. As the pair's eccentricity goes from 0.05 to 0.20, the birth at the island's centre moves from 7.155 to 7.445, while the period kept there, read from the same bookkeeping, falls only from 7.022 to 7.010.

Mardling's harmonic and her pendulum's test, in her notation and at every eccentricity read, are in Appendix C.

6. The edge, the door, and slow chaos

Every catch has an edge, where the flyer's timing is too far off for the tugs to correct. The island's edge is not alike all round. Stars leave through the side nearest the pair, and at the pair's higher eccentricity what looks like an island over hundreds of orbits turns out, over thousands, to be slow chaos divided into rooms.

Beyond the stability limit the island holds, and one side of its edge is a door: with the star's eccentricity at 0.20, 163 of the 166 recorded escapes start on the side toward the pair, a side the published stability rules do not see. Watched ten times as long, the island at the pair's eccentricity 0.25 turns out to be slow chaos divided into rooms, and the rocking's rule measures the island together with its chaotic layer.

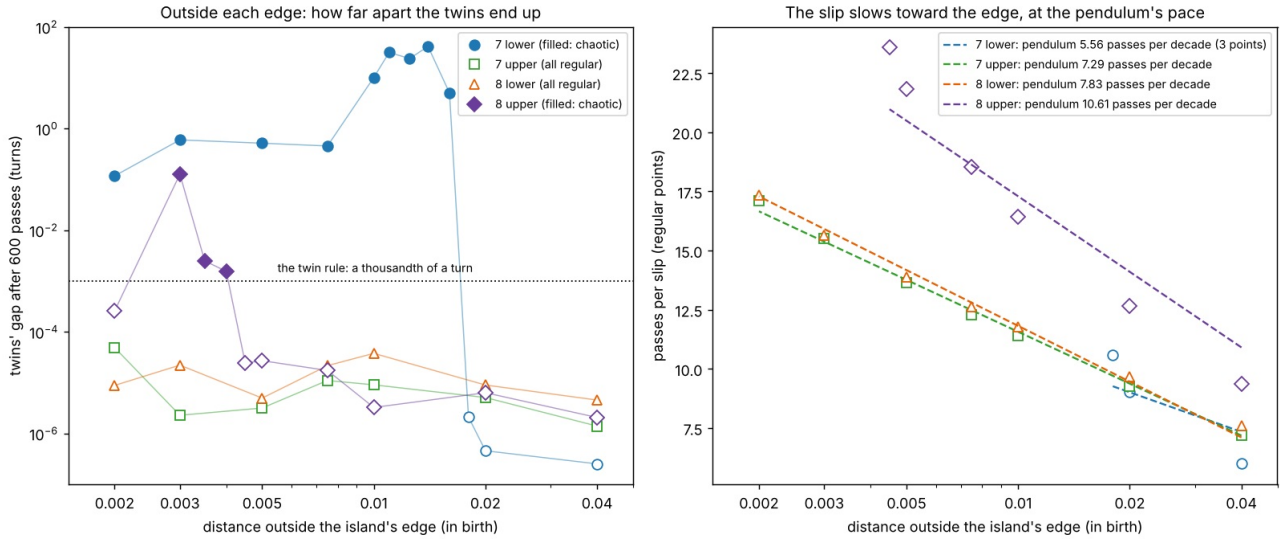


Figure 8. The edge. Islands 7 and 8 at the pair's eccentricity 0.10 and the star's 0.20, read from outside each edge. Left, how far apart twins started 10^{-10} apart end up after 600 passes (the median gap over the last 50 passes), against the distance outside the edge; filled points are chaotic by the twin rule (a thousandth of a turn, dotted). Below island 7 the chaos reaches 0.016 and is gone at 0.018; above island 8 it reaches 0.004 and is gone at 0.0045; the other two edges are regular to 0.002. Right, on the regular points, the passes a star takes to slip once round against its distance outside the edge, with the pendulum's slope from each island's width drawn through them: 7.57 passes per tenfold step measured against 7.29 forecast above island 7, 7.37 against 7.83 below island 8, and 13.6 against 10.6 above island 8, where the points bend away from a straight line beside the chaotic layer. From probes 23 and 24 (runs 20261006_201311 and 20261006_204230).

Chaos with a shape

Chaos here means something narrow: two stars started a hair apart, 10^{-10} in birth, end up in different places, so no run can say where any one star will be after many passes. It does not mean that anything can happen, or that the motion has no shape. Think of leaves on a river. Where the water runs over rocks, no one can say where a given leaf will be a minute later, yet the rapids sit at the same rocks every day, and the leaves leave them by the same bank. The chaos around these islands is like that. It has a place: a thin layer on one rim of the island and almost none on the other. It has speeds: in some places twins part within a few hundred passes, in others only after thousands. It has a direction: of the chaotic stars below island 7 that go round at all, 17 of 18 go round the way the stars above it do. And it has a door: with the star's eccentricity at 0.20, stars leave almost only from the side nearest the pair. Each of these is read from many runs at once, as counts and fractions, never from the fate of a single star.

The island's rim is chaotic. Twins started inside the edge part by up to tens of turns, while twins far outside stay within 1.4×10^{-5} turns and twins deep inside within 10^{-7}. The layer is lopsided. At island 7 it reaches 0.016 in birth beyond the lower edge (regular again at 0.018) and less than 0.002 beyond the upper. Island 8 has it the other way round, 0.004 above and under 0.002 below.

Where the motion is regular, outside the layer, the edge follows the pendulum's law. The star's slip period grows by the same number of passes for each tenfold step nearer the edge, and the island's own width predicts that number (Figure 8). At island 7's upper edge it is 7.57 passes per decade against a forecast of 7.29; at island 8's lower edge, 7.37 against 7.83. A damped lock, which creeps out as one over the square root of the distance, would have grown the slip period nearly twice as much over the same span. Read a step at a time, though, the slope is not constant. Between 0.02 and 0.04 outside the edge it runs 5% and 11% below the forecast; nearest the edge, between 0.002 and 0.003, it runs about a quarter above it (1.23 and 1.24 times). Beside a chaotic layer it is steeper still: at island 8's upper edge, beside its layer, it is 1.76 times the forecast between 0.005 and 0.0075. The forecast is met by the ladder as a whole, not at every distance. Near the edge a star slips more slowly than a frictionless pendulum would, and most slowly next to a chaotic layer.

On the slice the atlas reads, island 7's chaotic layer at the pair's eccentricity 0.10 stays bound: every chaotic birth stayed bound for all 600 passes, about 4,200 of the pair's periods, though it did not stay held.

The layer grows with the pair's eccentricity and stays on the same side. Mapped with twins at births 0.005 apart over 600 passes, counted as chaos touching the edge, held rim births included, it measures nothing at the pair's eccentricity 0.05, then 0.030, 0.085 and 0.120 in birth at 0.10, 0.15 and 0.20, the late parters counted (Figure 1). Almost all of it is on the lower rim; counted as chaos touching the edge, the upper rim carries 0.010, at 0.15 only,

though at 0.20 a few chaotic births lie just beyond the upper edge without touching it (Figure 1). At 0.20 the layer no longer stays bound: eleven of its births escaped within 600 passes. This is the picture Mudryk and Wu (2006) found for planets inside a binary, where neighbouring sub-resonances overlap further as the companion's eccentricity grows, seen here from outside the pair. Around a circular binary, the overlap of whole-number resonances, driven by the small body's own eccentricity, makes a continuous chaotic zone close to the pair (Shevchenko 2015), and for planets around binaries the overlap of the N/1 resonances matches the size of the chaotic region (Gallardo, Beaugé and Giuppone 2021). Our layer lies beyond the inner unstable region, around a single resonance (below). It is not the overlap of island 7 with island 8, nor of its multiplet's members, which lie inside it (Appendix B), though at the star's eccentricity 0.28 it joins the sea below the island (below). The held island does not grow with the layer. Its width stays between 0.26 and 0.31 in birth while the layer eats its lower side, where Mardling's pendulum would have it about double from 0.05 to 0.20. Counted together, island and layer grow from 0.26 to 0.375. Turning the star's eccentricity instead, from 0.12 to 0.28 with the pair held at 0.10, the chaos within 0.15 outside the island's edges lies below the island at every step but 0.16, where it is even on the two sides, and never shrinks: 0.005, 0.020, 0.020, 0.125 and 0.250 in birth, while the island widens from 0.080 to 0.370. At 0.28 the chaos below the island has no edge of its own. Walked on below where the scan stopped, every birth is chaotic down to 6.435, and 57 of the 63 walked escape within 600 passes; held, regular stars begin again only at 6.430, the top of the next island down. The chaos fills the whole 0.735 in birth between the two islands: there the island's layer has joined the sea between them. The ragged side follows the pair, not the star. In that scan, at 0.24 all 31 escaping runs leave from below; at 0.28, 135 of 139 do, and four leave from above, the second door already ajar.

Chaos that holds its stars, and chaos that lets them go

Not all chaos is the same. Both kinds scramble a star's path so that no run can say where it will be, but one kind lets its stars go, while the other, just as scrambled, keeps them about twenty-five times longer. The leaky kind sits on the island's side nearest the pair. With the star's eccentricity at 0.20, 163 of the 166 escapes we recorded, with the light star and the heavy one, started on island 7's low-birth side, the side nearest the pair: below the island, or, in the runs watched for 6,000 passes, inside its lower half. Within 600 passes none started above island 7, and in the heavy star's scans none started beside island 8; the escapes from above came after 5,904 and 14,278 passes, and one more, at the pair's eccentricity 0.25, left from 0.001 past the middle of the island's core as registered from 600 passes, 7.400 to 7.540 (Appendix C). One leak has turned up inside the island too: the chaotic thread through island 7 at the pair's eccentricity 0.25 (Appendix C), whose runs left just after the 600 passes the scans read. Yet there is chaos on those other sides too. At the pair's eccentricity 0.25, 18 of the 23 births within 0.15 above island 7 are chaotic, and none escaped in 600 passes. With the heavy star, island 8's ragged rim is its lower one, and stars that wander off it come back and hold again (Section 11). The chaos below island 7 lets its stars go; the chaos elsewhere holds them, though not forever. The chaos below also turns its stars the wrong way. A regular star born below the island circulates one way round and a regular star born above it the other. Of the 18 chaotic stars born below the island that circulate in the scans of 600 passes at the pair's eccentricities 0.10 to 0.20, 17 turn the way the stars above the island turn, as if carried outward past the island, and all 13 that escaped from below in those scans were turning that way when they left. The one exception is the chaotic star born farthest below, 0.10 under the island's edge. The two sides differ less in whether a star can leave than in how long it takes: about twenty-five times longer above.

The door stays on that one side only while the star's own orbit is moderately eccentric. In the atlas, with the pair at 0.10 and seven starting angles, escapes begin at the star's eccentricity 0.24, all 15 of them below the island. At 0.26 and 0.28 they still come almost only from below: 35 of 43 and 111 of 117. The rest start among stray locked births on the island's ragged lower rim, within 0.035 of its lowest lock, except three at 0.28 that leave from above. From 0.30 a second door opens above the island. It lets out 21 of 177 escapes at 0.30, 114 of 355 at 0.32, and 231 of 476 at 0.34, just short of the island's doubling (Section 10), nearly as many as the 243 that leave from below (two more leave from inside it). The door faces the pair while the star's orbit is moderately eccentric; as the star's own eccentricity grows toward the doubling, the far side opens too.

Over 6,000 passes the island at 0.25 dissolves

Most of our maps watch each star for 600 orbits; here we watched ten times as long, to see whether an island that looked solid really keeps its stars. Every birth from 7.375 to 7.580, 0.005 apart, was run for 6,000 passes with its twin, after we registered the guess that the island's core would hold. It did not. Of the 28 births from 7.400 to 7.540 held for the first 600 passes, only six still held at 6,000, all between 7.465 and 7.505, with one more lower down at 7.395 (Figure 9). Every pair of twins parted, held or not, and parted the way chaos does: on a log-log plot the gap between them climbed with slopes of 5 to 12, the mark of exponential parting (Appendix A.4), where twins on a regular orbit drift apart only in proportion to the pass count. Most gaps first crossed a thousandth of a turn between passes 500 and 1,050, around or beyond the end of the 600 passes the scans read. Ten stars escaped, all born at 7.460 or below, below the island or inside its lower half, after 442 to 5,292 passes; none of the 24 born from 7.465 up escaped, though 18 of them lost their hold. Which births these are is not itself a result. Run again with every tolerance a hundred times tighter, one birth that had lost its hold lost it again but at a different pass, and another that had lost it held throughout (Appendix A.1). Beyond the time over which twins stay together, a run is one sample of the chaos, not the birth's fate;

what carries over is fractions and patterns across many births. The pattern is this. How long a birth keeps its hold dips sharply in three places. Two are the thin escaping threads already found over 600 passes, at 7.385 and 7.420 (Appendix C); the third, near 7.450 to 7.460, is seen only in this run, and a finer grid has since resolved it (Appendix C). Between the dips the old island is divided into rooms. The median time a birth keeps its hold is 4,122 passes in the room between the first two threads, 2,018 passes from 7.425 to 7.460, the whole 6,000 in the room from 7.465 to 7.505 (six of its nine births held throughout, a share the finer grid cuts to two in five), and 1,345 on the upper flank. The walls are not all alike. Each birth's twin is a second sample of the same chaos, and counting both, the first two threads are the fastest leaks (7.385 escapes after 442 and 462 passes, 7.420 after 712 and 641). The third leaks slowly and, on this grid, marks where the leaking stops: of the 36 runs born from 7.375 to 7.460, 25 escape, and of the 48 born from 7.465 up, none. Inside the rooms the stars stay longest in the middles; below the third thread only three births lose no star in either run, 7.405 and 7.410 in the middle of the first room and 7.435 in the middle of the second. Read over the first 600 passes, the island still draws the mark of a regular centre: each run's swing shrinks toward a birth near 7.525 and grows again past it. Over the long runs that centre is not where the island survives. On this grid the births nearest it lost their hold, and the narrowest swings sit 0.02 to 0.05 lower, across the room that keeps its stars. So at 0.25 there is no lasting island on this grid, only chaos slow enough to look like one for a few hundred passes, partitioned by thin threads into rooms that empty at different rates, and the one-sided door now runs through the middle of where the island was.

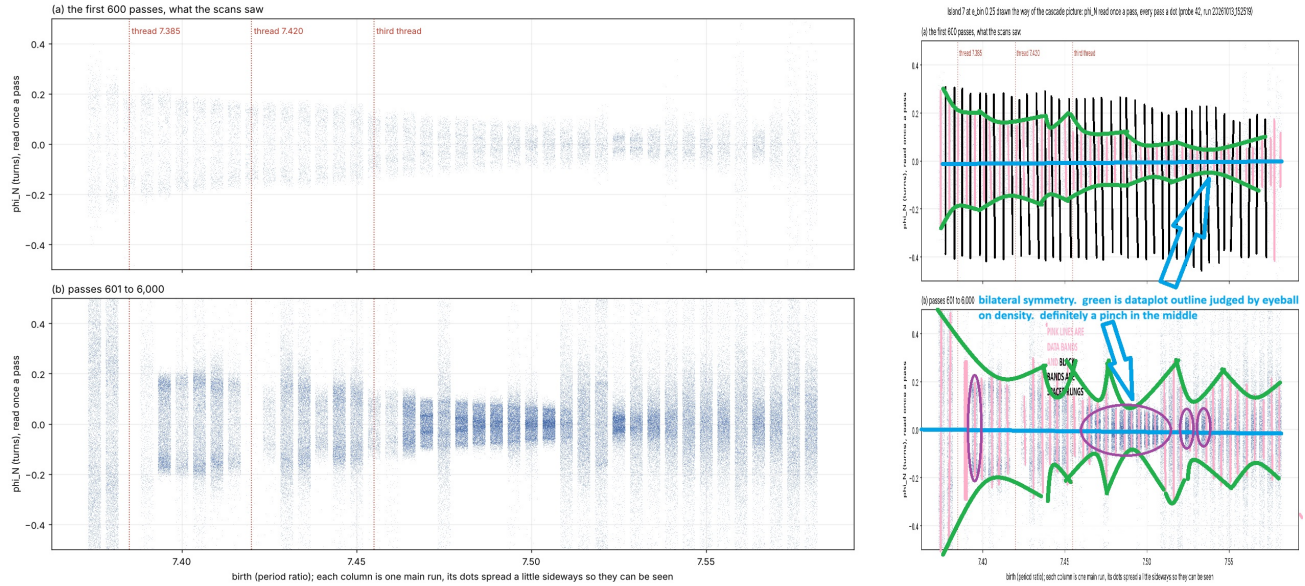


Figure 9. Slow chaos that looks like an island. Island 7 at the pair's eccentricity 0.25 read once a pass, every pass a dot, birth across: above, the first 600 passes, where the band runs smoothly across the threads; below, passes 601 to 6,000, where the threads show as gaps and the rooms between them take different shapes. From probe 42's main runs (run 20261013_152519). Right, a working copy of the same picture, squeezed sideways and marked by eye by the author before anything was measured from it: a line of symmetry (blue), an outline drawn by the density of the dots (green) and the crowded rooms (purple), with a working note at its foot left off. The three guesses written from these marks, that the island is symmetric about φ_N = 0, that it pinches in the room that keeps its stars, and that its waist sits a little lower in birth over 6,000 passes than over 600, all held when measured (Section 3). The two halves set side by side by triple_paper_figure7_marks (run 20261021_052614).

What the published rules and maps see

Astronomers already have rules of thumb for which orbits around a pair of stars are safe; here is what those rules say about our island and its one-sided door. Three published stability rules weigh the island. Mardling and Aarseth's (2001) puts all of it on the unstable side: at the island's lower edge and centre the star's closest approach is 0.93 to 0.97 of its critical distance. Vynatheya, Hamers, Mardling and Bellinger's (2022), which adds the pair's eccentricity, puts all of it on the stable side, at 1.07 to 1.15 of its critical value. Its stable side also takes in the eleven births in the layer that escaped at the pair's eccentricity 0.20 (1.063 to 1.071). Holman and Wiegert's (1999), fitted for test particles started on circular orbits around a binary, puts the edge of the inner unstable region at period ratios 4.03 to 5.36 at our pair's eccentricities 0.05 to 0.25, so it too puts the island, near 7, on the stable side. But beyond that edge they found islands of instability at single whole-number resonances, at the innermost one beyond it, from the 3:1 to the 9:1 as the masses and the pair's eccentricity varied, and thought instability might arise farther out, at higher resonances. For our pair the innermost is the 5:1 or the 6:1. We take island 7's chaotic layer, and the escapes from it, to be akin to that instability, at a resonance beyond the innermost; our star, unlike their particles, starts on an

eccentric orbit. The first two rules are smooth fits, and the second was trained on third bodies of at least 1% of the pair’s mass and judged over 100 outer orbits; the island and its rim are the kind of resonant step such a fit smooths over, as Holman and Wiegert’s islands are for theirs. In the layer, twins part with an e-folding time of about 12 passes at the pair’s eccentricity 0.10, about one rocking period, falling to about 8 and 5 passes at 0.15 and 0.20. That is the direction of the regime Batygin, Mardling and Nesvorný (2021) describe where resonances overlap, in which the Lyapunov time approaches the orbital period. Mapping planets started on circular, coplanar orbits around binaries over 10^5 of the pair’s periods, Quarles, Satyal, Kostov, Kaib and Haghighipour (2018) report the mean-motion resonances beyond the limit as unstable gaps, with islands of stability about halfway between them, as earlier maps had found. Doolin and Blundell (2011), mapping test particles around binaries over 5×10^4 of the pair’s periods, had found striations of instability they took to be resonances between particle and pair, and Quarles and Lissauer (2016), mapping circumbinary test bodies around α Centauri AB, found the pair’s resonances bringing instability at select intervals.

Our star, started at eccentricity 0.20, holds an island at the resonance and keeps its instability to one side. Started on a circular orbit instead, with the pair’s eccentricity at 0.20 and the star’s mass as before, it neither escapes nor is pumped anywhere across the 7:1. Over about 4,500 to 5,000 of the pair’s periods (660 of the star’s laps), none of 82 runs from 6.80 to 7.60, 0.02 apart at two starting directions, escapes, and none is pumped: the largest eccentricity any reaches is 0.053, where we had set 0.15, three-quarters of the island’s starting 0.20, as the mark of a pumped orbit. We read that 0.053 as the pair’s short-period pull on a round orbit. The test was registered, with the guess that no gap would open (run 20261018_201130; a planted check in its tool had kept the bar from before the trial run, so the same runs were read again by the mended tool, run 20261018_203454). Started at 0.20, on a grid 0.005 apart at one starting direction, 11 of 91 runs escape within 600 passes, from births 7.175 to 7.255 (run 20261008_070206); eight of the circular runs lie in that window. Run on for 10^5 of the pair’s periods, Quarles and colleagues’ length, the same circular starts at the first starting direction, 41 births from 6.80 to 7.60, all stay, and none is pumped: the largest eccentricity is again 0.0527, reached early and never passed (four runs, 20261018_205711, 20261018_221712, 20261018_231408 and 20261019_001051, read together in run 20261019_010756; registered, with the same guess). At this pair, then, at the 7:1 and over the maps’ own length, a star started circular finds no door: the door, and the island’s one-sided edge, belong to a star with an eccentricity of its own. That fits Holman and Wiegert’s finding that, for particles started circular, the islands of instability beyond the edge sit at the innermost whole-number resonance, which for our pair is the 5:1 or the 6:1.

The rocking measures the island with its layer

Mardling’s pendulum links how fast a held star rocks to how wide its island is, so the rocking can be used to measure an island; this part asks what exactly it measures. Mardling’s relation between the centre’s rocking and the island’s width measures the island together with its chaotic layer, not the held island alone. At the pair’s eccentricities 0.15 and 0.20 twice the centre’s rocking rate exceeds the held island’s half-width by 16% and 34%; with the chaotic layer counted into the island, the two come within about 10% at every eccentricity from 0.05 to 0.20, which was noticed after the run. Registered forward, the rule missed narrowly at 0.25 (13%, against the 12% registered) and held at 0.225 (9.4%, inside the 10% registered); with the star’s eccentricity turned it holds at 0.20 and 0.24 and fails at 0.28, where the layer has joined the sea below the island (Appendix B). A resonance sized from its libration rate may therefore be sized to where its chaos ends rather than to where its stable core ends; that is tested here in one system.

The island at the pair’s eccentricities 0.225 and 0.25 over 600 passes, the door’s escapes over 6,000 and 30,000 passes, the rims that lend and take back their stars, the islands at 0.15 and 0.20 over 6,000 passes, and the finer grid at 0.25 are in Appendix C.

7. How a lock rests, and the beat

On the trapeze, a flyer held in the catch rocks about it, arriving early and then late. A locked star swings back and forth about a resting angle. This section shows that it swings like a frictionless spring, not like a damped door that settles, and then listens closely to the motion at the island’s very centre.

Drawn in the plane of the star’s eccentricity vector, every locked run is an arc swinging back and forth across one axis (Figure 10). Its resting angle lies within 1.1° of that axis for swings from 0.06 to 0.46 turn. A damped lock behaves differently: its resting point slides a quarter turn toward the edge as it nears the edge. These locks rest like springs: conservative, not damped.

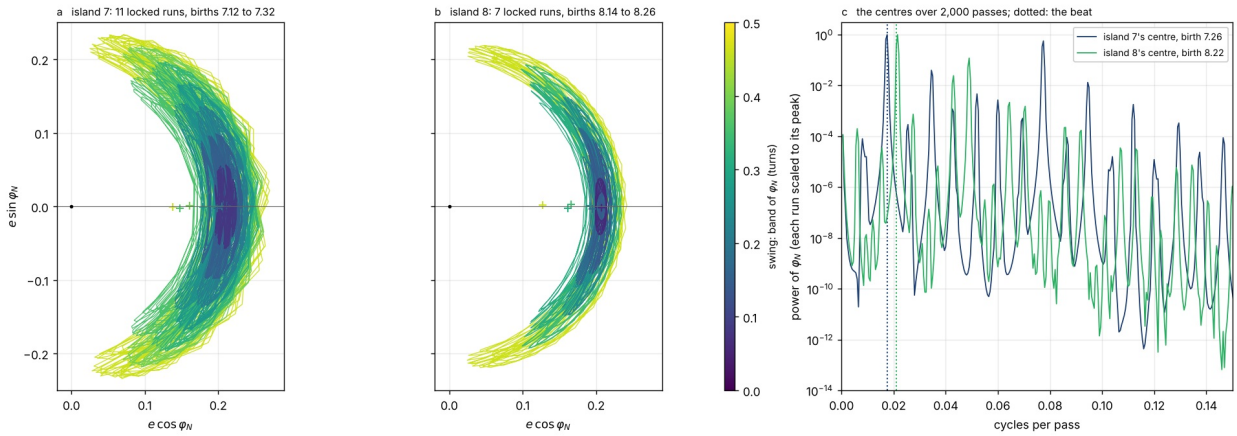


Figure 10. A lock rests like a spring. The light star, with the pair at 0.10 and the star at 0.20. Left and middle, every locked run of the coarse scan at islands 7 and 8 (11 and 7 runs, 300 passes each) drawn in the plane of the star's eccentricity vector measured from the holding angle, one arc per run, coloured by its swing; the arcs' centres (crosses) all lie on one axis, within 1.1° of it. Right, φ_N's spectrum at each island's centre over 2,000 passes, its slow line beside the beat (dotted). From probe 22 (run 20261006_190322), as probe 23 drew it (run 20261006_201311), and probe 24 (run 20261006_204230); the runs were recomputed as run 20261015_075527, which matches all three records to their last printed digit.

At the island's centre, where the swing has almost vanished, φ_N still wanders slowly. Measured over 2,000 passes, the wander sits at 0.99 and 1.02 of twice the star's apse turn, at islands 7 and 8. That is the beat: everything the main pull does not hold turning past it at twice the ellipse's turn. In Mardling's terms it is the splitting between neighbouring harmonics of the pull, set apart by the slow turning of the orbits, which she calls the resonances' "Zeeman splitting".

Two notes

The motion at the island's very centre is quiet but not simple: it is two rhythms played together, like two notes in a chord. Drawn on its own, without a flank's wider swing drawn over it, the centre's angle is a strikingly regular pattern. It was first seen by eye, as a slow, regular wave under the flank's swing on a shared plot, traced at 53.5 passes a wave, between the two centres' beats of 57.2 and 46.5 passes. Its spectrum says what it is. Every line at island 7's centre and at island 8's (8 and 5 lines above a thousandth of the peak) is a whole-number mix of two notes, the beat and the rocking: the beat's harmonics, and the rocking with sidebands spaced by the beat. Tested forward across the middle of island 7 at the star's eccentricities 0.165 and 0.17, at 52 births over 2,400 passes, the median fraction of lines so explained is 1, among a median of 19 lines per birth down to a hundred-thousandth of the peak, and the two notes explain them better than the beat with a second note chosen at random does at 44 of the 51 births with a second line (registered; with only the lines above a thousandth of the peak, 5 to 8 per birth, a random second note did as well at half of them, so we registered the stricter test before running it). The centre is two clocks tied together, the beat and the rocking, keeping time without locking: their ratio slides with the star's eccentricity (Section 8).

8. The rocking: quickening, locks, and two dials

The flyer's early and late keep a rhythm of their own, and that rhythm can itself fall into step. Inside the island the star rocks to and fro, and the rocking likes to lock onto simple fractions, a third, a fifth or a seventh of a swing per orbit, odd fractions about three times as often as even ones; and the island's two sides rock at different speeds.

Inside the island the rocking locks to simple fractions of a cycle per pass, at the core of the lock at a fifth to within a few billionths of a cycle, while the births around a core lose the lock; locks at fractions with odd denominators fill about three times as often as even ones, wherever the runs start; and stars of equal swing rock differently on the island's two flanks, so the island has two actions, not one.

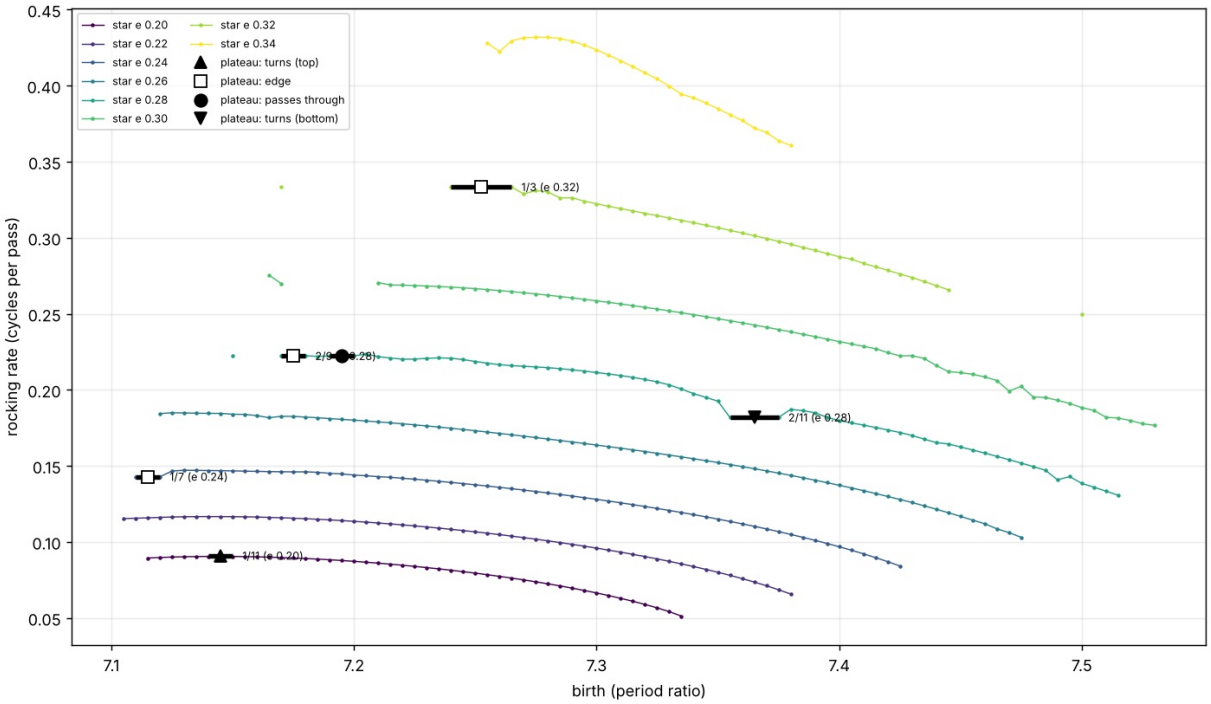


Figure 11. The rocking across the island. The rocking rate against birth for eight eccentricities from 0.20 to 0.34, the runs started at φ_N = 0, plateaus marked by kind: where the rate turns, where it passes through, and at an island's edge. From probe 30's atlas (run 20261007_004206), with the plateaus as the atlas look found them (run 20261007_224331).

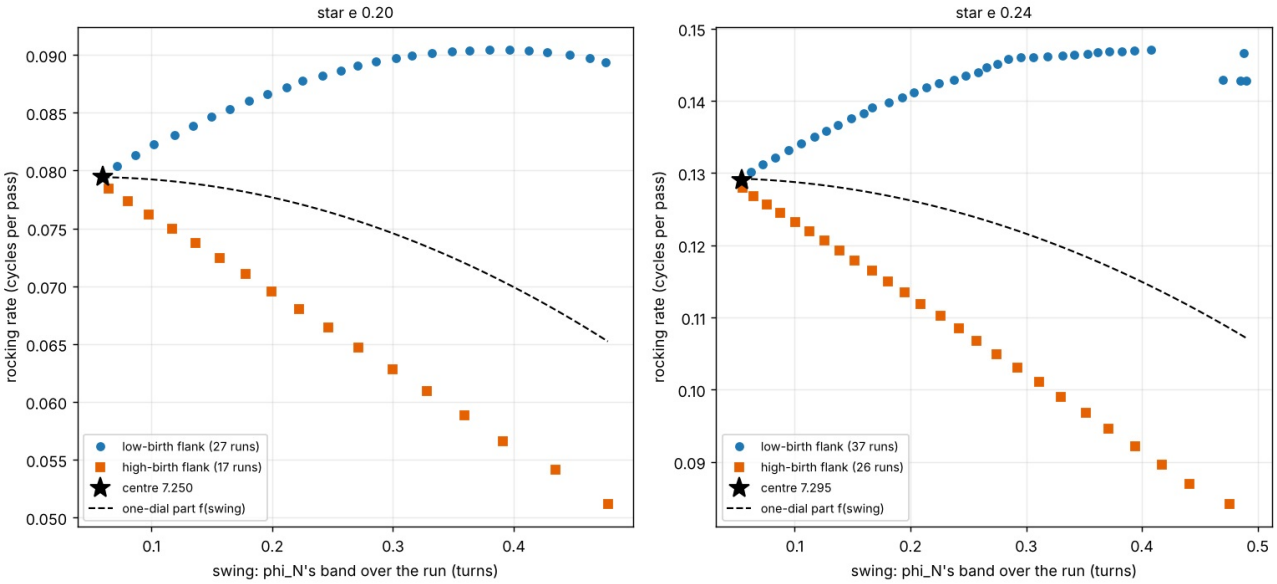


Figure 12. Two dials. At equal swing, the locked runs on the low-birth and high-birth flanks of the island at the star's eccentricities 0.20 and 0.24, started at φ_N = 0, two branches from one root at the centre, with the one-dial part drawn between them; the low-birth flank rocks faster at every matched swing (20 pairs of 20 and 28 of 28). At 0.24 the flank's three widest swings (births 7.110 to 7.120) break from the rest: they rock at a seventh, the edge plateau of Figure 11. From the same atlas, with the one-dial part as triple_two_action_look fitted it (run 20261008_063548).

Quickening

As the star's orbit grows more eccentric, the rocking quickens steeply: the quickest swing takes 11.0, 6.8, 4.5 and 3.0 passes at eccentricities 0.20, 0.24, 0.28 and 0.32, a log-log slope of −2.77 (Figure 11).

A slide, not a lock

When the star's orbit is nearly round, its rocking slows below what our first reader could hear, so we listened again with a better reader; what it heard is a rocking that keeps step with the beat at a ratio that slides, not one that locks. Our rocking reader takes the strongest rhythm of φ_N above 0.05 cycles per pass, a floor set so that it would skip the beat of Section 7, which near the island's centre is often the stronger motion. Where the star's orbit is nearly round the rocking itself falls below that floor: at 0.12 the strongest rhythm of every held, locked star lies between 0.016 and 0.026 cycles per pass, while the reader reported weaker lines at 0.049 to 0.072, the lowest a peak found at the floor and refined to just under it. So the rates first reported there (about four times the star's apse turn, nearly the same across the island), and a guess at the star's eccentricity 0.14 scored with the same reader, say more about the reader than about the island. Read again with a second reader that sees below the beat and names it, the rocking at 0.12 runs below the beat (0.6 to 0.9 of it, falling straight with birth) and at 0.16 near twice it (1.9 to 2.0 times), both read after the fact. Forward, at 0.13 and 0.15 over 2,400 passes, neither of our guesses held: at 0.13 the rocking is not one straight line on birth (the band carries 22% of what a line leaves, where we allowed 20%), and at 0.15 it does not lock at twice the beat. What the runs show instead, seen after scoring, is that the middle of each island keeps step with its own beat at a steady ratio, 0.95 of it at 0.13 over five births and 1.60 times it at 0.15 over six, while beat and rocking both change by several percent. So 0.16's 'twice the beat' might be one point on a slide rather than a lock.

We registered where the ratio would sit at three more of the star's eccentricities, and all three held: 1.26 times the beat at 0.14 (over five births), 1.78 at 0.155 (seven) and 2.006 at 0.16 (six), each run of births flat within 1%. From the star's eccentricity 0.13 to 0.16 the middle's ratio climbs smoothly, 0.95, 1.26, 1.60, 1.78, 2.006, and reaches twice the beat at 0.16 without pausing on the way; the climb even quickens toward it. Whether it rests there, a lock, or keeps climbing, a slide, these runs could not say, so we ran the same island at the star's eccentricities 0.165 and 0.17, registering first that a lock would keep the ratio within 1.5% of 2 and that above 2.1 is a slide. At 0.165 the middle keeps 2.233 times its beat, flat within 0.9% over seven births, and at 0.17 it keeps 2.486, flat within 0.8%. It is a slide. From 0.13 to 0.17 the ratio climbs 0.95, 1.26, 1.60, 1.78, 2.006, 2.233 and 2.486, quickening as it goes, and passes twice the beat without stopping. What holds steady across each slice's middle is the ratio, not the rate: across the seven births at 0.165, beat and rocking each fall by about 6% while their ratio holds within a percent.

A real lock

A lock is when the rocking settles on an exact simple fraction, such as exactly one swing every three orbits, and stays there across a whole range of starts; here is one, and what it holds and what it does not. At eccentricity 0.32 and birth 7.25, fourteen adjacent starting angles from −0.035 to +0.030 turn all rock at 0.333311 to 0.333339 cycles per pass: exactly a third, over a window 0.070 turn wide. Its edges are ragged, with stray thirds among starts that rock slower or break loose. Across the atlas's seven starting angles, 31 of its 2,909 locked runs whose rocking the second reader resolves rock within 10^{-4} of a third, every one of them at eccentricity 0.32. The lock does not hold the star's ellipse still. Those 31 runs turn it anywhere from 0.0050 of a turn per lap backward to 0.0029 forward; only the six on the starting angle 0 at the island's lower edge, births 7.240 to 7.265, all turn it by 0.0007 or less, because along that line the beat passes through zero twice there, near births 7.242 and 7.257. Below the island, among births 7.165 to 7.235, 77 of the 105 runs escape, and the only 7 that stay locked all rock at a third: over 600 passes, the only survivors in the chaos under the island are runs locked to the third. The refuge is not for good. Run for 2,400 passes on the starting angle 0, the survivor at 7.170 escaped at pass 850; of the six locked at a third at the island's lower edge, 7.240 escaped at pass 881, 7.260 and 7.265 drifted off the third (to 0.333516 and 0.332400 cycles per pass), and only 7.245 to 7.255 still rocked at exactly a third. Below the island, how long the refuge lasts goes with where the star was born. Of the atlas's six other runs locked at a third there, each run for 2,400 passes, the three born at 7.190, on three starting angles, escaped (at passes 693, 810 and 862); the three born just above, at 7.195, 7.200 and 7.205 on the angle −0.02, still rocked at exactly a third at the end. We had guessed the refuge temporary everywhere, registering that at least three of the six would escape. Three did, which meets the guess as written, but not as meant.

Most flat stretches are not locks

A rate that looks flat in a coarse map may be a real lock, or may only be turning near a fraction without touching it, so every flat stretch was run again on a finer grid of births. An atlas of 5,656 runs (101 births, 7 starting angles, 8 eccentricities) found 20 flat stretches of the rocking rate near simple fractions. Re-run at births 0.001 apart (0.002 in the windows at the star's eccentricities 0.20 and 0.24, where the run's time ran out first), 11 of the 20 are real locks over 1,000 passes: at least three neighbouring births within 5×10^{-5} of the fraction and flat; over longer records the three at two ninths hold none (Appendix C). The other 9 are places where the rate turns near the fraction without touching it. The locks sit where the rate stands still. Six are at the island's edge, four at a turn of the rate, and only one where the rate is passing through. At a third, four lines lock at the island's edge, their births rocking within 3×10^{-8} to 2×10^{-6} of a third. At eccentricity 0.20 the crown of the rate rises through an eleventh as the starting angle

turns, and only the line whose crown grazes 1/11 locks, at the crown. Two of the flat stretches were two rocking lines of nearly equal strength trading places, so the reader heard one or the other depending on the run's length. Counted over the whole atlas, the locked runs whose rocking lies within 10^{-4} of a fraction with denominator 2 to 12 number 121, against an average of 34.5 at decoys moved one step of a 600-pass spectrum to either side; 31 of the 121 are the third, and without it the excess is 90 against 34.0. The odd denominators hold their runs far more often than the even: 74 runs on 149 crossings of fractions with denominators 5, 7, 9 and 11, against 16 on 88 for 4, 6, 8, 10 and 12, 2.73 times as many per crossing.

Followed for 2,400 to 12,000 passes, each of the two longest locks splits into a core and a shell: at a fifth, seven core births sit within 3.5×10^{-9} of the fraction over 6,000 passes, and at a seventh five sit within 2×10^{-8}, while the births around each core slide off, lose their lock or escape (Appendix C). Run again with every tolerance a hundred times tighter, the shells' median slide between 2,400 and 6,000 passes falls to 1.18×10^{-7} from 7.96×10^{-7}, the fifth's seven core births stay within 4.8×10^{-9}, and at the seventh four of the five core births part from their twins, slow chaos, the slowest after 14,795 passes.

So the shells' slide was the arithmetic's; the losses at a lock's edge are partly the arithmetic's, and those in the seventh's core are chaos; and the fifth's core is a lock in clean arithmetic too.

Odd chains, off the mirror too

Odd fractions lock about three times as often as even ones; this part checks that the preference belongs to the island and not to where we happened to start the runs. Every start line of the atlas lies within 0.05 turn of the mirror line $\varphi_N = 0$, so the parity might belong to where the runs start rather than to the island. We ran the atlas's grid again on four start lines well off the mirror, at 0.06, 0.08, 0.10 and 0.12 of a turn (3,232 runs of 600 passes), registering first a guess that there the even chains would fill in, the odd holding no more than 1.5 times as many runs per crossing as the even. They did not. Off the mirror the odd denominators hold 51 runs on 67 crossings and the even 10 on 40, 3.04 times as many per crossing, so the parity is not the start's. The locks keep to their addresses there too: 61 runs within 10^{-4} of a fraction, against 16.0 at the decoys.

Two dials, not one

In a simple island, how wide a star swings would alone decide how fast it rocks; island 7 needs a second dial as well, which side of the island the star sits on. An action, here, labels the ring a star traces inside the island: with one action, a star's swing alone would set how fast it rocks, and the twist is how the rocking rate changes from ring to ring.

In an island with one action, two starts of equal swing lie on the same ring and rock alike. Island 7's do not. At every eccentricity from 0.20 to 0.34, the star started on the island's low-birth flank rocks faster than the star of equal swing on its high-birth flank, by 0.013 to 0.076 cycles per pass, in every matched pair (Figure 12). A two-part rule fits each line of the atlas. One part slows with the swing, like a pendulum. The other drops steadily with birth. The fit is within 0.0005 to 0.0033 cycles per pass, and on every line but 0.28 its maximum lands within 0.025 in birth of the fastest run. The "fastest line" inside the island is therefore where the two parts balance: the highest point of the slice, not a ring where the twist vanishes. With the runs on the lock at a third set aside, the one-dial part's twist at the centre is negative at every eccentricity from 0.20 to 0.34. The centre's rocking rises from 0.08 to 0.42 without the vanishing twist that a one-action island would need (Dullin, Meiss and Sterling, 2000). At eccentricity 0.28 the fit is good once the flat stretches are set aside (within 0.0017). Its maximum still lands 0.055 from the fastest run, which sits right beside the two-ninths lock. At births 0.001 apart that neighbourhood is rough: between two short locks at two-ninths the rate jumps between neighbouring births by about eight times as much as on a smooth line. Run with twins for 1,000 passes, 7 of the 22 births between the two locks are chaotic and none of the 9 lock births is: the rough stretch is chaos, and the short locks are regular islets inside it. A one-harmonic model in which the star's eccentricity and period are tied by the quantity the harmonic conserves gives the low-birth flank the faster rocking at every matched swing (Appendix B).

The slide's ratio across the atlas, and two cautions on the count of odd and even chains, are in Appendix C.

9. The hold drags on the star's ellipse

Each tug lands on the flyer's own oval path too, and that oval slowly turns. Holding the star also slows the slow turning of its orbit's long axis, the more the firmer the hold, until near one eccentricity of the star's orbit the turning stops at the island's centre and then runs backward.

The hold drags on the turning of the star's ellipse, in a straight line with how firmly it grips, by close to the amount Mardling's coefficient gives with nothing fitted over 600 passes, until near the star's eccentricity 0.28 it stops the centre's ellipse and beyond it turns it backward.

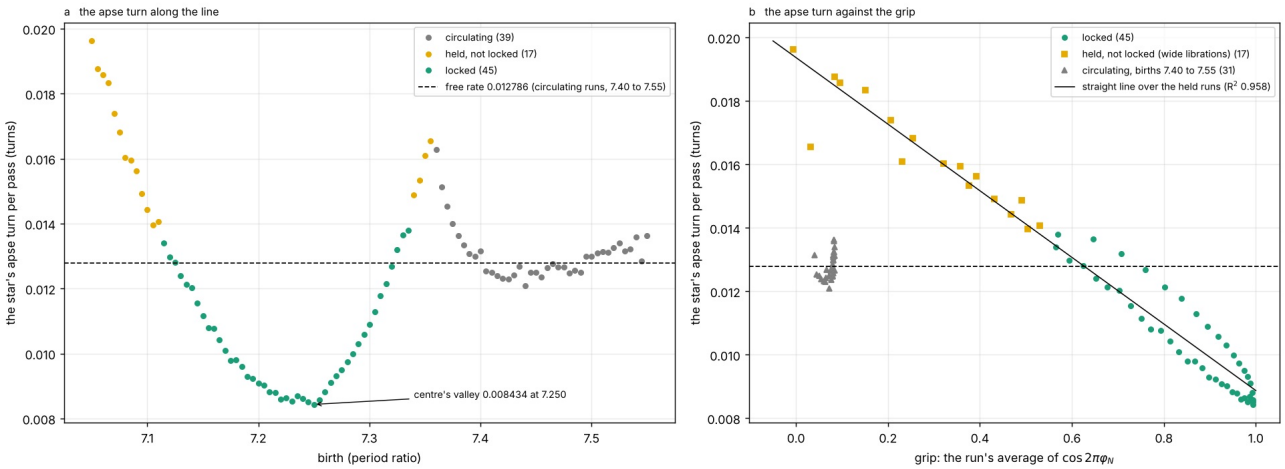


Figure 13. The drag on the ellipse. Left, the apse turn per pass against birth along one line, a valley at the centre rising to peaks near the edges, the free rate drawn across; right, the apse turn against the run's grip, the held runs on a straight line (R^2 0.958) and the locked runs on two branches that meet at the centre. From probe 30's atlas at the star's eccentricity 0.20, the runs started at φ_N = 0 (run 20261007_004206), the right panel from its saved angles as the grip was first read (run 20261008_065732), redrawn as run 20261015_140349, which matches that reading to its last printed digit.

Free of the island, the star's ellipse turns about 0.0128 of a turn per pass under the pair's steady pull. Inside the island it does not turn freely. At the centre it turns 0.0084 per pass, two-thirds of free (Figure 13). Toward the edges it turns faster, and in the widest swings just inside the rim it reaches 0.0196 per pass, half again faster than free. At the island's upper edge the curve comes to a point, like a fin. Because the hold is a lock of coupled clocks (Section 5), the star's period follows. Across the held runs the apse turn tracks half the lap ratio's excess over seven, with a correlation of 0.94.

Where the passes land

What sets the drag is where the passes land. Each pass twists the ellipse a little, and whether the twist holds it back or pushes it on depends on φ_N. We call a run's average of cos(2π φ_N) its *grip*. A star at the centre has a grip near 1: every pass lands on the holding phase. A rocking star's passes spread over phases, and its grip falls. Across the held runs the apse turn falls in a straight line with the grip (R^2 0.958). Drawn, the locked runs fall on two branches that meet at the centre: at equal grip the low-birth flank's ellipse turns more slowly than the line (26 of the 27 locked runs born below the centre) and the high-birth flank's faster (15 of the 17 born above it; the centre itself sits just below the line), the second dial of Section 8 seen in the drag (noticed when the figure was drawn). The two branches are the island's habit, not one line's: on all 28 of the atlas's start lines at the star's eccentricities 0.20 to 0.26, seven at each, 936 of the 957 locked runs born below the centre fall under their line and 604 of the 668 born above it over theirs, and the gap between the branches widens as the star's orbit grows more eccentric.

The pair's pull

If the drag comes from the same pull that holds the star, it should grow in step with that pull as the pair's orbit grows more oval; it does, until the island starts to break up. The drag belongs to the same pull that holds the island. Measured at four eccentricities of the pair, from 0.05 to 0.20, the line's slope is −0.0055, −0.0105, −0.0160 and −0.0222 of a turn per pass per unit of grip: 0.105 to 0.111 times the pair's eccentricity each time, and the holding pull is itself proportional to the pair's eccentricity (Mardling 2013). At 0.25 it steepens to −0.0305, 0.122 times the pair's eccentricity, past the −0.0255 to −0.0274 that the harmonic's scaling gave beforehand; there the island has halved and chaos crowds both sides (Section 6; Appendix C). So at the island's centre the ellipse turns more slowly the more eccentric the pair: 0.0112, 0.0084, 0.0064 and 0.0048 per pass.

The ellipse stands still, then turns back

The star's own eccentricity does more. As the star's orbit grows more oval, the turning of its ellipse at the island's centre slows to a stop and then runs backward. Along the atlas's lines (the pair at 0.10), the slowest locked ellipse, at the island's centre or a little below it, turns forward by 0.0084 of a turn per pass at the star's eccentricity 0.20, then 0.0067, 0.0045 and 0.0022 at 0.22, 0.24 and 0.26; at 0.28 it stands almost still (−0.0005), and at 0.30, 0.32 and 0.34 it turns backward (−0.0029, −0.0053, −0.0078). At 0.34 every locked star's ellipse turns backward. At 0.28 the beat crosses zero twice across the island, at births 7.251 and 7.333, and the second crossing is the island's centre itself (its narrowest swing, at 7.335). On the atlas's six other starting angles at 0.28 the narrowest swing sits within 0.005 in birth of a zero crossing on three, and up to 0.024 from one on the rest, but on all seven its ellipse turns by no more

than 0.00045 of a turn per pass, against 0.0024 to 0.0029 forward at 0.26 and 0.0022 to 0.0027 backward at 0.30: at 0.28 the centre's ellipse is all but still whichever line crosses it. So the beat, twice that turn, passes through zero near the star's eccentricity 0.28, where the harmonics that Mardling's splitting sets apart share one frequency again. Over 2,400 passes the centres' beats at 0.20 to 0.26 fall on a straight line, within 2.3% of their mean size, that would reach zero at 0.286; beyond 0.26 the beat falls faster than the line, and between the centres at 0.28 (+0.0005) and 0.30 (−0.0054) it crosses zero at 0.282. The crossing at 0.282 was read after the runs; what it does to the island is open (Appendices B and C).

The zero has since been pinned at the star's eccentricity 0.2811, and over 24,000 passes at a tolerance a hundred times tighter the ellipse at 7.340, just beside the centre, stays still (Appendix C).

One term, seen twice

The pull that holds the star and the drag on its ellipse turn out to be one term of Mardling's formula seen from two sides, and with nothing adjusted the formula gives the drag's size within about 6% at the star's eccentricities 0.20 to 0.26. In what follows e_i and a_i are the pair's eccentricity and semi-major axis, e_o and a_o the star's, and [n′:n](m) names a harmonic of the pair's pull as Mardling does (Appendix C).

Mardling's harmonic predicts the drag's form. Her term for island N is a function of the pair's eccentricity, $X_1^{2,2}(e_i)$, times a function of the star's, $X_N^{-3,2}(e_o)$. Through Lagrange's equation for the star's ellipse, the drag's slope should follow the first and the slope of the second, each averaged over a run into the grip. Against the pair's eccentricity, the slope divided by $X_1^{2,2}$ stays between 0.0352 and 0.0378 at all four eccentricities. Against the star's eccentricity, along the atlas's lines from 0.20 to 0.28, the slope grows 1.23, 1.40, 1.80 and 2.40 times, within 12% of the harmonic's 1.28, 1.59, 1.93 and 2.29. The hold and the drag are one term of the expansion, seen twice.

Her coefficient gives the drag's size as well as its shape. For a light star outside the pair the harmonic is $(3/8)(a_i^2/a_o^3)$ $X_1^{2,2}(e_i)$ $X_N^{-3,2}(e_o)$ cos φ_N: the pair's pull, shrinking as the cube of the star's distance, times one function of each eccentricity, times the cosine of the holding angle. It gives back her pendulum (her equation 40), and through Lagrange's equation it predicts the drag's slope with nothing fitted. Taken at the star's starting eccentricity, the prediction is 0.74 to 0.90 of the measured slope on every line, a tenth to a quarter short on all of them. Taken at the held runs' own mean eccentricity, which lies above the starting value while the harmonic grows steeply with it, it is 0.945 to 1.039 of the measured slope on the atlas's lines at the star's eccentricities 0.20 to 0.26. At 0.28 it is still 0.822. The pull's next terms do not make up the difference. The next member of the harmonic's family could hold still there, because the beat passes through zero, but it gives at most 0.7% of the slope; the next order of the pull adds about 3% on every line alike. What the 0.28 line has is spread: its held runs' own eccentricities run from 0.2531 to 0.3164 and lean on the grip, while the drag grows steeply with eccentricity, so one slope a line averages unlike runs together. Taking each run's drag at its own eccentricity instead, the predicted slope there times the run's grip, with one scale fitted a line, the prediction is 0.867 to 1.007 of the measured drag on all five lines, inside the band we registered (0.85 to 1.18), and the scatter about each line falls to 0.27 to 0.82 of what one slope a line leaves. This was our fifth registered try at this number, so it weighs less than a first try would, and 0.28 still reads lowest (Appendix B). Over 2,400 passes, on the runs that keep their hold, it reads 0.82 to 0.87 on every line, and on four of its five lines it rises from the island's middle toward its rim (Appendix B).

The fin at the island's upper edge in detail, what carries the two branches, how far the drag law carries across start lines, and where its level near the rim misses are in Appendix C.

10. One doubling, then escape

This is the act pushed until it breaks. As the star's orbit grows more eccentric the island's centre rocks faster, until its rhythm doubles, a swing in two steps, once; a step further, every nearby star escapes.

Following the island's centre in small steps of the star's eccentricity, its rocking climbs 0.310, 0.331, 0.354, 0.381, 0.418, 0.479 and 0.497 cycles per pass from 0.320 to 0.350. It crosses a third between 0.325 and 0.330 without pausing and climbs to just under a half by 0.350 (Figure 14). At 0.348 the centre rocks at 0.4964 cycles per pass, still short of a half; of nine starts around it only two alternate steadily, both barely, and none escapes within 450 passes, so the doubling lies between 0.348 and 0.350. Mardling's pendulum for this resonance, which gets the island's width within about 30% at eccentricity 0.20, would reach a half only at 0.41: the real island's rocking quickens faster than a pendulum's as the star's orbit stretches. Near the real half, between 0.348 and 0.350, the centre loses its stability by doubling. A run beside it alternates φ_N by 0.045 turn on alternate passes, a steady left-right-left. A frictionless cascade would go on doubling. This one does not. All nine starts around the doubled centre at 0.350 escaped within 1,600 passes, while all nine around the centre at 0.348, followed for the same 1,600 passes, held, so island 7's cascade ends at its first doubling, by escape. On a known map (Section 4), a loop pushed by a slow beat doubles once and then locks to the beat instead of doubling again. The stars' newborn does not lock that way: its alternation swells and fades at its own rocking's distance from a half, which does not follow the beat (Appendix B), so why the cascade stops at its first doubling remains open.

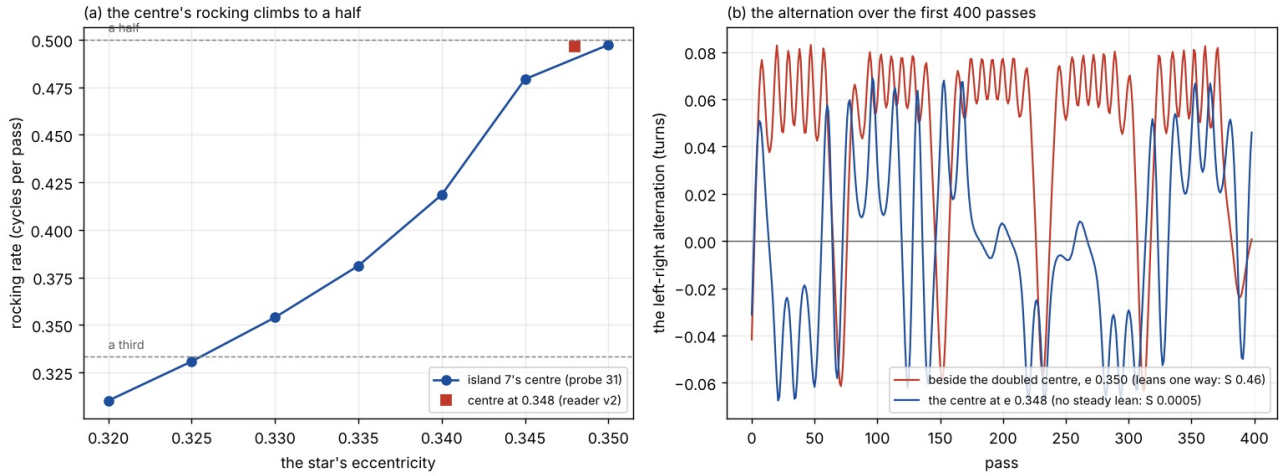


Figure 14. One doubling. Left, the centre’s rocking rate against the star’s eccentricity, climbing past a third without pausing to just under a half, with the centre at 0.348 read by the second reader; right, the left-right alternation over the first 400 passes, beside the doubled centre at 0.350, where it leans one way, and at the centre at 0.348, where it still swings through zero. Left, probe 31’s centre follow (run 20261007_151231), with 0.348 read by the second reader (run 20261014_085115); right, the doubled centre from probe 32’s starts run again with their series kept (run 20261013_200621), and the centre at 0.348 from the series a later look kept (run 20261014_054907).

11. With a heavy third star

With a heavy flyer the act becomes a pair of partners: flyer and bar pull on each other and pass the swing back and forth. A heavy third star keeps the same islands but moves them just below the whole numbers instead of just above, trades eccentricity back and forth with the pair, and at each close pass bends any measurement of the pair taken at that moment.

In the field’s terms, a heavy third star moves the islands below the whole number by the same rule, trades eccentricity with the pair through the same harmonic, and bends any reading of the pair taken at its own closest approach by the short-period response Georgakarakos worked out.

Everything so far is for a star too light to move the pair. Make the third star heavy, 0.8 of either star, and the pair stops being a fixed clock. Its ellipse turns forward, by 0.031 to 0.075 of a turn each time the star goes round (read on its mean ellipse, below), and its eccentricity is traded with the star’s. One scan asks whether the islands survive: 82 runs at births 6.60 to 8.60, 0.05 apart, each started either at the holding angle or half a turn from it, and each followed for 200 passes.

They survive, on the other side of the whole number. Eight runs started at the holding angle lock φ_N, at births 6.85 to 7.05 and 7.95 to 8.05, with bands as small as 0.034 and 0.066 turn at the two centres. A ninth, at 7.10, seems to hold too, above seven, at 7.0944. Read at the star’s closest approach its pair’s eccentricity falls to 0.011, too low to read the direction of the pair’s ellipse well, so it was not counted with the eight. Read on the pair’s mean ellipse it is not held (below). The angle that holds is the same φ_N that holds the light star. With the light star the pair goes round a little more than seven times per orbit, because the star’s ellipse turns forward and the pair’s does not. With the heavy star both ellipses turn forward and the pair’s turns faster. Measured from the pair’s ellipse, the star’s ellipse now turns backward, and the pair goes round a little less than the whole number: 6.9332 times per orbit at the centre of island 7 and 7.9245 at the centre of island 8 (births 7.00 and 8.00, read at the star’s closest approach). The rule is the same in both cases. The island sits off the whole number by twice the star’s turn measured from the pair’s ellipse. On the finer scan of Appendix C, births 0.005 apart over 600 passes, with the pair’s apse read from its mean ellipse, island 7’s centre (birth 7.010) has the pair’s apse turning 0.03604 of a turn per orbit and the star’s 0.00395. So 7 + 2 × (0.00395 − 0.03604) = 6.93581, against 6.93570 turns counted. Island 8’s centre (8.015) gives 7.92516 against 7.92522. We had registered that the rule would match the turns counted to within 0.01 at each of the eight births we read this way, the two centres among them; it did at seven, and missed only at the birth that seemed to hold but is not held on the pair’s mean ellipse.

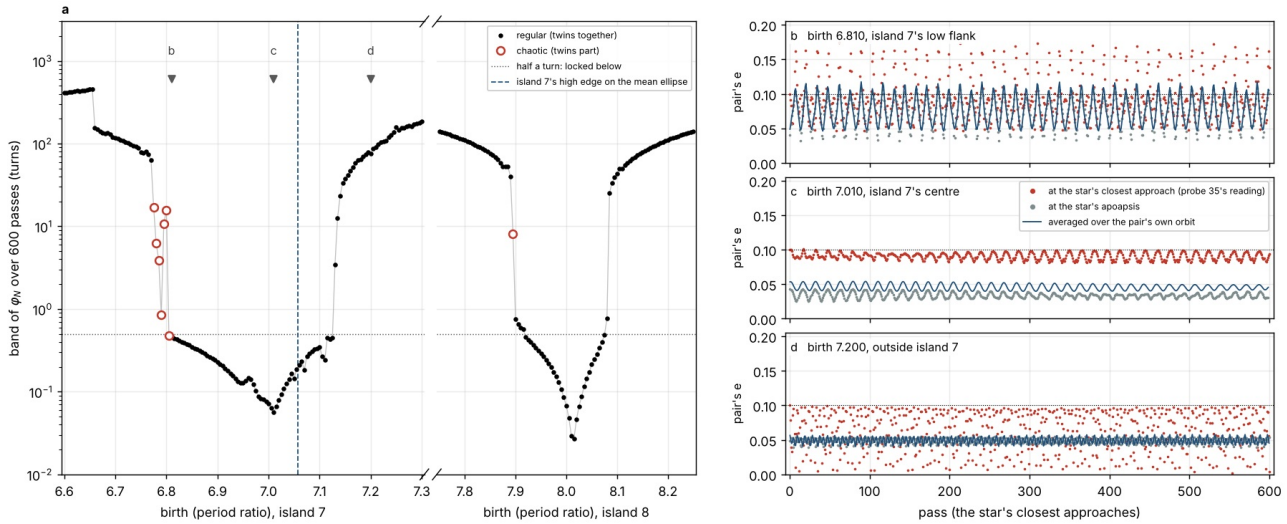


Figure 15. Islands below the whole numbers. The heavy third star, 0.8 of either star of the pair. (a) φ_N's band over 600 passes against birth, read at the star's closest approach, births 0.005 apart around islands 7 and 8 (probe 35, run 20261008_131041): two wells, at births near 7 and 8, where the pair turns a little less than 7 and 8 times per orbit. Open circles are chaotic births, whose twins part; all lie on the wells' low-birth sides. Below the dotted line the band is under half a turn. Dashed: island 7's high edge read on the pair's mean ellipse, between 7.055 and 7.060 (run 20261009_025920); the births held above it are held only as read at closest approach, where the star's tide bends the reading. (b to d) The pair's eccentricity pass by pass at three of those births, on island 7's low flank (6.810), at its centre (7.010) and outside it (7.200), read at the star's closest approach, at its apoapsis and averaged over the pair's own orbit, dotted at the 0.10 the pair was started with; recomputed as run 20261015_074509, which matches the first drawing (run 20261009_022616) and probe 35's records to their last printed digit. Panel d's birth was also followed for 6,000 passes (run 20261009_080853, whose first 600 passes repeat this one's within 3.7×10^{-9} in the pair's eccentricity), and its scattered readings are not noise: on the map of φ_N against ψ, each reading's bend from the pair's orbit average lies close to its neighbours', by a median 0.0603 of the bends' spread over the 6,000 passes, against 0.1126 over the first 600.

The pair's eccentricity, and a reading that bends it

How you measure matters: a reading taken just as the star swings closest catches the pair mid-stretch, and can show changes that are not really there. Figure 15's right-hand panels show it pass by pass at three births marked on its left panel: the reading at closest approach, the reading at the star's farthest point, and the average over the pair's own orbit. Every heavy-star run reads the pair's eccentricity once per pass, at the star's closest approach. There the star's tide is strongest, and it bends the pair's osculating ellipse by an amount set by where the pair stands in its orbit at that moment, which is φ_N. As the star swings closest, it pulls harder on the nearer of the pair's two stars than on the farther, so for that moment it stretches the pair's orbit along the line toward itself. The two stars are alike, so the pair looks the same after half a turn and the stretch comes round twice in each of its turns, once with each star in front; φ_N says where in that rhythm, and where along its own oval, the pair is when the star swings by (Section 2). Off the islands, on the regular runs where φ_N circulates slowly enough for each pass to sample it, the per-pass change of that reading follows $\sin 2\pi\varphi_N$ closely (correlations of 0.708 to 0.984 in size). Its coefficient grows with φ_N's rate, with a sign set by the way φ_N turns. A force acting through the harmonic would give a fixed coefficient and a swing that shrinks as the rate grows; a fixed function of φ_N gives what is seen. That holds cleanly below island 7 and around island 8. Above island 7 the reading passes close to zero eccentricity, φ_N's turns are not well counted, and the coefficient carries a large offset. Read again at the star's farthest point, where its tide is 0.296 as strong, the off-island swing is 0.159 to 0.450 as large. Averaged over one period of the pair from each farthest point, it is 0.0072 to 0.0188, against 0.0801 to 0.1002 at closest approach. The runs at island 7's high end that seemed to drain the pair (to 0.003 at 7.125) are not drained: the two averaged first, 7.100 and 7.125, keep a mean eccentricity between 0.028 and 0.056 throughout their runs, and on the mean ellipse they are not held either. The mean eccentricity also sits below the nominal 0.10 in every one of the twelve births averaged, at run means of 0.041 to 0.076. The 0.10 is the reading at closest approach at the start, not the pair's eccentricity. Read at closest approach, the pair's eccentricity seemed to hold still at the islands' centres and swing outside them, and to drain at 7.10; both were the bend. The bend is the short-period response of a pair's eccentricity to an outer body that Georgakarakos (2002, 2003) works out, for triples more widely separated than ours and with the pair started circular, read once per pass at closest approach, which turns it into a function of φ_N. Checked against his formula for an eccentric outer orbit (his 2003 equations 6 to 15,

with the pair's masses equal), on the three births of Figure 15's right-hand panels followed for ten of the star's laps at 64 readings a pair period, the pair's short-period eccentric vector is 1.115 to 1.117 of his prediction, with correlations of 0.982 to 0.993; at the star's closest approaches the bend is 1.076 to 1.288 of it, with correlations of 0.966 to 0.975. That is at a fictitious period ratio, measured from the star's closest approach, of 5.0 to 5.4, half the 10 and more he tested; in his own regime, a wider triple with the pair started circular, the same reading gives 1.015, with a correlation of 1.000.

Starting the pair on its mean ellipse

Because the closest-approach reading misled us about the pair's true shape, we mapped the heavy star's islands again with the pair started at its true average shape, so that heavy and light stars could be compared fairly. The scans above set the pair's eccentricity at 0.10 as read at the star's closest approach, not on its mean ellipse. In the new maps, at the star's first farthest point, the pair's eccentricity averaged over its own orbit is set to 0.10 or 0.05, pointing where the earlier start pointed. Births are 0.02 apart, then 0.005 apart with twins across each edge, for 600 passes. With the pair at 0.10 and the star started at the holding angle, island 7 holds from 6.920 to 7.210 (centre 7.100, the pair turning 6.90387 times per orbit) and island 8 from 7.965 to 8.160 (centre 8.080, 7.92107). With the pair at 0.05, island 7 holds from 6.800 to 7.040 (centre 6.980).

Now the heavy and light stars can be set side by side at the same pair. At 0.05 the light star's island 7 has no chaotic layer, and the heavy star's has one 0.030 wide in birth (6.775 to 6.800). At 0.10 the light star's layer is 0.030 wide. The heavy star's is at least 0.065 wide below its held island, with 0.005 above, and it does not stop at the edge: births at 6.900, 6.910 and 6.920 hold for all 600 passes while their twins part. Island 8 is ragged 0.015 below against 0.005 above. Started a quarter turn from the holding angle instead, both islands move and narrow (island 7 holds from 6.965 to 7.155, island 8 from 7.995 to 8.140), but the wider layer is still the lower one at both: 0.085 against 0.005 at island 7, and 0.020 against 0.005 at island 8. The starting angle does not choose the ragged side.

Held for 600 passes is not held for good (Figure 16). Nine births held every one of 600 passes while their twins parted. Most sat on an island's low-birth side, in all three of these starts; one each sat near the high-birth edges of islands 7 and 8. Followed for 6,000 passes, all nine leave their island, six of them in the second stretch of 600 passes, and two escape outright. Two more runs, a main and a twin, end the same way without being counted as escapes: the star is flung onto an orbit so wide that one lap spans thousands of the pair's orbits, and the run's time is spent before the next pass. In both, and in the escape from 6.910 beside them, the pair's eccentricity, about 0.14 on average before, stood at 0.42 to 0.45 when the star's laps first lengthened, and reached 0.50 to 0.76 by the end. Several hold again later, among them all four runs at island 8. The three on island 7's low side, with the pair at 0.10 and the star at the holding angle, never do. Just inside them the chaos stays. Every birth from 6.925 to 6.980, 0.005 apart, holds for all 6,000 passes, and so does its twin. At 6.925 to 6.940 the twins part by a tenth of a turn after 967 to 2,231 passes, yet neither leaves. From 6.945 inward they stay together, never more than 0.0082 turn apart. So that rim has three parts: chaos that leaves (6.920 and below), chaos that stays held (6.925 to 6.940), and regular motion (6.945 inward). The line between leaving and staying lies between 6.920 and 6.925.

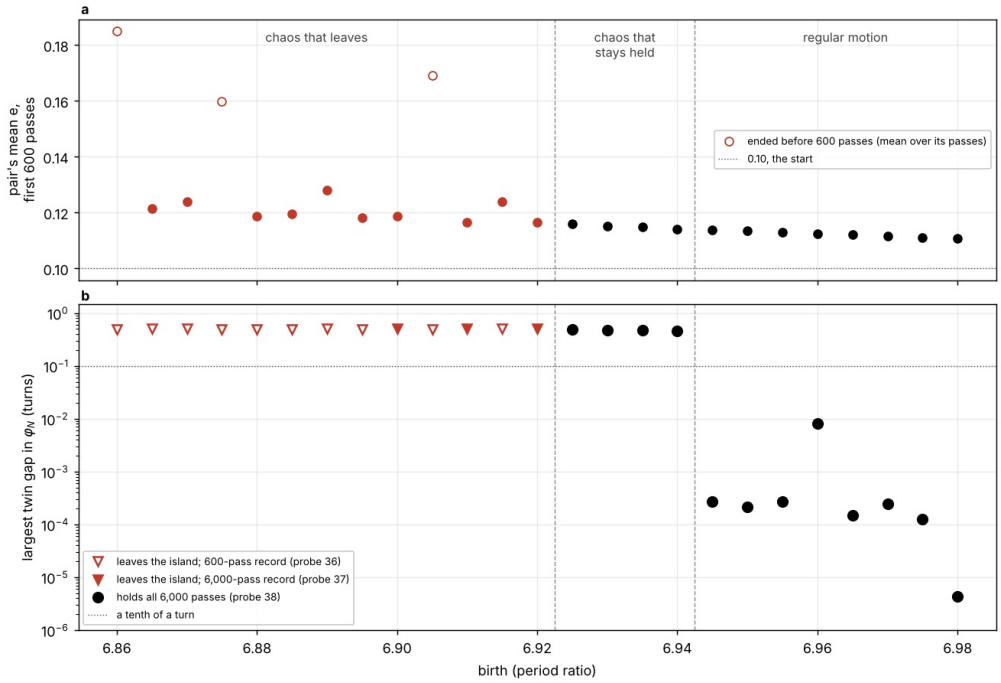


Figure 16. Chaos that leaves, chaos that stays. Island 7's lower rim with the pair started on its mean ellipse at 0.10 and the star at the holding angle, births 6.860 to 6.980, 0.005 apart, each run read from the longest record kept (probe 36, run 20261011_153059, 600 passes; probe 37, run 20261012_020012, and probe 38, run 20261012_175845, up to 6,000). Above, the pair's mean eccentricity over each main run's first 600 passes (hollow where the run ended sooner, by escape or on a lap too wide to finish). Below, the largest gap in φ_N between each run and its twin, marked by whether the run leaves the island. Chaos that leaves (6.920 and below), chaos that stays held (6.925 to 6.940: the twins part by nearly half a turn, yet both hold all 6,000 passes) and regular motion (6.945 inward); the two boundaries are read from the runs by the figure's code.

The sections on the light star are untouched by the bend: a star of a millionth of either star's mass raises no tide on the pair. Reading the heavy star needs two changes, described in Appendix A.2. The pair's turns are counted as they happen rather than predicted from its period, because the heavy star stretches and squeezes that period by several per cent within a run. And the pair's eccentricity is averaged over its own orbit, away from the star's closest approach.

The heavy islands' edges, where the star and the pair trade, and the rim's eccentricity are in Appendix C.

12. Why it is hard, and the findings in full

What the islands say about why it is hard

We opened the three-body problem to see why it is so hard, and the islands give a concrete answer. The trouble is not that the motion has no order. It is that order and chaos sit side by side, even on the finest grids we ran, and that a short look cannot always tell them apart. A light star is held by nothing more than timing, a catch made afresh at every pass. The catch has an edge, lopsided, with a door on one side that the published rules smooth over. What reads as a steady catch over hundreds of orbits can be slow chaos over thousands, in rooms whose calm sits to one side of where the short look pointed. Inside the catch the rocking locks to exact fractions, yet most stretches that look locked are not. The catch drags on the star's own ellipse until, at one eccentricity, the ellipse stands still and then turns back. Pushed harder, it doubles once and lets go. A heavy flyer changes the bar that catches it. And both the arithmetic and the eye can make patterns that are not there, so the patterns here were written down before they were measured wherever they could be, those seen only afterwards are called descriptions, and every miss is kept. Each of these has a number, and each number a run anyone can make again. The findings follow in full.

The findings in full

The findings in full, each with the sections that carry it. They are grouped a little differently from the front page: the bench has its own section (Section 4), and the edge and a lock's rest have items of their own here. Where an item's longer records moved to Appendix C, they stand there under the section's name.

1. **Held.** A light star orbiting an eccentric pair of equal stars, about seven (or eight) of the pair's periods per orbit, is held in an island. One angle stays put while every other angle goes round. The angle that holds, φ_N, belongs to the pair's eccentricity, and the hold is a lock of coupled clocks, the pair's orbit and the star's lap keeping time together. The star's period runs a little more than seven of the pair's, and the extra fraction of a turn per orbit equals twice the slow turning of the star's own ellipse. (Section 5; Figure 7.)
2. **Chaos that holds its stars, and chaos that lets them go.** Beyond the stability limit the island holds, but one side of it is a door, and the published stability rules smooth it over: one calls the whole island unstable, the other two call it stable, and the eleven stars that escape from its layer at the pair's eccentricity 0.20 sit on the stable side of the refined rule. The chaos around the islands comes in two kinds. With the star's eccentricity at 0.20, 163 of the 166 escapes we recorded, with the light star and the heavy one, started on island 7's low-birth side, below it or inside its lower half, the side nearest the pair; two left from above, after 5,904 and 14,278 passes, one from just past the island's middle, and chaotic threads through the island itself leak too. At the pair's eccentricity 0.25, over 6,000 passes, ten stars left from below the island or inside its lower half and none from its upper half; their twins, a second sample of the same chaos, did the same. On a grid five times finer, 26 of the 54 runs born from 7.445 to 7.471 escaped and none of the 138 born from 7.472 to 7.540; the highest, at 7.471, lies 0.001 past the middle of the island's core as registered from 600 passes, 7.400 to 7.540. The chaos above island 7, and beside island 8, passes the same twin test for chaos, yet it holds its stars far longer: run ten times as long in a second set of runs, eleven of twelve stars born below the island left within 1,186 passes, while eight of nine above stayed for all 6,000 and the ninth left at pass 5,904; watched for 30,000 passes, twelve of fourteen above stayed bound, the other two leaving at passes 5,904 and 14,278. The door stays on the pair's side while the star's own orbit is moderately eccentric; from the star's eccentricity 0.30 a second door opens above the island. Started on a circular orbit, as the published stability maps start theirs, the star finds no door at all: none of 41 starts across the 7:1 escapes in 10^5 of the pair's periods, the maps' own length. On the way out the chaos below the island turns its stars the way the stars above it turn: 17 of the 18 chaotic stars born below it that circulate do. (Section 6.)
3. **What reads as an island can be slow chaos.** Watched ten times as long, the islands at the pair's eccentricities 0.15 and 0.20 last, each giving up about 0.02 in birth at its eroded rims. At 0.25, by contrast, what reads as an island over 600 passes is chaos too slow for that length of run to show: watched ten times as long, 22 of its 28 inner births lose their hold, and every pair of twins parts exponentially. What remains is a row of rooms divided by chaotic threads, the third of which, on a finer grid, is a band about 0.01 wide. Stars born in the island leave only from below a sharp edge just above that band, and the room that keeps its stars, where about two births in five hold for all 6,000 passes, lies 0.02 to 0.06 in birth below the centre that the first 600 passes point to. (Section 6; Figures 1 and 9.)

4. **Mardling's pendulum, tested piece by piece.** The angle that holds is the harmonic Mardling (2013) names [N:1] (2), found here by working out which angle the strongest term of the pair's pull carries and counting its whole turns, before her paper was read. Her pendulum's relation between the small-swing rocking rate and the island's width holds within 1.1% at the pair's eccentricity 0.10, and the island sits off the whole number by twice the star's apse turn, as her splitting has it. Outside the conditions her model was built for, the pendulum is off by amounts we measured. Its widths run up to about 30% wide. From the pair's eccentricity 0.15 the rate–width relation fails for the held island, and holds instead, within about 10%, for the island and its chaotic layer counted together. Against the pair's eccentricity the centre's rocking quickens less than the pendulum's (1.435 times from 0.05 to 0.20, against 1.98), and against the star's it quickens more, reaching its doubling at 0.348 to 0.350, where the pendulum would only at 0.41. (Sections 5, 6 and 10, and Appendix B.)
5. **The arithmetic can fake resonance physics.** At the tolerance used for most runs (relative 10^{-9}), every unlocked rocking rate falls steadily as a run goes on, by about 2.8×10^{-8} cycles per pass in each pass on the line where it was measured most closely, while a rate locked on a fraction does not move with it. Over records of different lengths that fall made the shells around the locks seem to slide, and wore part of their edges. With every tolerance a hundred times tighter it shrinks to 0.0093 of itself, the shells' slide disappears, and the losses that remain in the seventh's core are chaos, read by the twin test. With the heavy star a near-cancelling slow beat magnifies the same drift, shortening by a third over 30,000 passes at birth 7.200; at a second birth, 7.300, a test found the slow beat keeping pace with the pair's falling eccentricity and drifting 7.2 times faster than at 7.200, and holding at the tight tolerance. Ten times tighter than that, over 60,000 passes, the drift is about a hundred times smaller again; at looser tolerances it moves the orbit's two rates in the proportion a nudge in birth does, leaving the eccentricity's second line almost still, though it is no nudge in birth: it drains the energy where a nudge outward adds to it, and it keeps its sign where the drifts of the energy and of the angular momentum both change theirs. The size of the energy's drift shrinks with nearly the same steepening powers. Made again on a second machine, the long runs gave the same verdict in every test registered before they were sent there. (Sections 3 and 8; Appendices A.1 and A.5.)
6. **The edge is thin, chaotic and lopsided.** At the island's rim the motion is chaotic in a thin layer. The layer is wide on one side and almost absent on the other. Beyond it the star's drift slows the way a pendulum does near its tipping point, with the slope the island's width predicts (within 6% at the two clean edges, over the ladder as a whole). The layer widens as the pair grows more eccentric, from none at 0.05 to 0.120 in birth at 0.20, always on the same side, and at 0.20 stars in it escape. Turning the star's own eccentricity instead leaves it on that side too: the ragged side follows the pair. (Section 6; Figures 1 and 8.)
7. **A lock rests like a spring.** Every locked star swings symmetrically about the same resting angle, whatever its swing, and the island's centre wanders slowly at twice the turning rate of the star's ellipse. Seen on its own, the centre's angle is a regular pattern of two notes, that slow wander and the rocking: every line of its spectrum is a mix of the two. (Section 7; Figure 10.)
8. **The rocking has two dials.** The rocking quickens steeply as the star's orbit gets more eccentric. It locks on simple fractions of a cycle per orbit, almost only where the rate turns or the island ends. Over 2,400 passes the middles of the longest locks, at a fifth and a seventh, close in on their fractions, to within 1.3×10^{-7} and 8×10^{-7}, while their end births fall off and the shorter locks do not hold. Over 6,000 passes each middle splits. A core keeps closing in, seven births at a fifth to within 3.5×10^{-9} and five at a seventh to within 2×10^{-8}, while the births around it lose their lock or escape; and a lock at an eleventh that 2,400 passes could not hold appears. The shells' slide and the eleventh's lock carry a slow drift of every unlocked rate that comes from the arithmetic (Appendix A.1). Over 12,000 passes the fifth's core holds within 4.1×10^{-9} but closes no further, and the seventh's loses one of its five births. With every tolerance a hundred times tighter the shells' slow slide disappears, so it was the arithmetic's; three of the seventh's five core births lose their lock, and all three are chaotic by the twin test; of the two that stay locked over 12,000 passes, 7.120 parts from its twin at pass 14,795 and escapes at 29,323, slow chaos too; the fifth's core stays a lock; and over 12,000 passes the eleventh is one unbroken lock from 7.127 to 7.135, its births from 7.128 up within 1.2×10^{-8} of their fraction and 7.127 within 2.3×10^{-8}. At equal swing it also depends on where in the island the star started. The island has two actions, not one, and its "fastest line" is only the highest point of the slice we read it on. Where the star's orbit is nearly round, at its eccentricities 0.12 to 0.16, the rocking was slower than our first reader could read; read again, it climbs with the star's eccentricity from below the beat to about twice it, and in the middle of each island it keeps step with the beat at a ratio that climbs smoothly with the star's eccentricity, from 0.95 at 0.13 through twice at 0.16 to about two and a half at 0.17, without resting on any whole number: a slide, not a lock. Its locks favour fractions with odd denominators about three to one per crossing, on start lines well off the mirror as near it, so the start does not make them. (Sections 5 and 8, and Appendix B; Figures 11 and 12.)

9. **The hold drags on the star’s ellipse.** At the island’s centre the star’s ellipse turns at about two-thirds of its free rate. Near the island’s edges it turns half again faster than free. The more the star rocks, the less the hold holds the ellipse back, and Mardling’s coefficient, taken at each line’s mean eccentricity with nothing fitted, gives the drag’s slope against the grip over 600 passes within about 6% at the star’s eccentricities 0.20 to 0.26. As the star’s own orbit grows more eccentric the drag grows until it wins: near the star’s eccentricity 0.28 the centre’s ellipse stands still, over 24,000 passes at the tight tolerance as over 2,400, and above it turns backward. The law that ties the drag to each run’s grip, the swing’s second harmonic, rocking and eccentricity, fitted on one start line, holds within about 0.05 turn of it and fails beyond; with the swing’s band added, one law fitted on all eleven start lines describes every one, and fitted near the mirror it carries to lines well off it at the star’s eccentricities 0.20 and 0.22, though not at 0.24 and 0.26. (Section 9; Figure 13.)
10. **One doubling, then escape.** Pushed to higher eccentricity, the island’s centre doubles once (a swing in two steps, as its rocking nears half a cycle per orbit), between the star’s eccentricities 0.348 and 0.350. Then every nearby start escapes, where a frictionless cascade would have gone on doubling. (Section 10; Figure 14.)
11. **A heavy star moves the island below the whole number.** Make the third star heavy and the same angle still holds it, but its island sits just below seven and eight instead of just above, because the pair’s own ellipse now turns faster than the star’s. Each heavy island is chaotic mainly on its low-birth side; at island 7 that is where the star and the pair trade the most eccentricity, through the same harmonic that holds the star. Read at the star’s closest approach, the pair’s eccentricity seems to drain outside the islands; that is the star’s tide bending the reading. That bend is the pair’s short-period response to the star as Georgakarakos (2003) works it out: measured over ten of the star’s laps, the pair’s short-period swing is about 1.12 times the size his formula gives, and the bend read at closest approach 1.08 to 1.29 times, at half the period ratio he tested. Started on the pair’s mean ellipse, the heavy star’s ragged rim is wider than the light star’s at the same pair, stays on the low-birth side whatever the starting angle, and reaches inside the held island: at island 7’s lower edge, chaos that leaves gives way to chaos that stays held within 0.005 in birth. (Section 11; Figures 15 and 16.)
12. **How the eye found it.** Several of these findings began as pictures read by eye: the author’s marks on plain drawings of the runs, each written down in his words as a test before anything was measured. Zoomed out ten times as long at his bet, a scatter of readings became layered, regular structure on a smooth surface of two angles, which holds, read at relative tolerance 10^{-12}, out to near returns 31,063 passes apart over 60,000, where a twin nudged by 10^{-10} in birth drifts off only in a straight line; his clockwork star polygon on the maypole is the inscribed-angle theorem; his dead zones hold on runs he never drew on; the slow swell he traced in the centre’s rocking is the beat; his squeezed and tilted strobe picture found the room that keeps its stars below the early centre; his line that the ratio stays fixed while the rates spread holds over 2,400 passes at the star’s eccentricities 0.20 to 0.24, and cannot hold near 0.28, where the beat passes through zero. Some reads missed, among them a straight-sided narrowing that a finer grid dissolved and lanes that a tool could not see. Sorted blind, picture-reads that name a relation held 19 times in 30 and those that name a shape 11 in 28. Of 490 predictions written before their runs, 236 came out as written, 211 did not and 43 could not be read; every miss is kept. (Section 3; Figures 4, 5 and 9.)

13. What is left for theory

Most questions this work raised were run to an answer (Appendix B). What remains needs theory rather than more runs, and each item says where it is told.

Which side a slowly modulated island tears: the chaotic layer sits on island 7’s lower side and island 8’s upper side, while both members of the island’s multiplet lie wholly inside it (Appendix B, ‘What makes one rim ragged’). Why, from the pair’s eccentricity 0.15 on, the drag’s line carried to zero grip lies above the free stars near the island’s rims: the pair’s secular pull, worked out at each free star’s own period and eccentricity, does not settle it (Appendix B, ‘Why the ellipse turns faster than free near the rim’). Why the centre’s rocking outruns the held island’s width: the second fundamental model of resonance does not explain it (Appendix B, ‘Why the rocking outruns the held width’). How long a hold lasts for good, and what sets the walls between the rooms of the island at the pair’s eccentricity 0.25 (Appendix B, ‘How far a hold over 600 passes is a hold for good’). Why the cascade stops at its first doubling, where the bench’s pushed loop locks instead (Section 10; Appendix B, ‘The newborn’s rocking before escape’).

Why the heavy islands are ragged where the star and the pair trade most, a place where two threads Mardling announced meet (Appendix B, ‘Why the heavy islands are ragged where they trade’), and why the kick’s size falls across the island, which one term at first order does not carry (Appendix B, ‘The kick’s size’). Why the drag’s size from Mardling’s coefficient matches the measured drag near the island’s rim and falls short of it inside (Appendix B, ‘The drag’s size at 0.28’). Whether the pair’s push on the star is odd in the swing, as the fuller odd chains suggest (Appendix B, ‘Odd chains and even chains’). And what sets the slow line near 0.001 cycles per pass beside the zero of the beat at the star’s eccentricity 0.28 (Appendix C).

14. Who came before

These ideas have owners, and they are credited where each is used. Mardling's general expansion of the three-body disturbing function (MNRAS 435, 2187, 2013, DOI 10.1093/mnras/stt1438) is the framework our islands live in. Its [N:1](2) harmonic is our φ_N with the sign reversed, librating about zero with a width that vanishes for a circular inner binary. Its "Zeeman splitting" is our island's offset from the whole number and our beat. Her expansion has since been applied to a test particle beyond a circular perturber, where the overlap of a chain of 2:j resonances (the family our ψ belongs to, by our identification) sets the closest approach below which the distant scattered disk turns rapidly chaotic (Batygin, Mardling and Nesvorný, ApJ 920, 148, 2021, DOI 10.3847/1538-4357/ac19a4), and the Mardling–Aarseth (2001) stability formula has been refined with, among other things, the inner eccentricity (Vynatheya, Hamers, Mardling and Bellinger, MNRAS 516, 4146, 2022, DOI 10.1093/mnras/stac2540). Overlap of neighbouring resonances is, in her treatment, what makes unstable triples chaotic (Mardling, IAU Symposium 246, 2007, DOI 10.1017/S1743921308015615); in the [N:1](2) language of her 2013 paper, the resonant harmonic and its non-resonant neighbour [N+1:1](2) (her Sections 2.2 and 2.3). The second fundamental model of resonance, where an island inside need not behave as a pendulum, is Henrard and Lemaître's (1983). The short-period and secular response of an inner pair's eccentricity to an outer body, for coplanar triples of comparable masses, is Georgakarakos's (MNRAS 337, 559, 2002, DOI 10.1046/j.1365-8711.2002.05936.x; MNRAS 345, 340, 2003, DOI 10.1046/j.1365-8711.2003.06942.x); we take the bend of Section 11 to be its short-period part, read once per pass at the star's closest approach (checked against his 2003 formulas, which are for wider triples with the pair started circular: within about 12% in size; Section 11).

Mudryk and Wu (2006, ApJ 639, 423, DOI 10.1086/499347) explained the instability boundary for planets orbiting one star of a binary as the overlap of sub-resonances. These are moved apart by the companion's secular pull without being widened, and their overlap brings chaos and ejection. Ours is the mirror case, a star outside the pair. Their twin test is the ancestor of ours. The mirror case has a literature of its own. Holman and Wiegert (AJ 117, 621, 1999, DOI 10.1086/300695), mapping where test particles started on circular orbits survive around an eccentric binary, found beyond its inner unstable region an island of instability at the innermost whole-number resonance with the pair, from the 3:1 to the 9:1 as masses and eccentricity varied, the one in their table at the 7:1; for our pair that innermost resonance is the 5:1 or the 6:1, and our islands 7 and 8 lie beyond it (Section 6). Around a circular binary, the overlap of the whole-number resonances, driven by the small body's own eccentricity, makes the continuous chaotic zone through which a small third body diffuses until it is ejected (Shevchenko, ApJ 799, 8, 2015, DOI 10.1088/0004-637X/799/1/8, by the theory of the separatrix map); the harmonic that holds our islands needs the pair's eccentricity instead (Appendix E). At that zone's border the whole-number resonances are unstable teeth, and a planet can survive in the resonance cell between two of them, as Kepler-16b does near the half-integer 11/2, between the unstable 5/1 and 6/1 (Popova and Shevchenko, ApJ 769, 152, 2013, DOI 10.1088/0004-637X/769/2/152). A semi-analytical model of resonance widths, applied to the N/1 resonances of planets around binaries, finds their overlap matching the size of the region where orbits are chaotic (Gallardo, Beaugé and Giuppone, A&A 646, A148, 2021, DOI 10.1051/0004-6361/202039764). Maps of test particles and planets around binaries show the mean-motion resonances beyond the limit as striations of instability or unstable gaps (Doolin and Blundell, MNRAS 418, 2656, 2011, DOI 10.1111/j.1365-2966.2011.19657.x; Quarles, Satyal, Kostov, Kaib and Haghighipour, ApJ 856, 150, 2018, DOI 10.3847/1538-4357/aab264, whose planets start circular), and for circumbinary test bodies around α Centauri AB the pair's resonances bring instability at select intervals (Quarles and Lissauer, AJ 151, 111, 2016, DOI 10.3847/0004-6256/151/5/111). In our setting island 7 has a held core: at the pair's eccentricity 0.20, nine of the eleven births run from its lower half kept their hold for 30,000 passes, the slow chaos taking the two nearest its lower rim, and at 0.25 the island dissolves over 6,000 passes (Section 6 and Appendix B).

Secondary resonances, a rocking commensurate with another frequency, are Érdi, Nagy, Sándor, Süli and Fröhlich's (2007) around L4, and Franklin and Soper's (2012) inside first-order resonances. Twistless tori and their reconnections in the restricted problem are Dullin and Worthington's (2014; arXiv 1309.1280). They sit within the non-twist literature: Howard and Hohs (1984), Howard and Humpherys (1995), Simó (1998), Wurm, Apte, Fuchss and Morrison (2005), Dullin, Meiss and Sterling (2000), and Dullin and Ivanov (2005). Stability changes along families of resonant periodic orbits are Antoniadou and Libert's (2018). Overlap as the criterion for chaos is Chirikov's (1979), and his review sets out the pendulum as the model of a nonlinear resonance, which ties a lock's width to its swing (Appendix B). Frequency analysis reads chaos in a frequency that drifts from one stretch of a record to the next (Laskar, Froeschlé and Celletti, Physica D 56, 253, 1992, DOI 10.1016/0167-2789(92)90028-L); our records of different lengths apply a form of that idea to the rocking rate (Sections 3 and 8; Appendices B and C). Chaos that clings to an island, holding orbits near it for long stretches and so making correlations decay slowly, is stickiness (Karney, Physica D 8, 360, 1983, DOI 10.1016/0167-2789(83)90232-4), reviewed with transport through broken invariant curves by Meiss (Rev. Mod. Phys. 64, 795, 1992, DOI 10.1103/RevModPhys.64.795); we take our rims that lend and take back their stars, and the births beside a lock's core that lose the lock or escape after thousands of passes, to be instances of it (Sections 6 and 8; Appendix C). The doubling cascades and their constants are Feigenbaum's (1978, 1979), and the crossover between the frictionless and the damped constant as the damping is turned, one curve for every damping, is van der Weele, Capel, Post and Calkoen's (Physica A 137, 1, 1986, DOI

10.1016/0378-4371(86)90062-2); the frictionless constant, for area-preserving maps, was first seen by Benettin, Cercignani, Galgani and Giorgilli (1980) and Bountis (1981) and computed by Greene, MacKay, Vivaldi and Feigenbaum (1981).

Every citation was checked against its publisher's record, field by field: authors and their order, year, journal, volume, pages, title and DOI, from the metadata each publisher deposits with Crossref or from the publisher's own page, and Franklin and Soper (2012), which exists only on arXiv, from arXiv.

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Records checked against their publishers on 14 October 2026, again on 15 October, and the last seven open points on 16 October, field by field against the publishers' Crossref records or pages, and Franklin and Soper on arXiv (TRIPLE_CITATION_CHECK_2026-10-14 and its addenda). The seven added on 17 October were checked the same way, six against their Crossref records and Gallardo and colleagues against the publisher's page (TRIPLE_PRIOR_ART_SEVENTH_2026-10-17). The four added on 18 October, Doolin and Blundell, Quarles and Lissauer, Quarles and colleagues, and van der Weele and colleagues, were checked against their Crossref records (TRIPLE_HEADLINES_2026-10-18).

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Appendix A. Methods

A.1 The integrator and the run. Each run integrates the three bodies directly, with an eighth-order Runge–Kutta method, DOP853 (Hairer, Nørsett and Wanner 1993), through SciPy's solve_ivp (Virtanen and colleagues 2020), at relative tolerance 10^{-9} and absolute tolerance 10^{-11}, in units where the pair's semi-major axis and each of its stars' masses are 1. A pass is the star's closest approach to the pair's centre of mass, found as an event of the integration; approaches closer together in time than 0.3 of the star's period count as one, the deeper kept. A run is given the time its passes need and a tenth more. The star has escaped, and the run stops, when it is more than 40 from the pair's centre and unbound from it. A star that completes its passes and leaves within the extra tenth is recorded as escaping at its last pass. Three of the 81 escapes in the light star's runs of 6,000 passes did so (the twins of 7.380 and 7.453, and 7.466, all at the pair's eccentricity 0.25): they left just after the 6,000 passes, not within them. We count them as escapes, and none of them moves an escape edge. The arithmetic has a floor: with every tolerance a hundred times tighter, the centre of island 7 at the pair's eccentricity 0.10 moves by 1.1×10^{-4} turn in net turns over 600 passes, while at 0.25 two births move by 1.3×10^{-3} and 0.22 turn by pass 600, and one of them, which had lost its hold over 6,000 passes, keeps it. The arithmetic also lets the pair's eccentricity fall steadily: by 6.8×10^{-6} over 600 passes at the pair's eccentricity 0.10, 1.3×10^{-5} at 0.20 and 1.1×10^{-5} at 0.25, a few parts in 10^5. The fall is the integrator's, not the star's: at the pair's eccentricity 0.25, with the star's mass set to 10^{-12} it is the same within 0.3%, and with every tolerance a hundred times tighter it shrinks to 2.3% of itself. It moves the rocking too. On the line where the star's eccentricity 0.24 crosses an eighth, every unlocked rocking rate falls by about 2.8×10^{-8} cycles per pass in each pass, steadily over 12,000 passes; with every tolerance a hundred times tighter the fall shrinks to 0.0093 of itself, and the pair's eccentricity falls 5.1×10^{-11} a pass where at the kit's tolerance it falls 1.13×10^{-8} (Appendix B). A rate locked on a fraction does not move with it. Every comparison of unlocked rates read over records of different lengths carries this fall, and a steady sweep can carry a rate into or out of a lock. The 60,000-pass runs at relative tolerance 10^{-12} were made in 12 segments, each a twelfth of the run's time (5,605 or 5,606 passes, since that time, set from the birth with a tenth to spare, holds more passes than the run reads), each started from the state the last one saved, because a segment keeps every step of its integration until it ends; the series built from them is deposited, and so is the first of the twelve segments of the run at birth 7.200 (its file stamped 20261028_082609, in run 20261028_102627), which the energy's check reads; made in two segments instead of one, 600 passes at that tolerance agree within 1.19×10^{-9} turn in φ_N.

A.2 Reading at the close approach, and the whole-turn count. Every angle is read at the star's closest approach and tracked by counting whole turns of the pair between passes from the lap ratio, so that an angle that slips by more than half a turn between readings is never misread (Section 3). With a heavy third star, the pair's turns are counted from its axis crossings as they happen, and the pair's eccentricity is averaged over one period of the pair from each farthest point of the star, because at closest approach the star's tide bends the osculating value; re-run with the extra readings, twelve of probe 35's births reproduce its closest-approach readings within 7.3×10^{-8} in eccentricity and 1.5×10^{-5} turn in φ_N; and whether φ_N is held is checked on the pair's mean ellipse, its apse taken from the orbit-averaged eccentricity vector, because where the pair's mean eccentricity is small against the bend the apse read at closest approach is the bend's.

A.3 Reading the rocking. The rocking rate is read from the spectrum of φ_N over a run. The first reader takes the refined strongest peak at or above 0.05 cycles per pass, a floor set to keep it off the beat. Shown answers known by construction, it reads a single line at 0.080 as 0.08000, and a line at 0.021, below its floor, as 0.05016: it is blind there, which is why the rates at the star's eccentricities 0.12 to 0.16 were read again (Section 8). The second reader looks from three cycles per run upward, counts a line only when it stands 20 times above the spectrum's median power and above a thousandth of the strongest line, drops each line's sidelobes, and names the beat, twice the star's apse turn per lap. It reports the rocking as resolved when the strongest line lies more than 2/N from the beat, N being the number of passes; as 'beat strongest' when it does not; and as an overtone when the rocking lies within 2/N of twice the beat. It passed eight known answers: a rocking strong or weak beside the beat (0.080 read 0.08000 both ways), one under the first reader's floor (0.021 read 0.02101), one inside the beat's window over 600 passes and clear of it over 2,400 (0.0240 read 0.02400), one on twice the beat, the beat alone, and a featureless control, in which it found no line. On the 169 held, locked runs that both readers read, they agree within 5.0×10^{-7} cycles per pass. That agreement checks the second reader's code, not its accuracy: both compute the same spectrum and refine its peak the same way, so they should agree whatever the true rate. Their accuracy was checked apart, on lines planted between the spectrum's bins beside a beat and rounded to single precision as saved series are: over 600 and 2,400 passes the first reader gives each planted rate within 10^{-8}, and the second within 10^{-8} but for a line below the first reader's floor over 600 passes, read within 9×10^{-7}. On a clean line both are accurate far below their agreement. A second line a tenth as strong beside the rocking moves the second reader's rate by at most 1.9×10^{-6}, three bins away over 600 passes, and by less farther off or over longer runs; on a run, what limits a rate is how clean its line is.

A.4 The twin test. Every start can be run again with its birth larger by 10^{-10}, its twin. Over a run of 600 passes a birth is called chaotic when it escapes, lasts under 100 passes, meets a different fate from its twin, or keeps a thousandth of a turn or more from its twin in φ_N (the median gap over the last 50 passes). The test has a floor (Section 3): twins started 10^{-10} apart need about 16 e-foldings to reach a thousandth of a turn, so in 600 passes only chaos that e-folds

faster than about once every 37 passes can be seen. Runs of 6,000 passes read how the twins' gap grows instead: the slope of its logarithm against the logarithm of the pass, from pass 100 until the gap first reaches a hundredth of a turn. A gap that drifts in a straight line, as on a regular orbit, gives a slope of 1, and an exponential gap a large one. Shown planted gaps, the reader gave 1.00 for a straight-line drift, 1.00 for an oscillating drift and 6.34 for an exponential; our registered tests call a slope below 2 a regular drift, and the exponentials of Section 6's runs at the pair's eccentricity 0.25 ran above 4. The rule has an edge: a pair that parts within roughly the last 25 passes of a run still has a small median gap over its last 50, and reads regular. Searched for in every series we kept, it happened in five of the light star's pairs over 600 passes: birth 7.420 at the pair's eccentricity 0.25 (apart from pass 587 and 1.43 turns apart at the end); 7.275 at 0.20, in the scan of the pair's eccentricities (a median gap of 5.5×10^{-4} turn over its last 50 passes, 3.6×10^{-3} at the end); and 7.265, 7.055 and 7.175 in the scan that turns the star's eccentricity (at 0.16, 0.24 and 0.28). Four of them move a number we print by one birth. The fifth, 7.275, sits just inside the island's lower edge; counted chaotic, it joins the lower rim's layer to two chaotic births beyond it and widens that layer from 0.105 to 0.120 in birth. Every number in this paper counts all five chaotic. Over 6,000 passes it happened once (7.360 at the pair's eccentricity 0.20), which the long runs' own reader, the first pass at which the gap reaches a thousandth, already counts. With the heavy star two pairs lie within the stored series' precision of the bar, and neither touches a number we print; the heavy star's edge scan has none among its 243 pairs, and its zoom-out kept no twins to compare. The test is not blinded by the arithmetic's own floor (A.1), because twins share their arithmetic: on regular orbits at the pair's eccentricity 0.10 they end 600 passes a median 1.5×10^{-6} turn apart. Not every held pair below the bar is that close. At 0.10 two of the ten end 3.0×10^{-5} and 3.8×10^{-5} apart (7.070 and 7.075, near the lower rim), and at 0.20 eight of the nine end between 5.9×10^{-5} and 8.6×10^{-4} apart, on the lower flank and at the upper rim (7.295 to 7.305 and 7.515 to 7.535); we had registered that none would. That is slow chaos already parting them, too slow for the 600-pass test, and the reason the long runs read how the gap grows (7.300, whose twins end 8.6×10^{-4} apart, lost its hold at pass 10,771; Appendix B). The converse does not hold. Of the six pairs under 10^{-5} at pass 600 that the 6,000-pass runs also watched, at the pair's eccentricities 0.15 and 0.20, three parted later (7.175, 7.185 and 7.290, at passes 659, 2,389 and 934), while the three between the bars that they watched all parted, by pass 1,354: slow chaos can hide under 10^{-5} at pass 600, and a small gap there does not make a birth regular.

A.5 Fidelity checks. Before any test of a new run is read, the run must reproduce the runs of record it shares with earlier work. A long run, for example, is compared at pass 600 with the scan of 600 passes that first read the same births: within 2×10^{-6} turn of net turns it passes; within 10^{-3} it passes on a looser tier and its verdicts stand with a note; beyond that, every verdict of the run reads no reading. The island's runs at the pair's eccentricities 0.15 and 0.20 for 6,000 passes matched their 600-pass scans within 2×10^{-6} in all 67 main runs. The same check is made after any change to a tool. What it tests is reproducibility, not accuracy: with every tolerance a hundred times tighter, a regular orbit's net turns move by about 10^{-4} turn over 600 passes (A.1), fifty times the 2×10^{-6} the check asks for, so the check confirms that a tool repeats a run on the same kind of arithmetic. Across two machines, two inner births of those islands, run again in the assistant's workspace, matched the laptop's runs at pass 600 within 4.8×10^{-7} and 1.7×10^{-6} turn. The long runs of 28 October were made again from nothing on the laptop: the run of 60,000 passes and its twin, the run at 3×10^{-12}, the twins at 7.300 and the three energy runs, read there by the same tool that read them in the assistant's workspace, triple_surprise_look v1.6 (run 20261028_132326, by the laptop's clock; its run folder run_TRIPLE_CROSSCHECK_20261028_095855). Every verdict came out the same, as we had registered: of its lines, 83 are the same but for the time stamps in file names and 20 give the same verdict with different digits; the three tests of these runs registered after they were sent, on the angular momentum and on the run at 3×10^{-12} (Section 3), were not part of it. Where the digits part is itself a reading. The energy's drift agrees to four digits at 10^{-10} and 10^{-11} and only to two at 10^{-12}, where the drift turns to rising (Section 3) and the two machines' own rounding, which differs, already shows in the third digit. The twins' gaps differ by up to about a sixth, while the proportion between their two angles is the same on both machines, -0.490 at 7.200 and -0.492 at 7.300.

A.6 The bench on known maps: whether the curve turns back, and maps with one hump. Van der Weele and colleagues' crossover curve (Section 4) is, in their abstract's word, non-monotonic. Ours seems to turn back too. Pooled, the ratios fall below the damped constant, to 4.6639, and climb back to 4.6689. Read against each level's own damped value (that of det 0.30), the departure passes the damped value by at most 0.0012 of the span between the frictionless and the damped ratio and comes back, under the 0.01 we had registered as the mark of a turn back; by that mark it does not turn back, and the pooled dip is the approximants' own convergence. Their abstract does not give the size of theirs, so the two are not compared here. On maps with a single hump, flattened or sharpened, the cascade's ratio changes with the hump's shape (3.259608, 3.800556, 4.669199 and 7.284707 for humps of power 1.25, 1.5, 2 and 4), rising with the power.

A.7 Runs made on a laptop. The long runs were made on the author's laptop, at most eight processes at a time, each started by a launcher that looks in its folder for every file the run needs, uses only a copy whose SHA-256 fingerprint matches the one shipped, and refuses to start if any file has none. It copies the matching files under their proper names into a new run folder and runs there, so each run folder carries the exact tools that made it, beside its table, series, ledger and pictures. A run stops starting new integrations after 5.75 hours. Before a laptop tool was sent, it ran

in the assistant's workspace: its self-test on planted answers, a short trial of the real integration, a small batch through the real loop, a fake integration through the whole loop and through a time-limit stop, and its launcher against a tampered duplicate of a needed file, which it refused.

Appendix B. What running the open questions settled

Each question below began as open, and each says why. Most now carry a measured answer from runs made to settle them; what they leave is mostly a question for theory, and each says which. Several are questions Mardling's framework leaves open in print, and each says where.

What makes one rim ragged

One rim of each island is ragged with chaos and the other is clean, and the ragged side follows the pair, not the star. This part asks why. The answer so far: not because two separate resonances overlap, since the nearest members of the holding term's family lie wholly inside the island; which side tears is left for theory.

The layer sits on island 7's lower side and island 8's upper side, and at island 7 it stays on the lower side at every eccentricity of the pair from 0.10 to 0.20, and at every eccentricity of the star from 0.12 to 0.28 with the pair at 0.10 but 0.16, where it is even (Section 6): the ragged side follows the pair, not the star.

A weaker neighbouring term was priced by counting powers of eccentricity and could not explain both islands. Mardling (2013, Sections 2.2 and 2.3) traces the chaos of unstable coplanar triples to the resonant [N:1](2) harmonic interacting with its non-resonant neighbour [N+1:1](2), and to the overlap of their resonances. Islands 7 and 8 lie too far apart to touch, so if overlap makes this layer, it is overlap inside one island's multiplet: the harmonics her Zeeman splitting sets apart.

We weighed the multiplet. For a light star the pair's ellipse stands still, so the holding term's two nearest members sit at period ratios offset by twice the star's apse turn per lap: [7:1](0) below when the ellipse turns forward, and the next order's [7:1](4) above. Each member's half-width follows from the island's own half-width and the square root of its strength against the holding term's. On all nine slices we have, the pair's eccentricity 0.05 to 0.20 and the star's 0.12 to 0.28, both members lie wholly inside the holding island, so the layer is not the overlap of separate resonances. Inside the island they modulate it slowly: the beat runs at 0.001 to 0.485 of the centre's rocking. The stronger member is [7:1](4), on the high side, on every slice, and it grows with the pair's eccentricity as the layer does (0.0051 to 0.0814 of the holding term as the pair goes from 0.05 to 0.20).

At island 8 the ragged rim is the high one, beside its stronger member; at island 7 it is the low one. Which side a slowly modulated island tears is a question for the theory of such resonances, not for one more run.

Whether a lock's width follows its fraction once the rate's slope is taken into account

Simple fractions should hold wider locks than complicated ones, a lock at a fifth wider than one at an eleventh. This part set out to test that and found something else. On the smoothest line we have, no lock at an eighth, a ninth, a tenth or an eleventh is wide enough to see over 2,400 passes; the shapes first seen at those crossings could not be told apart from the integrator's arithmetic sweeping the rates down; and read at the tight tolerance, the passage through a tenth is the pendulum of a resonance seen from outside its islands, regular, with a gap where no birth comes near the fraction.

With only one lock at a crossing, a fair test of width against denominator needs the slope of the smooth line each lock interrupts. We tried it on the saved runs, reading each lock's width in rate rather than in birth: from the curve of the line beside a lock at a turn, and from its slope beside a lock at a crossing. The guess, registered first, was that the width falls with the denominator. It did not hold: about 1.4×10^{-3} cycles per pass at a fifth, 1×10^{-6} to 4×10^{-5} at two ninths, and 1.8×10^{-5} at an eleventh. But the line beside the two-ninths locks is rough. Its scatter about any smooth curve is 100 to 800 times the widths it should measure, and even at a fifth and an eleventh the scatter matches the width. The saved runs cannot settle the question. A fair test needs locks of several denominators on smooth lines at one eccentricity of the star.

So we ran one smooth line, the atlas's start-0 line at the star's 0.24, for 2,400 passes, first on births 1.5×10^{-4} apart around each place it crosses a simple fraction and then on births 10^{-5} apart around each crossing. It passes an eighth, a ninth, a tenth and an eleventh without resting: no two neighbouring births rest on any of them, so over 2,400 passes no flat stretch wider than 2.9×10^{-6} to 6.5×10^{-6} cycles per pass shows there. Nothing that thin can be ranked, and the test reads no reading, where we had registered that the widths would fall with denominator. At a seventh, where the line turns, it flattens instead. Its slope falls twentyfold, from 0.24 to 0.55 per unit birth at the other four crossings to 0.011. The rate stays within 2.5×10^{-5} of a seventh over about 0.0036 in birth, and on the finest grid it still slides through the fraction without resting. On births ten times closer, 10^{-6} apart, the rate still rests on none of the four over 2,400 passes: any flat stretch at the fraction is narrower than 4.8×10^{-7} cycles per pass at an eighth, 8.9×10^{-7} at a ninth, 4.5×10^{-6} at a tenth and 1.7×10^{-6} at an eleventh, and at a tenth, on births 10^{-7} apart, narrower than about 1.8×10^{-6}. The test still reads no reading.

But resting takes time. Near a fraction p/q the slow angle swings like a pendulum, the model of a nonlinear resonance set out in Chirikov's (1979) review, and a lock's half-width in rate times its swing's period is about 2/q, so a lock can rest within a record only if its swing is shorter than the record. Over 2,400 passes at a tenth, that means a lock wider than about 8×10^{-5}. So these runs say the smooth stretch holds no lock that wide; thinner ones they cannot see. A longer record can show what a shorter one cannot: the lock at an eleventh in the atlas, at the star's 0.20, kept no two neighbouring births on the fraction over 2,400 passes and holds all four within 2×10^{-7} of it over 6,000 (Appendix C).

The fractions still leave a mark. At an eighth and a ninth the rate falls evenly through the fraction, birth after birth, at the same slope on both sides. At a tenth it holds about 5×10^{-5} above the fraction over some 10^{-4} in birth, then drops through it within about 2×10^{-5} in birth to 6×10^{-5} to 9×10^{-5} below, crossing at 4.5 per unit birth where the line around it falls at 0.29 on average, and beyond the crossing the line falls at a quarter of its slope before (0.11 against 0.44 per unit birth). At an eleventh it crosses three times, down, up and down again, the first and last crossings 6×10^{-5} to 8×10^{-5} apart in birth, and its slope also falls to a quarter (0.13 against 0.54). These marks belong to the record as much as to the orbits. Read over the first 1,200 passes, the drop through a tenth is three to seven times less steep than over 2,400, so over that length the shape is a resonance the record does not resolve. Our reader does not make these shapes: on made-up series built from each crossing's own harmonics it moves a rate by at most 6×10^{-7} and a slope by at most 6%, and a second reader draws the same shapes at a ninth, a tenth and an eleventh.

Run for 12,000 passes on the laptop, with an eighth as the control, the question met the arithmetic first. On both lines every birth's rocking rate falls steadily as the run goes on: on the eighth's line by 6.61×10^{-5} cycles per pass from each window of 2,400 passes to the next, at every one of its 41 births, about 2.8×10^{-8} in each pass, and on the tenth's line, past the crossing, by about 2.6×10^{-8}. So a longer record reads every unlocked rate lower. The eighth's line moved down whole and kept its slope, 1.000 of its slope over 2,400 passes at both 6,000 and 12,000. The tenth's line moved below the fraction at all 41 births by 12,000 passes, so the test we registered there reads no reading.

Run again with every tolerance a hundred times tighter, a birth on the eighth's line keeps its rate within about 2×10^{-6} over four windows of 600 passes, where at the kit's tolerance it falls by 5×10^{-5}, and the pair's eccentricity all but stops falling (5.1×10^{-11} a pass against 1.13×10^{-8}): the fall is the arithmetic's, the floor of Appendix A.1 seen in the rocking. Near the tenth the tight arithmetic does not settle the rate. A birth just past the crossing jumps toward the fraction by 5×10^{-5} between its second and third windows of 600 passes, as φ_N's swing narrows, where at the kit's tolerance it falls steadily. So over records longer than a few hundred passes the kit's arithmetic carries every unlocked rate down through any fraction near it, and the crossing shapes above, and how they change with the record's length, could not be told apart from that sweep.

Read again with every tolerance a hundred times tighter, over 6,000 passes, the passage through a tenth has no drift to carry it, and it shows what it is. The 41 births cross the fraction between 7.3925998 and 7.3926098, about 2.5×10^{-4} in birth above where the kit's arithmetic had carried the crossing over 6,000 passes. On 27 births 5×10^{-7} apart about the crossing, none comes nearer the fraction than 1.4×10^{-5}: between two neighbours the rate steps from 1.42×10^{-5} above to 1.96×10^{-5} below, and on each side it leans toward the fraction more steeply the nearer the step. We had registered a gap, and it reads gap. Each birth was run with a twin, and none of the 41 pairs parts, where we had registered that the births nearest the fraction would be chaotic: the passage is regular.

What the windows had shown is a slow swing of the rate. Near a fraction p/q the slow angle, q times the rocking's phase less p times the pass count, turns at q times the rate's distance d from the fraction, so the rate should swing once every 1/(10d) passes near a tenth. Read in windows of 600 passes, it does: 11 of the 13 births we registered to read swing within 25% of that period, and every birth whose period lies between 800 and 2,300 passes matches it within 1.2%. The swing grows from 2×10^{-6} far from the fraction to about 7×10^{-5} beside it, and the jump toward the fraction in the tight run above was one such swing. So the passage through a tenth is the pendulum of a resonance seen from outside its islands. Births on either side of the fraction circulate past it, more slowly the nearer they start, and those nearest it swing more slowly than 6,000 passes can follow, which is why their mean rates come no nearer than the gap.

What the second action is

Section 8 found that the rocking needs a second dial besides the swing. This part asks what that dial is. The answer: the star's own mean eccentricity, which labels the quantity the holding harmonic conserves; birth labels it only roughly.

The star's free eccentricity is the obvious candidate. Across island 7 at the star's eccentricity 0.20, its mean eccentricity falls from 0.229 to 0.193 with birth, and the rocking quickens steeply with eccentricity, so the 'birth' dial may be the star's own eccentricity. We ran stars of equal birth at five eccentricities of the star and read each run's mean eccentricity. The star's eccentricity is the better second dial: with the swing, it leaves a scatter of 0.031 cycles per pass where birth leaves 0.043, and birth adds nothing once it is in.

That first comparison used rates at the star's eccentricities 0.12 and 0.16, below the slowest rate our reader was built to read, and at 0.28, where every one of these births rocks at two ninths. So we stepped the star's eccentricity finely instead, from 0.19 to 0.25 in steps of 0.01, at six fixed births from 7.15 to 7.20, where the rocking is free and our

reader sees it: 42 runs, every one held and locked. There the star's mean eccentricity carries the rocking. With the swing, it leaves a scatter of 0.0032 cycles per pass where birth leaves 0.0135, about a quarter as much, and adding birth lowers that by 1.2%, as our registered guess asked. Across the seven eccentricities the rocking grows as the star's mean eccentricity to the power 2.6, near the 2.5 that Mardling's pendulum gives, whose strength grows as the fifth power at small eccentricity.

What this cannot settle is the dial inside one slice. There the star's mean eccentricity falls with birth so closely (correlations of −0.98 to −0.997) that the two cannot be told apart from these runs, so these runs alone cannot say whether the island's second action is the star's own eccentricity. The atlas's seven start lines loosen that tie a little, since at each birth they give seven swings. Over all of them, 318 to 506 runs a slice, the star's mean eccentricity carries the rocking beyond birth and swing: at the star's eccentricities 0.20 to 0.26, a fit in birth and swing leaves 1.65 to 5.20 times the scatter of a fit with as many terms in birth, swing and the mean eccentricity, with the swing read from φ_N's root-mean-square (1.95 to 4.93 with it read from the band).

That is what the one-harmonic model below expects once our starts are read as they are. Its second action, the quantity the harmonic conserves, is fixed by the start's period and eccentricity, but in mean elements, and our starts are osculating. Read at every closest approach in fourteen runs at two births, it differs from one start line to the next at the same birth, rising steadily with the starting angle, and it follows the run's mean eccentricity with a correlation of −0.9997. At three more births it does the same: at 7.150 and 7.300 it rises strictly with the starting angle on all seven start lines, as we had registered, and at 7.350, held on only three of them, on those three. Within each run it drifts by about a quarter of that spread (0.239 and 0.245 of it at the two births, under the quarter we registered). So the star's own eccentricity is the better dial because it labels the conserved quantity, which birth labels only roughly; the small drift is the one place a second harmonic may still act.

At the star's 0.28, over 2,400 passes, that drift is a slow swing rather than a creep: it keeps one sign in only 77% of the 31 runs there, where we had registered at least 80%, and over the whole record it is about half its size over the first 600 passes. It is also some twenty times the drift at 0.20. Over 2,400 passes the same holds at 0.20, 0.22, 0.24 and 0.26: the drift keeps one sign in only 52% to 68% of the runs, and at three of the four lines it is smaller over the whole record than over the first 600 passes. At every eccentricity of the star we ran, the drift is a swing, not a creep.

In the one-harmonic model the conserved quantity ties the star's eccentricity to its period. Our stars all start at one eccentricity at φ_N = 0. A star born low in the island starts at the short end of its swing and swings out to longer periods and, by the tie, higher eccentricities, so it carries a higher mean eccentricity than one born high, and the harmonic, whose strength grows as the fifth power of that eccentricity, rocks it faster. Run with nothing fitted, the model gives the low-birth flank the faster rocking at every matched swing at every eccentricity of the star from 0.20 to 0.34, as registered; its gaps lie inside the measured range at 0.20 to 0.28 and above it at 0.30 to 0.34, where few matched pairs remain and the model's neighbouring islands begin to overlap, so the registered guess that they would lie inside at six of the eight eccentricities missed: they lie inside at five.

Why the ellipse turns faster than free near the rim

Near the island's rim, held stars turn their ellipse faster than free stars far out, and from the pair's eccentricity 0.15 on, the drag's line carried to zero grip lies above even the free stars just outside the rims, which a simple brake should not do. This part asks why. The measurements narrow the question, and the answer is left to theory.

The grip explains the drag's shape, and the drag's slope follows the pair's eccentricity. Its level is half explained: free stars just outside the lower rim turn half again faster than those far out, and at the pair's eccentricities 0.05 and 0.10 the zero-grip level falls between the free rates just outside the two rims, but at 0.15 and 0.20 it lies above both, and at 0.25 above the one free reading there (at 0.25, 0.0326 against 0.0120 above; the single 'free star' below, 0.0065, turned out to be the chaotic thread inside the island at 7.420, so there is no free reading below at 0.25 yet).

We then read free stars at matched distances, 0.030, 0.060 and 0.090 outside each rim, at five eccentricities of the pair. Matched, the free star just outside the lower rim turns faster than the one three times farther out at 0.10 (0.0223 against 0.0199) but not at 0.05 (0.0207 against 0.0222). From 0.15 on, the star 0.030 below the lower rim is chaotic, or at 0.25 held: the chaotic layer has taken the free star's place, so there is nothing free to compare with there. Where a free star exists 0.030 above the upper rim, at 0.15 and 0.20, the zero-grip level still lies above it (0.0223 against 0.0130, and 0.0267 against 0.0123). Why the drag's line, carried to zero grip, lies above the free stars near the rims stays open; the drag's size (Section 9) does not settle it, since the harmonic's term vanishes at zero grip.

Two pieces of it could be run, and were. The pair's secular quadrupole pull, worked out at each free star's own period and eccentricity, overstates the free stars: far above the island, more than 0.10 in birth past its edge, they turn at 0.89 of it (median over 218 runs), where we had registered within 5%. Nearer the rim their turn's excess over it grows with one over the distance, but loosely (rank correlation 0.75, against the 0.8 registered). And the zero-grip level's lead over that rate does not rise in order with one over the centre's beat across the atlas's five lines at the star's eccentricities 0.20 to 0.28 (rank correlation 0.70; it dips at 0.24). What a near-resonant term adds at second order in the harmonic's strength needs a derivation that our tools cannot make; it is for theory.

Why the rocking outruns the held width

Mardling's pendulum ties how fast the centre rocks to how wide its island is. Ours rocks faster than its held width allows, and this part asks why. Counting the chaotic layer in with the island closes most of the gap, within about 10% from the pair's eccentricity 0.05 to 0.20, tested forward at 0.225; the second fundamental model of resonance does not explain the rest.

At the pair's eccentricities 0.15 and 0.20, twice the centre's rocking rate exceeds the held island's half-width by 16% and 34%, and the centre's rocking quickens 1.435 times from 0.05 to 0.20, where Mardling's pendulum gives 1.98. With the chaotic layer counted into the island, the two come within about 10% at every eccentricity from 0.05 to 0.20; that was noticed after the run, so it was registered as a guess and tested at 0.25. There, with the chaos within 0.15 of each edge counted in, twice the centre's rocking exceeds half the width by 13%: close, but outside the 12% registered.

That count used the island cut short by its one-birth thread. Read across single gaps whose neighbours are held and regular (7.390 to 7.570, 0.185 wide), with the chaos within 0.15 of each of those edges counted in, the two agree within 1.1%. That reading's top birth, 7.570, held for the scan's 600 passes but chaotic by the twin test, lost its hold in a run of 6,000 passes (Section 6); counted with the chaos instead of the island, it leaves the sum unchanged, since the rule counts island and chaos together. That reading came after the guess was scored, so it became a new guess, tested forward: at the pair's eccentricity 0.225 the two agree within 9.4%, inside the 10% registered.

With the pair at 0.10 and the star's eccentricity turned, the bridged reading agrees within 4% at 0.20 and within 8% at 0.24, but not at 0.28 (+29%). We had read that miss as the layer's reach, since the lower layer runs farther than the 0.15 counted, and registered that counting all of it would bring the rule within 10%. It does not. Below the island at 0.28 the chaos runs unbroken for 0.735 in birth, down to the next island (Section 6), and counted whole it takes the rule to −34%. There the island's lower edge borders the sea, and the island has no layer of its own to count. At 0.12 and 0.16, read with the second reader (Section 8), it agrees within 2.3% and 3.5%, read after the fact; the first reader, blind below 0.05 cycles per pass, had put it 149% and 51% off.

The twist, how fast a free star's slip changes with birth, stays between 1.05 and 1.25 at the three eccentricities where both sides could be read (0.12 to 0.20), so the pendulum's own correction is small there. In deriving widths, Mardling assumes the eccentricities are not so small that the turning of the ellipses adds significantly to them, and leaves that case for elsewhere (Mardling 2013, Section 3.7; for φ_N's own width the like assumption is in her Section 2.3); our islands, whose offset and beat are both set by that turning, are that case.

She also names the second fundamental model of resonance (Henrard and Lemaître 1983) as the alternative to the pendulum (Mardling 2013, footnote 9). Our harmonic is of fifth order in the star's eccentricity, not first, so her normal form does not apply as it stands; we ran the nearest model of that kind. It keeps the one harmonic with its strength's full dependence on the star's eccentricity, adds the steady turning of the star's ellipse, and is reduced to one degree of freedom by the quantity the harmonic conserves (in the usual Delaunay actions, the star's Γ less (N − 2)/N of its Λ, which ties the star's eccentricity to its period). Births are read as mean elements, which the integrator's starts are not. It keeps the pendulum's sizes. At the star's eccentricity 0.20 it makes island 7 33% wider than measured and its centre rock 25% faster, and from 0.20 to 0.34 the centre's rocking rises 3.4-fold against 5.3 measured, though by 0.34 the model's islands 6, 7 and 8 overlap, and one harmonic no longer describes them. The strength's growth with eccentricity moves the pendulum's rate by only 1 to 2% and its width by about 5%, and it makes twice the centre's rate fall short of the half-width, by 3 to 13% as the pair's eccentricity grows from 0.05 to 0.20, where ours exceeds it by up to 34%. So it is not why the rocking outruns the width.

What the model adds, with nothing fitted, is shape. Its island is lopsided as ours is, the centre about twice as far from the lower edge as from the upper (2.10 to 1 against 2.18 measured at island 7; 1.66 against 1.75 at island 8). The star's mean eccentricity falls with birth across it, −0.156 per unit birth against −0.175 measured; that follows almost wholly from the tie and from every star starting at one eccentricity at φ_N = 0, so it tests the harmonic's angle more than its strength. And it gives the second dial (the second action, two items above).

How far a hold over 600 passes is a hold for good

A star held for 600 passes is not always held for good. This part asks how long a hold lasts, and what divides the island at the pair's eccentricity 0.25 into rooms. At 0.20 the slow chaos empties the island's lower half from its outer edge inward, two births in 30,000 passes; at 0.25 the walls between rooms are not set by where the rocking passes a simple fraction, and what sets them stays open.

At the pair's eccentricity 0.25 a hold over 600 passes is not a hold for good: watched for 6,000 passes, the island read over 600 passes dissolves into slow chaos (Section 6), and the rocking rule, which counts island and chaos together, has no lasting island there to be tested on. At 0.10 the two outermost held births kept their hold for 6,000 passes. At 0.15 and 0.20 the islands last 6,000 passes, less about 0.02 in birth at their eroded rims, though at 0.20 slow chaos has reached into the island's lower half (Section 6).

What stays open is how long ‘for good’ is. Six thousand passes is more than 40,000 of the pair’s periods, and the slow chaos in the 0.20 island’s lower half could still empty it over a longer run. So we ran the births 7.300 to 7.400 of that lower half, with twins, for 30,000 passes, more than 200,000 of the pair’s periods. Two of the eight born 7.300 to 7.370, where twins part slowly, lost their hold, last within a turn of their start at passes 10,771 (7.300) and 23,267 (7.310), and both escaped, at passes 11,977 and 28,569; the twin of 7.300 escaped too, at pass 22,094, while the twin of 7.310 held. We had registered that at least three would lose their hold. The other six born 7.300 to 7.370 and all three born 7.380 to 7.400 held through all 30,000 passes, their twins with them, and nothing born from 7.380 up escaped, as registered. The slow chaos does empty the lower half, but from its outer edge inward and slowly: in 30,000 passes it took the two births nearest the lower rim.

At 0.25 the long runs divide the old island into rooms along thin threads (Section 6). We guessed, registering it first, that the threads sit where the rocking keeps a whole number of beats, as secondary resonances inside an island would. They do not, and the reason is that the beat is no steady drum here: across the island it falls tenfold, nearly to nothing in the room where no star escapes. The rocking holds steadier, near a ninth of a cycle per pass. Each thread lies between births whose rocking falls on either side of a ninth, while the middles of the rooms keep to one side of it. We saw that after the fact, so we registered it as a guess and ran a finer grid (Section 6).

The third thread is real, a band of fast chaos about 0.01 wide, and the escapes stop sharply at 7.472, 0.015 above the band. The edge lies 0.001 from a pair of neighbouring births whose rocking straddles a ninth, while the nearest pairs straddling a tenth and an eighth lie 0.010 and 0.013 away. That passes the test as we wrote it, but weakly, because the rocking is near a ninth almost everywhere.

Inside the band it wanders between about 0.09 and 0.15 of a cycle per pass. In the room above it most runs ring louder at the beat than at their rocking, and of the thirteen whose rocking can be read, ten sit between 0.104 and 0.120, near a ninth. It is not a rate the room keeps. Read over 2,400 passes, the 23 births of the room held that long rock anywhere from 0.055 to 0.138, their median 0.107, below a ninth (we had registered that it would stay above 0.1117), and read stretch by stretch a birth’s rate wanders, 7.477 reading 0.136 in its third stretch of 600 passes and 0.058 in its fourth; by this reader every stretch’s median lies below a ninth (0.093 to 0.105). A few stray readings and every small wander cross a ninth, so half the births from 7.445 to 7.501 lie within 0.003 of a pair straddling it.

A classical picture of resonances inside an island suggests that a wall where the rocking passes a ninth would be a chain of small islands locked at a ninth, with fast chaos between them. We registered that too, and no birth in the band reads at a ninth. So we looked ten times closer, where the 600-pass rocking had stepped across a ninth between two neighbours, 7.473 and 7.474, after registering the guess that three or more births 0.0001 apart would lock at a ninth with fast chaos beside them. None did. Of 31 births from 7.4720 to 7.4750, each run with its twin for 600 passes, 20 could be read, none rocks within 0.5% of a ninth, and no pair of twins parts within the run.

The spectrum of each run shows why the step had looked like a crossing. It holds 7 to 17 lines of comparable strength between 0.08 and 0.15 cycles per pass, the same lines recurring from birth to birth (seven of them at 27 to 31 of the 31 births), with only their heights reshuffled: the second-loudest line has a median of 0.84 of the loudest’s power. The reader’s single rate hops between neighbouring lines rather than sliding through a ninth. We then guessed, after seeing the hops, that each run’s lines form an even comb about one line just above a ninth; 4 of the 31 do, too few for the guess to hold. Near this edge, over 600 passes, the rocking is a cluster of lines, not a rate, so a wall cannot be placed by where a rate passes a ninth, and what sets the walls stays open.

The newborn’s rocking before escape

Section 10 found one doubling at the island’s centre and then escape. This part asks what the newborn rocks at before it escapes. Its swell is the rocking seen once a pass, and the escape comes with the doubling, not before it.

Read over the stretch each run holds before it breaks loose, the alternation is too weak to read: the slow drift that carries the star out swamps it. Over the first 400 passes, where four of the eight runs that hold that long alternate steadily, the alternation swells and fades at 0.0121 to 0.0126 cycles per pass in all four, about 1.6 times half the beat, not at half the beat as the loop bench would have it (a guess registered first and not borne out). What sets that common line is read below.

We read the same way at the star’s eccentricity 0.348, two fifths of the way back toward 0.345, to see whether the line stays put or moves with the beat. There the question had nothing to read: none of nine runs escaped over 450 passes, every band was narrower than at 0.350, and only two alternated steadily, both barely. Their lines spread from 0.0035 cycles per pass at the centre to 0.057 a short way out. An alternation watched on alternate passes swells and fades at the distance of the rocking from half a cycle per pass, so the centre’s line suggests a rocking 0.0035 short of a half, and the doubling just above 0.348. Reading the rocking directly from the same runs bears this out: in 15 of 16 runs the swell’s rate equals the rocking’s distance from half a cycle per pass, most within 0.00005.

The envelope line is not a separate motion; it is the rocking seen once a pass, and the common line at 0.350 says only that the steady alternators there rock at one shared rate, about 0.4875 cycles per pass. Followed for 1,600 passes, the nine starts at 0.348 all hold, their φ_N bands 0.1658 to 0.1845 turn, where all nine at 0.350 escaped within the same

1,600 passes: the escape comes with the doubling, not before it.

Why the heavy islands are ragged where they trade

With the heavy star, each island's chaos sits on its low-birth side, where the star and the pair trade more eccentricity than at the other edge. This part asks whether the start chose that side, and what the trade looks like there. The start did not choose it; each chaotic layer is a band of high trade; why the trade is largest there needs theory.

The heavy star's chaos sits on the low-birth side of both islands, while the light star's island 8 is ragged above. In the scans started at the star's closest approach, island 7's ragged side is where the star and the pair trade the most eccentricity, while island 8's two edges trade about equally. Those runs all start at $\varphi_N = 0$, and the first swing goes up on the low-birth flank of both islands and down on the high-birth flank in every held run but one (φ_N at the third pass: positive in 42 of 43 and 23 of 23, negative in 23 of 23 and 13 of 13), so the start might have chosen the side. It does not: started a quarter turn from the holding angle, with the pair's eccentricity averaged over its own orbit, both islands are still ragged mainly below (Section 11).

With the pair started on its mean ellipse, the lower edge also trades more eccentricity with the star than the upper edge, at both islands and in every start: 1.20, 1.08 and 2.04 times as much at island 7, and 1.31 and 1.06 times at island 8. Each chaotic layer is a band of high trade. At its outer end the swing of the pair's mean eccentricity jumps between neighbouring births, from 0.030 to 0.087 at island 7 with the pair at 0.05, and the stars that escape swing it by 0.39 to 0.52. The same jump appears in an earlier set of scans started on the mean ellipse, at the outer end of every complete lower layer: 2.11 to 2.77 times between neighbouring births, at three islands 8 and one island 7. Even single chaotic births outside a layer swing the pair as the layer does, while their regular neighbours do not.

Across each island the trade follows the swing of φ_N in two of the three starts, at both islands in one and at island 7 in the other (rank correlation 0.90 to 0.99 over the island's births), but not in the third: there the trade stays between 0.051 and 0.061 across island 7 while the swing runs from 0.43 to 0.87 turn, and 4.06 turns at a chaotic birth on the rim (rank correlation 0.41), and follows only loosely at island 8 (0.85). That flat trade is made almost at once: over its first 50 passes each of the island's held births trades 82% to 99% of what it trades over 600, but for two near the upper rim, 7.130 and 7.135, at 75% (we had registered at least 80% at every birth). The trade is not simply the swing read another way.

What remains is why the trade is largest there, and we cannot run that question; it needs theory. Two threads Mardling announced meet here: the eccentricity a companion induces in the pair, derived in a paper she cites as submitted (her Paper II, which we could not find in print), and the 'modes of oscillation of a binary' excited by a triple companion, promised for a later paper in her series (Mardling 2013, Section 3.4).

The kick's size

The kick is the nudge the heavy star gives the pair's mean eccentricity at each close approach (Section 2). This part asks whether Mardling's coefficient gives its size. With nothing fitted it gives a median 1.029 times the measured kick at island 7 and 1.199 at island 8, but it does not carry the kick's fall across the island.

Across island 7 the pair's mean eccentricity is kicked through the holding harmonic (median correlation 0.957), and island 8's kick is 0.216 of island 7's (0.235 on island 8's four clean runs) against the harmonic's 0.268 to 0.297. The ratio was read before it was predicted, so that part of what follows is a check, not a test.

The absolute kick follows from Mardling's coefficient through Lagrange's equation for the pair, as the drag's size did in Section 9: the star's tide on the pair, $(3/4)\ m_3\ (a_i^2/a_o^3)\ X_1^{2,2}(e_i)\ X_N^{-3,2}(e_o)\ \cos\varphi_N$, taken at each run's own mean eccentricities with nothing fitted. Registered before it was computed, it gives the measured kick to a median of 1.029 over island 7's fourteen runs and 1.199 over island 8's four clean ones, inside the band we set (0.80 to 1.25), and island 8's kick 0.247 of island 7's against the measured 0.216.

Run by run on island 7 it falls from 1.350 of the measured kick on the low-birth flank to 0.598 at the high edge, as the pair's mean eccentricity across those fourteen runs falls from 0.0757 to 0.0341; one term at first order does not carry that trend, and what does is a question for the theory at the next order. The trend does not follow the swing of the pair's eccentricity across each run (rank correlation 0.35, where we had registered that it would follow the swing more closely than the mean, 0.78). It follows the star's mean eccentricity more closely still (0.86): across the island the predicted kick halves, from 0.01689 to 0.00821, as the star's mean eccentricity falls from 0.2245 to 0.1928 and the harmonic weakens steeply with it, while the measured kick stays between 0.01140 and 0.01867.

The drag's size at 0.28

Section 9 gives the drag's size from Mardling's coefficient with nothing fitted, and one line, the star's 0.28, kept reading low. This part tracks down why. The leftover is mostly the part of a slow cycle that a 600-pass record cuts off; read over 2,400 passes, the size is not one number: the coefficient matches the measured drag near the island's rim and falls short of it inside, about 0.7 of it on the slice middles.

Section 9 gives the drag's size from Mardling's coefficient, run by run, inside the band we registered on all five lines, but on our fifth try, and 0.28 still reads lowest (0.867). The tries in order: at the star's starting eccentricity, a tenth to a quarter short everywhere; at each line's mean eccentricity, within 6% at 0.20 to 0.26 but 0.822 at 0.28; then the terms that could add at 0.28, priced before any was built. The member of the harmonic's family whose angle is φ_N less twice the difference between the pair's apse and the star's, with coefficient $X_1^{2,0}(e_i)\, X_7^{-3,0}(e_o)$, could hold still with φ_N where the beat passes through zero, but even with its cosine held at one it gives 3.9% of the gap; the next order of the pull, with φ_N's own angle, adds 2.7% to 2.9% on every line, and the next member of that order less than 1%. Next, each run's eccentricity fitted beside its grip (a correlation of 0.61 at 0.28): the gap at 0.28 closed (1.012) and the 0.26 line overshot (1.191), outside the band. Last, each run's drag at its own eccentricity, which passed. The pass cuts the scatter about four of the lines to 0.27 to 0.56 of what one slope a line leaves, but at 0.28 only to 0.82, so something there is still left over.

We looked for it three ways. It does not split by side of the beat's zero at 0.2811, where the centre's ellipse stands still (0.7 standard errors). It does not sit in the chaotic layer at the rim: probe 41's twins call only four of the 76 held runs chaotic, and leaving them out changes nothing. Nor is it the island's locks: the runs locked at two ninths or two elevenths fit the drag law about five times better in mean square than the rest. It belongs to the unlocked runs inside the island, which rock out of step with the pass, so a 600-pass record ends partway through a slow cycle between the two and both its grip and its apse turn carry the part cycle. Longer records shrink it. Run again for 2,400 passes, the 31 middle births at 0.28 leave a mean square scatter about the law of 0.326 of what their first 600 passes leave, under the half we had registered, so most of the leftover is the part cycle of a short record.

The longer record also reads the drag's size far more tightly. Over their first 600 passes these births give the prediction as 0.975 of the measured drag, loosely (0.766 to 1.340 at one standard error); over 2,400 passes, 0.695, firmly (0.621 to 0.789), below the band we registered for the drag's size. At 0.20 to 0.26 the longer record does not move it. The same middle births give 0.71 to 0.77 over 600 passes and 0.70 to 0.78 over 2,400, and 0.28's 0.695 over 2,400 sits with them. So on the slice middles the size reads 0.70 to 0.78 at every line, about a fifth below what the atlas's runs across the island give at the same lines over 600 passes, each run's drag taken at its own eccentricity as in Section 9 (0.92 to 1.01 at 0.20 to 0.26, and 0.867 at 0.28). Across so narrow a band of births the star's eccentricity and its grip move together, with correlations of −0.92 to −0.95 against −0.21 to −0.86 across the island, so there the fit cannot cleanly part the eccentricity term from the drag.

So we ran the runs across the island again, all 369 of them, for 2,400 passes. The size does not hold as one number, and the test we registered, 0.85 to 1.18 at every line, reads not borne out. On the runs that keep their hold and lock for all 2,400 passes, 45 to 70 a line, it reads 0.82 to 0.87, three lines below the band. Read over their first 600 passes, the same runs give 0.80 to 0.87, so the longer record moves it by at most 0.015, within one standard error. What moves it is which runs are fitted. The runs that lose their hold or lock before 2,400 passes, 7 to 18 a line and all at the ends of the island's range of births, give 0.97 to 1.33 over their first 600 passes, and leaving them out brings every line down.

Split by place, the long-held runs farther from each line's centre read higher than those nearer it on four of the five lines (0.76 to 0.91 against 0.66 to 0.85, beyond their errors), while 0.20 reads the other way (0.85 against 0.95). So the prediction from Mardling's coefficient matches the measured drag near the island's rim and falls short of it inside: about 0.7 of it on the slice middles, 0.8 to 0.9 across the long-held runs, and 1 to 1.3 at the rim. In the halves at 0.26 and 0.28 the eccentricity and grip terms move together closely (correlations −0.91 to −0.94), so those sizes are the least certain. The size we give in Section 9 comes from all the runs across the island over 600 passes, which makes it a size nearer the rim's than the middle's.

Odd chains and even chains

Odd fractions lock about three times as often as even ones (Section 8). This part asks whether that is a rule of parity or only a preference for a few simple chains. Weighed chain by chain, it is both: simpler chains are fuller, and at equal size an odd chain is about three times as full as an even one.

Inside the island the rocking locks to fractions with odd denominators about three times as often per crossing as to even ones, near the mirror line and well off it alike (Section 8), so where the runs start does not make the parity. In a swing symmetric about its centre, a push odd in the swing feeds only the chains of odd denominator and a push even in it only the even ones, so the count may be reading which kind of push the pair gives the star. Two things kept that open. Most of the odd excess sits on two chains, the fifth and the seventh, so a count by parity could not tell a rule of parity from a preference for a few simple chains; and off the mirror the even crossings only just reach the floor we set for a reading.

Pooling the atlas's seven start lines with the four off the mirror crosses every chain from 4 to 12 at least 11 times, enough to weigh each by itself. Registered first, every odd chain holds more runs per crossing than the mean of its two even neighbours: the fifth 1.10 against 0.477, the seventh 0.673 against 0.284, the ninth 0.475 against 0.155, the eleventh 0.395 against 0.0744. Fitted together, the runs per crossing fall by a factor of 0.783 with each step in

denominator and rise 3.16 times for an odd one. Both hold, then: simpler chains are fuller, and at equal size an odd chain is about three times as full as an even one. Whether the pair's push on the star is odd in the swing, as that reading needs, is a question for theory.

Appendix C. The detail behind the sections

Each part below carries the full record behind a section's main text, moved here whole so that the sections can be read as a story. Nothing here is new; each part names its section.

For Section 5: Mardling's harmonic, and her pendulum's test in detail

Section 5 says our holding angle is Mardling's and that her pendulum passes a direct test. Here is that in her notation, with the test at every eccentricity we read: the rocking rule holds within 1.1% to 8% where her assumptions hold, and fails for the held island where a chaotic layer eats it.

The angle φ_N is the harmonic Mardling (2013) names φ_21N, the one that governs triples like ours near period ratio N, read at the star's close approach. Hers is M_bin − N M_star + 2(ϖ_bin − ϖ_star) (her equation 21, with M_star the star's mean anomaly and ϖ_star the direction of its ellipse's long axis); at a close approach M_star is a whole number of turns and θ is ϖ_star, so ours is hers with the sign reversed, which changes nothing about a swing around zero. In her notation it is the [N:1](2) harmonic: [n′:n](m) names the harmonic whose angle runs n′ times with the star's mean motion against n times with the pair's, at spherical-harmonic order m (her Sections 2.2 and 3.4). Her pendulum for it predicts that it librates about zero, as ours does.

It also predicts our islands' widths to within about 30%, and that the small-swing rocking rate in cycles per pass is half the island's half-width in period ratio. Ours satisfy that within 1.1% at the pair's eccentricity 0.10, and within 3% at 0.05, with the star's eccentricity at 0.20. With the star's eccentricity turned, it holds within 8% at 0.24. At 0.12 and 0.16 the island rocks near or below the slowest rate our first reader was set to read, and read with it the rule missed by 149% and 51%; read with a second reader that sees below the beat (Section 8), with the island counted the same way as at 0.24, it holds within 2.3% and 3.5%, a reading made after the fact.

Her formal domain is triples of 'moderate mass ratio', those whose stability the N:1 resonances govern, which in practice she puts at both mass ratios above about 0.05 (Mardling 2013, Section 1.1). Her expansion itself holds for any masses; our light star is a millionth of either star, and the expansion has since been applied to a test particle by Batygin, Mardling and Nesvorný (2021). At 0.15 and 0.20, where a chaotic layer eats the island's lower side, it does not hold for the held island (Section 6).

For Section 6: the island at the pair's eccentricities 0.225 and 0.25 over 600 passes

Section 6 says the island's edge changes as the pair grows more eccentric. These are the two scans that show it: at 0.225 the old picture holds, a ragged lower side and a nearly clean upper one; at 0.25 a chaotic thread runs through the island and its upper rim turns chaotic too.

Between the pair's eccentricities 0.20 and 0.25, at 0.225, the old picture still holds: the island is 0.220 wide in birth (7.325 to 7.540), the chaotic layer touching its lower edge is 0.125 wide and the one touching its upper edge 0.010, and all 27 escapes start below the island, 13 of them among the 30 births within 0.15 of its lower edge.

At the pair's eccentricity 0.25 the picture changes. The held island shrinks to 0.170 in birth (7.390 to 7.555), and a single chaotic thread runs through it at 7.420. Re-run at births 0.001 apart, the thread is narrower than 0.002: 7.419 and 7.421 are held and regular on either side of it. Its runs on another machine escaped soon after 600 passes, so the thread leads out. Read strictly as an unbroken held block, the island stops at the thread and measures 0.135 (7.425 to 7.555).

Below the island lies a chaotic band that no longer touches the island's edge. Of the 30 births in the 0.15 below 7.425, 24 are chaotic, the thread among them (Appendix A.4), and 15 escape within 600 passes, every escape from below the island. Above it, chaos begins at the very next birth and covers 0.090 of the next 0.15 in birth, all of it bound, with held births scattered inside. The lower side is still the ragged one, the side stars escape from, but the upper rim is now chaotic too. Counted the earlier way, as chaos touching the edge, the layer reads 0.005; counted within 0.15 of each strict edge (7.425 and 7.555), it is 0.120 below and 0.090 above.

For Section 6: the door's escapes over 6,000 and 30,000 passes

Section 6 says stars leave through the side nearest the pair. Watched for longer, the same holds: below the island eleven of twelve left within 1,186 passes, while above it fourteen stars lost two in 30,000 passes.

Run for 6,000 passes at the pair's eccentricity 0.25, eleven of twelve stars started below the island left, after 46 to 1,186 passes (half of them within 243). With one exception, the later a star left, the nearer to the island it had started. The exception, 7.385, started in a chaotic gap one birth wide between the island's edge and a small patch below it (7.375 to 7.380) that held for the scan's 600 passes, and it left at pass 442. Of nine stars started above, eight were still

bound after 6,000 passes, and one left at pass 5,904. Watched five times longer, for 30,000 passes, fourteen stars born 7.570 to 7.700 above the island, 0.010 apart, lost two: that one, and one more at pass 14,278. Both left after the 2,430 passes we had registered as the earliest, and fewer than half left, as registered; the other twelve circulated to the end.

For Section 6: held for a while, at the pair's eccentricity 0.25

At the pair's eccentricity 0.25 the island's rims are only lent: stars there hold for a while, drift off, and are often caught again. This part follows what each star does after its first move. Below the island the stars that leave do so soon after; above it the chaos keeps handing stars back, which the study of such systems calls stickiness.

Where the two sides differ is in what follows a star's first move. On both sides a star first stays within a turn of where it started, for anywhere from 1 to 923 passes; half of those below had moved a full turn away by pass 90, half of those above by pass 255. Below, ten of the eleven that escaped were gone within 273 passes of that first full turn, and the climb of the escape pass toward the island lies mostly in the first stay: over the ten started below the patch, its rank correlation with birth is 0.85, against 0.50 for the time after. Above, eight of the nine circulated for the thousands of passes that remained. The one that left had moved a full turn away first of the nine, at pass 14, and left last.

Four of these stars had read as held at the end of the scan's 600 passes, 7.375 below the island and 7.570, 7.590 and 7.630 above it, each chaotic by the twin test. Like the nine of Section 11, all four lost their hold, after passes 651 to 922, and none escaped. Five more births at the island's rims, held for the scan's 600 passes with their twins together, were run 6,000 passes the same way, after we registered the guess that all five would keep their hold. One did: 7.395, just inside the lower edge. The other four, 7.390 at the lower edge and 7.550, 7.555 and 7.565 at the upper rim, stood within a turn of their start for the last time at passes 1,602 to 2,615, and none escaped. Their chaos is too slow for twins to show in 600 passes.

Nor is a lost hold lost for good. Counted in stretches of 600 passes, all four were held again for whole stretches later, three of them rocking tightly enough to count as locked. So were seven of the nine stars above the island once they had first moved a full turn away, five of them locked; the one that escaped from above was locked through passes 4,800 to 5,400 and left at pass 5,904. Below the island, the one star that stayed, 7.375, was never held again in nine stretches. At this eccentricity the island lends its rims and takes them back, and the chaos above it keeps handing stars back. Chaos that clings to an island, catching stars for long stretches before letting them go, is what the study of transport in Hamiltonian systems calls stickiness (Karney 1983; Meiss 1992). Over 30,000 passes the chaos above the island handed back every one: each of the nine births from 7.570 to 7.650 was held again for whole stretches after it first moved a full turn away, and the two that escaped left 478 and 504 passes after a held stretch. At the pair's eccentricity 0.10, the outermost held births at both rims kept their hold through all 6,000 passes.

For Section 6: over 6,000 passes the islands at 0.15 and 0.20 last

The island at the pair's eccentricity 0.25 dissolves over 6,000 passes (Section 6); this part shows that at 0.15 and 0.20 it does not. Both islands last, giving up only about 0.02 in birth at their rims, though at 0.20 slow chaos reaches into the lower half.

Both islands were run the same way, every birth 0.010 apart from 0.030 below to 0.030 above the island read over 600 passes, each with its twin, for 6,000 passes, after we registered the guess that the inner births, those at least 0.030 inside both edges, would hold: all of them at 0.15 with twins drifting apart only in a straight line, and at least half of them so at 0.20. Both guesses held.

At 0.15 every inner birth, 23 of 23, held through all 6,000 passes and stayed locked in every stretch of 600, and its twins' gap grew as the pass count to a power between 0.6 and 1.4, a regular drift; no inner pair of twins ever came a thousandth of a turn apart. The island as a whole lasts from 7.185 to 7.445, against 7.165 to 7.445 over 600 passes: its lower rim gives up 0.02 in birth (7.165 escaped at pass 3,917, from inside the island read over 600 passes, and 7.175 lost its hold at pass 5,927), while its upper rim stands.

At 0.20 all 21 inner births held for 6,000 passes and stayed locked, 15 of them with a straight-line drift. The other six, scattered from 7.300 to 7.430, held too, but their twins parted faster than a straight line, as the pass count to powers of 2.1 to 5.8, and every inner birth from 7.300 to 7.370, five of them among the fifteen read as a straight-line drift, had its twins come a thousandth of a turn apart at some point between passes 518 and 5,456, which twins on a regular orbit, started 10^{-10} apart, never do in that time. That is chaos slow enough to keep its stars for 6,000 passes. Both rims of the island read over 600 passes give up about 0.02 here: two stars escaped from just inside its lower edge, at 7.270 and 7.280, after 2,881 and 2,036 passes, and at its upper edge 7.520 and 7.530 lost their hold. Every star that escaped in these runs left from below the island or from inside its lower rim.

So the long runs make a ladder in the pair's eccentricity: at 0.10 the rims hold, at 0.15 the island lasts with its lower rim slightly eroded, at 0.20 it lasts with both rims eroded and slow chaos reaching into its lower half, and at 0.25 it dissolves (Section 6).

For Section 6: on a grid five times finer, the third thread is a band

Section 6's rooms and threads were first seen on a grid of births 0.005 apart. Here the island at 0.25 is run five times finer: the third thread is a band of fast chaos about 0.01 wide, the leaking stops sharply at 7.472, and the room that keeps its stars is real but leakier than it looked.

We then ran every birth from 7.445 to 7.540, 0.001 apart, with its twin, for 6,000 passes (192 runs; the 23 that repeat runs of the 600-pass scan matched them within 2×10^{-6} turn at pass 600). The third thread is a band of fast chaos: from 7.446 to 7.456, and again at 7.459 and 7.467, twins come a thousandth of a turn apart within the first 600 passes. The leaking stops a little above the band, and sharply: 26 of the 54 runs born from 7.445 to 7.471 escaped, the highest at 7.466, 7.469 (a twin) and 7.471, in the lowest births of the room that keeps its stars, and none of the 138 born from 7.472 to 7.540.

The room that keeps its stars is real but leakier than the coarser grid showed. From 7.465 to 7.505, 17 of 41 births and 13 of their 41 twins held for all 6,000 passes, about two in five rather than six in nine; the coarser grid's nine births are among the 41 and held exactly as before, so the difference lies in the births between them. From 7.506 to 7.540 only 2 of 35 held.

At this spacing the fates speckle: neighbours 0.001 apart keep and lose their hold in no order a grid of 0.005 could show, and each run's swing speckles with its fate, narrow where the run held and wide where it did not. The straight-sided narrowing we read on the coarser grid, and the flat floor beneath it, were outlines of that speckle: we registered both before the finer run, and on it neither fits better than a smooth curve. What survives is the offset. Over the first 600 passes the narrowest swings sit at 7.524 to 7.531, where a regular centre would be; over 6,000 passes, 22 of the 24 births that hold lie at 7.505 or below, at least 0.02 under 7.525.

For Section 8: the slide's ratio across the atlas

Section 8's slide is a rocking that keeps step with the beat at a fixed ratio. This part asks how far across the atlas that ratio holds: over 2,400 passes it holds within 1% across the middle of the island at the star's eccentricities 0.20 to 0.24, and near 0.28, where the beat passes through zero, it cannot hold.

Read across the atlas at the star's eccentricities 0.20 to 0.34, over only 600 passes, the middle's ratio climbs on, 4.78 at 0.20, 8.00 at 0.22, 14.8 at 0.24 and 44.5 at 0.30, and falls back to 27.7 at 0.34. On a logarithmic scale the climb from 0.13 to 0.24 is a straight line, the ratio growing tenfold for every 0.096 of the star's eccentricity, the long runs at 0.13 to 0.17 and the atlas at 0.20 to 0.24 on the same line; at 0.26 the middle already reads 32 to 39, above the line's 22, as the beat heads for zero, and the two points beyond belong to the backward side. Most of that climb is the beat dying rather than the rocking quickening: from 0.20 to 0.30 the middle's rocking triples (0.087 to 0.258 cycles per pass) while its beat's size falls to a third (0.0186 to 0.0058), and near 0.28 the beat passes through zero as the hold stops the star's ellipse and then turns it backward (Section 9).

We had registered that at six of the eight eccentricities the ratio would hold within 1% across the middle while the rates moved at least twice as much. It held at three, 0.22, 0.24 and 0.34, where the rates moved 3.7 to 4.3% from end to end and the ratio 1.3%, every birth within 1% of the ratio's median. At 0.20 and 0.30 the rates barely moved across the four births whose ratio held; at 0.26, 0.28 and 0.32 no four births held it, and at 0.28 and 0.32 the rocking instead locks to the pass, at two-ninths and a third of a cycle (Section 8).

Over 600 passes the reader's resolution is close to the 1% the test asks, so we ran the middles of these slices for 2,400 passes, registering first that at 0.20, 0.22 and 0.24 the ratio would hold within 1% while the rates spread at least twice as much. It did: the ratio holds within 1% across seven or eight births (medians 4.84, 7.98 and 14.76) while the rates spread 2.7 to 3.2 times as much; at 0.22 it rests near 8 without locking to it, only 2 of its 7 births within 0.2% of 8. The longer runs also find plateaus at 0.26 (37.8), 0.30 (44.4) and 0.34 (27.6), and none at 0.28, where the beat passes through zero, or at 0.32, where the rocking locks to a third.

For Section 8: the locks over longer records

Section 8's locks were first read over 1,000 passes. This part follows them over 2,400, 6,000 and 12,000 passes, at both tolerances. Each long lock splits into a core that closes on its fraction and a shell that wears away; the core at a fifth holds within a few billionths, while the seventh's core is slow chaos; and at the tight tolerance the eleventh is one unbroken lock.

Run again for 2,400 passes, the real locks other than the third do not all hold. The middles of the two longest, twelve births at a fifth and ten at a seventh, come closer to their fractions with the longer record, to within 1.3×10^{-7} and 8.1×10^{-7} where over 1,000 passes they sat within 3.4×10^{-6} and 3.0×10^{-6}, which a slow slide of the size seen over 1,000 passes could not do. Their end births leave the fraction by 2×10^{-5} to 5×10^{-3} cycles per pass, though the last birth at a fifth comes back to it. The locks at two ninths and at an eleventh keep no two neighbouring births on their fractions. So the test we registered, one straight line through all the births of each lock, reads not borne out, because the ends of a lock seen over 1,000 passes fall away over 2,400 while its middle tightens.

Run for 6,000 passes, the middles do not stay whole either. Our second test, that each middle's largest distance from its fraction would shrink to under 0.4 of its 2,400-pass value, reads not borne out on all three. Each middle splits instead into a core and a shell. At a fifth, seven births, 7.455 to 7.461, sit within 3.5×10^{-9} of the fraction, ten times closer than the same births over 2,400 passes. Around them the rate slides off by 9×10^{-9} to 6×10^{-5}, and one birth inside the middle, 7.452, escapes at pass 3,077. At a seventh, five births, 7.120 to 7.124, sit within 2×10^{-8}, twenty times closer. Four of the other five lose their lock, φ_N's swing growing past half a turn, and drift off by 2×10^{-5} to 7×10^{-4}, and the birth just below the middle escapes. The shorter lock at a seventh, three births on another start line, loses its lock altogether. Meanwhile the four births at an eleventh, which over 2,400 passes kept no two neighbours on the fraction, all sit within 2×10^{-7} of it over 6,000, locked; two ninths still holds none. So a lock's middle is not exact as a whole: its core closes in while its edges wear away, and a longer record shows a lock that a shorter one could not.

Both readings carry the arithmetic's own drift. At the kit's tolerance every unlocked rate falls steadily as a run goes on (Appendix A.1), and on the lines where we measured it the fall moves a rate by about 5×10^{-5} between records of 2,400 and 6,000 passes, the size of the shells' largest slides; a sweep like that can also carry births into a lock, as the eleventh's appeared. The cores, held on their fractions, are not moved by the fall itself. Frequency analysis (Laskar, Froeschlé and Celletti 1992) reads chaos in a frequency that drifts from one stretch of a record to the next; our records of different lengths apply a form of that idea, and on them the cores close on their fractions while the shells move off.

Run for 12,000 passes, the cores do not keep closing. At a fifth the seven core births stay within 4.1×10^{-9} of the fraction, as over 6,000 passes (3.5×10^{-9}): the core holds its size and closes no further. At a seventh one of the five core births, 7.121, loses its lock between 6,000 and 12,000 passes, its swing growing past half a turn, and drifts to 1.7×10^{-7} off; the other four stay within 3.9×10^{-8}, and 7.131, outside the core, comes onto the fraction within 3.8×10^{-9}. The shells keep wearing: 7.451, held and locked over 6,000 passes, escapes at pass 10,538, and two of the shorter lock's three births escape too. At the eleventh, three of the four births close in further, to within 3.2×10^{-8}, while the fourth, 7.127, drifts to 2.4×10^{-7}, just past the 2×10^{-7} we registered for a lock that holds. Beside 7.452, which escaped, 7.4525 sits on the fifth within 9.2×10^{-8}, held and locked, while 7.4515 leaves just after its 12,000 passes. So our registered guesses, that both cores keep closing and stay whole and that the eleventh holds, read not borne out on all three.

These runs use the kit's tolerance, where the arithmetic's slow drift can carry a rate out of a lock. Run again with every tolerance a hundred times tighter over 6,000 passes, the shells' slides are the arithmetic's: their median slide between 2,400 and 6,000 passes falls to 1.18×10^{-7} from 7.96×10^{-7}, and the largest, 7.462's 5.8×10^{-5}, becomes 1.6×10^{-7}. Three of the eight births outside the cores that lose their lock or escape at the kit's tolerance keep their lock at the tight one, and none is lost at the tight tolerance alone. The losses are not all the arithmetic's, though. At the seventh, 7.121 has lost its lock by 6,000 passes at the tight tolerance, sooner than at the kit's, and over 12,000 it escapes at pass 8,007, while two more core births lose their lock and the other two stay locked within 6×10^{-9}. Two machines running 7.121 at the tight tolerance agree within 10^{-9} through pass 2,400 and part after pass 5,400: the core's middle is slowly chaotic.

The paper's own test bears that out. Started 10^{-10} apart at the tight tolerance, 7.121's twins part a thousandth of a turn at pass 5,515, and so do the twins of the two other core births that lose their lock, at passes 6,937 and 9,047, while the twins of the two that stay locked never do in 12,000 passes. Followed on to 30,000 passes, since 7.120's twins were still drawing apart, they part a thousandth of a turn at pass 14,795, and its main run then escapes at pass 29,323 while its twin stays held and locked (runs 20261019_053814 and 20261019_043245, read in run 20261019_062042; registered, with the guess that it is slowly chaotic). So the seventh's core is slow chaos through and through: four of its five births part, the slowest after 14,795 passes, and only 7.124, followed for 12,000 passes, has not been seen to part. We had registered both: that 7.121 is slowly chaotic, and that the chaotic pairs would be exactly the births that lose their lock. The losses at the seventh's core are chaos.

At the fifth the seven core births stay within 4.8×10^{-9} at the tight tolerance, as at the kit's. At the eleventh, over 6,000 passes at the tight tolerance, 7.127 lies 4.05×10^{-7} off while the other three are within 1.03×10^{-7}; over 12,000 all four sit within 2.3×10^{-8}, where at the kit's tolerance 7.127 had drifted to 2.4×10^{-7}. We had registered that last reading against a check across machines which 7.121's chaos failed, so it stands as reported, not as read. So we ran the births between and beside them at the same tolerance for 12,000 passes, after registering that the three between would lock. They do, all within 1.2×10^{-8}, and so does 7.135 beside them, within 7×10^{-9}, while 7.125 rocks a steady 1.5×10^{-4} off at every length of record; a second run of 7.129 gives back the first exactly. At the tight tolerance over 12,000 passes the eleventh is one unbroken lock from 7.127 to 7.135.

For Section 8: two cautions on the count of odd and even chains off the mirror

Section 8's count of odd and even chains off the mirror comes with two cautions: one about how thin the even count is, and one about a fault in the tool that read it, since mended.

The first is how thin the count is. The even crossings sit exactly on the floor of 40 we had set for a reading, and 37 of the 51 odd runs lie on two chains, the fifth (23 runs on 8 crossings) and the seventh (14 on 11). The second is a fault since mended: the count printed when the run ended read only the eight start lines at the star's eccentricities 0.20 and 0.22. The tool took a run whose star had been thrown wide, neither escaped nor held and out of time before its 600th pass, for a run cut short by the time budget, and left out every line that had one. The numbers here are the registered test read with that one fault mended, after the run's record had been read; nothing else in the rule changed.

For Section 9: the fin at the island's upper edge

Section 9's drag curve comes to a point at the island's upper edge, like a fin. This part asks how much of that shape is real: the tip sits at the hold's edge on all nine lines read, but the rest of its outline mostly does not repeat, and on a finer grid the point is a corner.

The fastest turn on that side, 0.0166 per pass, belongs to the last held birth (7.355); the next birth outward, no longer held, turns almost as fast (0.0163), and from there the turn falls back to the free rate within about 0.04 in birth, more gently than it climbed. Read the same way on eight more lines of the atlas (the other six start lines at 0.20, and the lines started at 0 at 0.22 and 0.24), the fin's tip sits at the hold's edge on all nine, on the last held birth or the next one out, and the turn is back within 3% of free 0.025 to 0.045 in birth beyond it on all nine. It falls more gently than it climbed on only four of the nine, so that part belongs to this line, and beyond the fin it dips below free and then rises above it on six.

Nor is the point the logarithmic cusp a pendulum's lingering would draw on both sides: over the four nearest births on each side, a logarithm in the distance from the tip fits better than a straight line on both sides on only one line. Taken side by side after that test, the atlas's spacing made the fall outside look logarithmic on eight of the nine lines and the climb inside straight on eight. Read again on Figure 13's line at a fifth of that spacing, 0.001 in birth, both sides are closer to straight, where we had registered that the split would hold: the point is a corner. The finer grid also shows the edge ragged over 0.004 in birth (7.358 is held between unheld neighbours) and the fastest turn, 0.0172 per pass, at 7.356, one step past the held block's last birth (7.355).

In the grip's terms (Section 9), these are the stars whose passes linger longest near the pendulum's balance point, half a turn from the hold, where each twist pushes the ellipse on; they sit on both sides of the edge, held and not. On the lower side the line begins at 7.050 still climbing, and the chaotic layer there blurs the edge.

For Section 9: what carries the two branches

Section 9's drag runs on two branches, one for each side of the island's centre. This part asks what carries them: not the star's eccentricity alone; where the island is widest, the rocking rate; and they are not a memory of the start.

The second dial is not the star's eccentricity alone. At equal grip the runs born below the centre do carry the higher mean eccentricity, on all 28 lines, and with each run's mean eccentricity and its average of $\cos(4\pi\ \varphi_N)$, the swing's next harmonic, added to the line, the line becomes nearly exact (R^2 0.996 on the start-0 line at 0.26); yet 808 of the 957 locked runs born below the centre still fall under it, and 581 of the 668 born above it over it.

Where the island is widest, what carries the branches is how fast each star rocks, the ring it lives on inside the island: on the runs whose rocking rate can be read, adding that rate to the line brings the branches to even at 0.20 (99 of the 192 born below the centre under the line, 63 of the 119 above it over), against 141 and 95 on the same runs without it; at 0.26 it changes little, and about seven in ten stay on their side. The branches are not a memory of the start: run for 2,400 passes on the starting angle 0, 57 of the 60 locked runs born below the centre in the four slices' middles fall under their line and 58 of the 60 born above it over theirs.

For Section 9: how far the law carries

Section 9's drag law was fitted near the mirror line, where the runs start. This part carries it to runs it has not seen: it holds on longer runs and near the mirror, fails farther off it, and the band of φ_N's swing carries most of what the start sets.

The line carries over to runs it has never seen. Fitted on the atlas's 600-pass runs with the grip, the swing's next harmonic and the rocking rate (the long runs do not keep the eccentricity), and applied unchanged to the 2,400-pass runs, it explains 96.1% to 97.8% of their spread in apse turn at each of the four slices, missing by 0.00014 to 0.00022 of a turn per pass where the grip line alone misses by 0.00064 to 0.00074. Carried to the atlas's six other start lines, the same line needs the eccentricity: unchanged, it explains 95.9% to 97.2% of their spread at each slice, falling from 98.5% to 99.2% on the start line at 0.01 to 87.2% to 91.3% on the line at 0.05, its misses one-sided; fitted again on the start-0 line with each run's mean eccentricity added, and carried over unchanged, it explains 97.4% to 98.6%.

Farther off the mirror it does not carry. On the off-mirror grid's four start lines, 0.06 to 0.12 of a turn (Section 8), the line without the eccentricity explains only 6.6% to 35.0% of the spread at the four slices, though it still misses by less than the grip line alone, and the line with it explains 44.3% to 71.3%. We had registered 90% for both, expecting the first to miss and the second to hold. Read start line by start line, the line with the eccentricity explains 99.4% to 99.7% on the line it was fitted on, 95.6% to 98.1% at 0.04, 91.4% to 95.7% at 0.05, 85.8% to 91.4% at 0.06, 60.0% to 78.3% at 0.08 and 29.9% to 58.4% at 0.10; at 0.12 it explains 12.2% to 33.7% at three slices, and at 0.26 less than the mean does. The fall is smooth and begins near the mirror; reported pooled over the atlas's six other lines, four of them within 0.03 of the mirror, it was hidden.

At 0.24 and 0.26 the line overstates the turn on all but a few of each start line's runs, and by more the farther the start lies off the mirror: from 0.0007 to 0.0015 of a turn per pass at 0.24, and from 0.0009 to 0.0023 at 0.26. The miss is not a reach beyond what was fitted: 745 of the 876 off-mirror runs have a grip inside the range of the line's own runs and miss all the same. At 0.20 to 0.24 it follows each run's band (correlations 0.67 to 0.88) far more closely than its start (−0.41 to 0.19); at 0.26 the two are alike (0.51 and −0.58). The drag law, as fitted, is a law of the mirror's neighbourhood. Within about 0.05 turn of it, it holds; beyond that, something the start sets takes over, something the grip, the swing's next harmonic, the rocking rate and the eccentricity do not carry.

The band carries it. Fitted on all eleven start lines at once with the band as a sixth term, one law explains 98.3% to 99.3% of the spread at the four slices and at least 96.1% on every start line, where we had registered 90% and 80%; with the start angle as the sixth term instead, 97.1% to 98.0%, and at least 82.8% on every line. That is a fit to all eleven lines at once. Fitted on the atlas's seven lines near the mirror and carried unchanged to the four beyond, the law with the band explains 96.2% and 93.3% of their spread at the star's 0.20 and 0.22, but 80.3% and 70.5% at 0.24 and 0.26, where we had registered 90% at every slice: the band carries most of what the start sets, and at the higher eccentricities not all of it. The same pooled fit without either leaves the farthest line at 53.3% at 0.26.

The law's terms are moments of one swing and are not separately identified: with the swing's third harmonic added, the share explained on Figure 13's slice stays at 99.70% while the grip's coefficient moves by half, and by many times at 0.24 to 0.28. So the law is read for what it predicts, not for its coefficients; the drag's size in Section 9 is read from the straight line in the grip alone.

For Section 9: the beat's zero pinned, and the slow line beside it

Section 9 finds that near the star's eccentricity 0.28 the centre's ellipse stands still, so the beat passes through zero. This part pins where, and follows a slow line that stays when the beat is gone. The zero sits at 0.2811; the ellipse there stays still over 24,000 passes at the tight tolerance; and beside the zero a slow line holds near 0.001 cycles per pass, the orbit's own, set by something theory must still name.

The zero has since been pinned. With centres at the star's eccentricities 0.27, 0.275, 0.285 and 0.29, run for 2,400 passes, the beat crosses zero once, at 0.2811, where we had registered 0.2817 within 0.003. At 0.30, where the centre's ellipse turns backward, the turn changes sign again high in the island: over 2,400 passes at births 7.415 to 7.450 it climbs through zero once, at 7.4284 (registered: 7.4276 within 0.005). So at 0.30 the island's upper flank still turns its ellipses forward while its core turns them back.

Watched for 6,000 passes at the centre of 0.28, the slowest strong line is the beat wherever there is a beat to hear, within one step of the spectrum (1/6,000 cycles per pass) at births 7.330 and 7.335. At 7.340, where the ellipse is all but still (a beat of 0.000035 cycles per pass), a slow line remains, at 0.000997. It lies within 0.000042 of the rocking's distance from a fifth (0.201039 against 0.2), as the strongest line outside the beat's window at 7.335 lies near its own (0.001374 against 0.001412). Near the doubling, the swell of an alternation is the rocking's distance from a half (Appendix B), and this might have been the same thing at a fifth.

It was seen after the run, on two births, so we registered it as a guess and ran eight more births 0.005 apart, from 7.310 to 7.360, for 6,000 passes each (run 20261018_203922). It does not hold. Of the eleven births read, only 7.335 and 7.340 have a slow line near the rocking's distance from a fifth, or half of it. At the eight new births the line lies at 0.000771 to 0.001356 cycles per pass, while that distance runs from 0.001320 to 0.008050; the nearest miss is at 7.345, whose line lies 0.000111 from half its distance, just outside the 10^{-4} we allowed. The first reading's 'within one step' and this test's 'not within 10^{-4}' are both true: the line sits less than a step from half that distance, but not as close as we asked. Seen after this run, the slow line follows the beat instead: it falls toward 7.340 from both sides as the beat does and rises away with it, at 1.16 to 1.32 times the beat at every birth but 7.340, where the beat all but vanishes and the line stays.

Registered in turn and run on six more births (run 20261018_220340), that reading holds in part, and half of the test could not have failed. Beside the beat's zero, at 7.3375 and 7.3425, the strongest line the reader can see sits at 0.000972 and 0.000937 cycles per pass, 3.61 and 2.76 times the beat; but the reader looks only above 3/6,000 cycles per pass, and the band we had asked about there lies below that floor, so that half reads no reading. The rocking's distance from a fifth there is 0.000320 and 0.000481, so the match at 7.340 was a coincidence of two small numbers. Farther out, the band we registered, 1.10 to 1.40 times the beat, holds at 7.300 and 7.305 (1.325 and 1.313) and fails

at 7.365 and 7.370 (1.085 and 1.039), where the line closes on the beat as the beat grows. So across the middle of the island at 0.28 the slow line rides just above the beat, at 1.039 to 1.325 times it, and at 7.3375, 7.340 and 7.3425, beside the beat's zero, the strongest line above the reader's floor holds near 0.001 cycles per pass.

Run for 24,000 passes, a record four times as long, at the kit's tolerance (run 20261019_012918), that test reads no reading: its check compared 7.340 with the laptop's run of record of that birth, and the two machines part by 2.2×10^{-5} by pass 6,000, past the 2×10^{-6} we had set. Over records that long at the kit's tolerance the rocking rate also falls steadily from one quarter of the record to the next while φ_N's band stays the same, the arithmetic's signature (Section 3), and the still ellipse seems to turn, so the long record has to be read at the tight tolerance. Run again at the tight tolerance for 24,000 passes (run 20261019_032214), four of these births hold their rocking rates flat from one quarter of the record to the next, changing by at most 1.04×10^{-9} cycles per pass in each pass, where at the kit's tolerance they fell 8.02×10^{-8} to 2.77×10^{-7}: that fall was the arithmetic's.

Over the long, clean record the centre's ellipse at 7.340 stays still, a beat of 0.000026, as we had registered, so the stillness near the star's 0.28 is not a feature of the shorter records. Away from the zero the slow line stays off the beat, at 1.307 and 1.214 times it at 7.320 and 7.355, also as registered: the offset is not the shorter record's resolution. Beside the zero it stays near 0.001 cycles per pass, at 0.001075 and 0.001108, but 0.000103 and 0.000111 from its 6,000-pass values, just outside the 10^{-4} we had registered; those values were read at the kit's tolerance, whose arithmetic moves the rocking rate there by more than 10^{-3} (0.200320 against 0.201703 over the same first 6,000 passes). Read again on the third birth beside the zero, 7.3425, in one clean run at two lengths (run 20261019_043449), the line is 0.001050 over the first 6,000 passes and 0.001048 over all 24,000, as we had registered: it does not move with the record once the arithmetic is clean, so it is the orbit's. Across the middle of the island at 0.28, then, the slowest strong line rides a little above the beat, and beside the beat's zero, where the beat itself all but vanishes, it holds near 0.001 cycles per pass. What sets it there, where neither the beat nor the rocking's distance from a fifth does, is a question for theory.

For Section 9: where the drag's level misses

Section 9's drag line, carried to zero grip, lands half again above the turning of free stars outside the island. This part shows that miss, and how much of it is explained.

The miss is in the drag's level. Extended to zero grip, the drag line gives an apse turn half again above the free rate measured on stars outside the island. So the drag's shape is the grip's, but its level near the rim is not yet explained (Appendix B). Read about each run's own wandering centre, the grip spreads a little further (to −0.116) and the line tightens (R^2 0.968). Its zero-grip value still stays half again above the free rate. This reading was itself registered, as the redesign of an earlier miss: we had registered that, read this way, the zero-grip value would come within 15% of the free rate, and guessed that it would not; it did not.

Part of the answer is the free rate itself. Stars not held within 0.030 below the island's lower rim, free or in the chaotic layer, turn on average at 0.019 to 0.021 per pass at each of the pair's eccentricities 0.05 to 0.20, against 0.013 to 0.015 within 0.030 above its upper rim. At the pair's eccentricities 0.05 and 0.10 the zero-grip level falls between the two; at 0.15 and 0.20 it lies above both. Against an ideal pendulum of the same swing, the island's grip runs high, by 0.04 on locked runs and 0.12 on the widest swings. Over the widest swings that excess grows with the pair's eccentricity, from 0.11 at 0.05 to 0.25 at 0.20. The star lingers less at the ends of its swing than a pendulum would: the same "quicker than a pendulum" that the rocking shows inside the island, where it outruns the held width (Appendix B).

For Section 11: the edges

Section 11 says the heavy star's islands sit just below the whole numbers. This part maps them to their edges: read on the pair's mean ellipse every birth of island 7 sits below seven, each island is chaotic on its low-birth side only, and the published stability rules bracket it.

A finer scan maps both islands to their edges: births 0.005 apart, 600 passes, each birth with a twin. It first reproduces the scan of Section 11, 12 of 12 births within 4.8×10^{-7} over their first 200 passes. Read at the star's closest approach, island 7 holds from birth 6.805 to 7.125, with its centre at 7.010 (band 0.056 turn, the pair turning 6.9357 times per orbit), and island 8 from 7.900 to 8.080, with its centre at 8.015 (band 0.027 turn, the pair turning 7.9252 times per orbit; Figure 15). Island 7's high-birth end, from 7.075 to 7.125, then holds above seven, the pair turning up to 7.1506 times per orbit. Read on the pair's mean ellipse (Section 11), island 7 ends between 7.055 and 7.060. From 7.060 up, the angle built on the mean ellipse drifts by 31.29 to 78.23 turns over the run, because there the apse the reading sees is the bend's, not the pair's. So every birth of the real island 7 sits below seven; the held runs above seven were the bend. Island 8 holds on the mean ellipse at both its edges (7.900 and 8.080).

Each island is chaotic on its low-birth side only: 0.035 in birth below island 7 (6.775 to 6.805), and one birth below island 8 (7.895). Both high rims are clean. So is everything more than 0.02 beyond either edge, except the two lowest births of island 7's layer (6.775 and 6.780). With the light star island 8 is ragged above (Section 6); with the heavy star

it is ragged below. Before the run we had guessed the light star's sides, and rims at least 0.010 wide at island 8; both guesses missed at island 8. None of the 486 runs escaped or came within 1.5 of the pair's centre.

The two stability rules of Section 6 bracket this island too. Mardling and Aarseth's (2001) calls every birth below 8.078 unstable. Vynatheya and colleagues' (2022) calls the whole grid stable, at 1.036 to 1.202 of its critical value, and this star's mass ratio (0.4 of the pair) lies inside that rule's training range. The second dial is here too. At equal band the low-birth flank rocks faster at all eight band levels at island 8, and at three of four at island 7.

For Section 11: where the star and the pair trade

With the heavy star, the star and the pair trade eccentricity. This part shows where: least at the islands' centres, most on island 7's low-birth flank, its chaotic side, and always through the same harmonic that holds the star.

The pair's mean eccentricity shows where the star and the pair truly trade. At the islands' centres it moves by 0.0144 and 0.0072, about as little as far outside (0.0072 to 0.0188). On island 7's low-birth flank it moves most, from 0.046 to 0.118 at 6.810, and the star's own eccentricity swings with it (0.1787 to 0.2255 at its farthest point). Near island 7's high edge (7.050) it moves 0.0422; near island 8's two edges, 0.0329 (7.920) and 0.0325 (8.075). At island 7 the chaotic side is the side where that trade is largest. At island 8 the two edges trade alike in these runs, so here the trade does not decide its side (started on the pair's mean ellipse, the lower edge trades a little more; Appendix B).

The trade runs through the same harmonic that holds the star. Across island 7 the mean eccentricity's change at each pass follows sin 2πφ_N at that pass, with a median correlation of 0.957 over 14 runs. At island 8 the median is 0.728 over 8 runs, because the runs at its centre barely swing (0.009 to 0.011) and their kick sinks below everything else. The sign is the same in all 22 runs. We had registered that this correlation would reach 0.8, first on the reading at closest approach (it read 0.702 and 0.360) and then on the mean eccentricity; island 8's centre is where the second missed. Island 8's kick is 0.216 of island 7's (0.235 on island 8's four clean runs), where the harmonic's two strengths give 0.268 to 0.297 (for the star's eccentricity 0.18 to 0.20): the same order, and short of it, noticed after the run (Appendix B). With a light star the hold drags on the star's ellipse (Section 9); with a heavy one the same hand also pumps the pair's.

For Section 11: the rim's eccentricity

One last view of the trade, from the starting line.

Across island 7's lower rim in Section 11 (6.925 to 6.980), the pair's mean eccentricity over the first 600 passes falls at every step inward, from 0.1158 at 6.925 to 0.1107 at 6.980, against the 0.10 it was started at and 0.097 over the whole run at the island's centre (7.100). The extra eccentricity on the low-birth flank is set by where the star starts, before anything leaves: it is the trade of the previous paragraphs, seen from the starting line.

Appendix D. Lessons learned the hard way

What fooled us is part of the result. Each lesson below is told in full where it happened.

A reader has a floor. The first rocking reader was set to skip the beat and could not see below 0.05 cycles per pass. Where the star's orbit is nearly round, at its eccentricities 0.12 to 0.16, the rocking lies below that, and the rates first reported there were read again with a second reader that sees below the beat (Section 8; Appendix A.3).

The arithmetic can fake physics. At relative tolerance 10^{-9} every unlocked rocking rate falls steadily as a run goes on, and over records of different lengths that fall made the shells around the locks seem to slide. A tolerance a hundred times tighter removed the slide and shrank the fall about a hundredfold, and the long frequency records here are read at that tolerance (Section 3; Appendix A.1).

A short run can call slow chaos an island. At the pair's eccentricity 0.25, what 600 passes read as an island did not last 6,000, and twins started 10^{-10} apart see chaos within 600 passes only when it is fast enough to part them (Sections 3 and 6).

A fidelity check tests reproducibility, not accuracy. It asks a tool to repeat a run on the same kind of arithmetic. On a slowly chaotic birth two machines part, and a check across machines then fails for a reason that is itself a finding (Appendix A.5; Appendix C).

A test can be unable to fail. Beside the zero of the beat at the star's eccentricity 0.28, half of one registered test asked about a band that lies below the reader's floor, so that half reads no reading (Appendix C).

A tool can mis-sort its own runs. The first count of odd and even chains off the mirror took a star thrown wide for a run cut short, and left out every line that had one; the count was read again with that one fault mended (Appendix C).

A planted check can keep a stale bar. The circular-start test's tool kept the bar from before the trial run, so the same runs were read again by the mended tool (Section 6).

Two small numbers can agree by chance. At two births beside the beat's zero a slow line matched the rocking's distance from a fifth; on the eight more births run to test it, it did not (Appendix C).

Many guesses miss. Of the 490 predictions written down before their runs, 211 did not come out as written and 43 could not be read. Every miss is kept, and 92 were redesigns of earlier misses (Section 3).

Appendix E. Approaches not pursued

Written as properties of the problem. Whole-number period ratios do not hold the star by themselves: at a whole number the third star is kicked or creeps. A flatness rule cannot tell a lock from a turning point of the rocking rate. The length of lines in a recurrence plot is not a clock reader here, because the fast inner spin cuts the lines short. A cascade without a beat cannot be had by making the pair circular: the [N:1](2) harmonic that holds these islands is proportional to the pair's eccentricity and vanishes with it (Mardling 2013, equations 15 and 41), and beside a star as light as ours a circular pair stays circular: started circular, its eccentricity stayed below 2×10^{-7} over 600 passes. So a circular pair has no island of this kind to double. (A heavier third body would stir it, as her caveat after equation 43 warns.) What remains near period ratio 7 is the [14:2](2) harmonic, which needs no eccentricity of the pair: the member of the chain of 2:j resonances that Batygin, Mardling and Nesvorný (2021) follow for a test particle beyond a circular perturber, where it is written 2:14, the perturber's count first (their Sections 2.1 and 2.3; the 1:j terms, they find, are smaller by a factor of the perturber's eccentricity, their Section 4); a beatless cascade would have to be sought in that different island.

Appendix F. Data and code

Every tool, run file and figure source, deposited with the paper; each figure names its run.

The figures as printed were redrawn without their working titles by triple_paper_plain (run 20261016_045736, and run 20261016_055053 for Figure 9's left half, beside which the author's marked working copy is set by triple_paper_figure7_marks, run 20261021_052614), which ran each figure's own tool again unchanged; every redrawn figure's ledger matches the ledger of its drawing of record line for line once run stamps are masked. Figure 6 is printed as drawn (run 20261019_164526). Figure 4 sets six of the author's marked working pictures side by side, each cropped to one panel and scaled, with no mark changed (triple_paper_figure_eye, run 20261025_033626), and Figure 5 sets his marked bend pictures above the zoom-out run, both scaled only (triple_paper_figure_zoom, run 20261025_180150).

Declaration

A preprint, with a limited literature pass

This is a preprint. It has had no peer review, and its literature pass is limited. Four of the works it cites were read in full: Mudryk and Wu (2006), Dullin and Worthington (2014, in its arXiv version), Mardling (2013), whose passages cited here were read a second time against each sentence that cites them, and Georgakarakos (2003, in its arXiv version). Seven were read through a summary of their full text, and the rest at abstract or index level, or through the reference lists of the papers we read; the table below says which. Every record was checked, field by field, against the metadata its publisher deposits with Crossref or against the publisher's own page; Franklin and Soper (2012), which exists only on arXiv, was checked there. Finding that others came before is the expected result of such a search. A work not cited here was not found, which is not the same as not there.

How each cited work was read

work	how it was read
Mudryk and Wu 2006	in full
Dullin and Worthington 2014	in full
Mardling 2013	in full; its cited passages again
Batygin and colleagues 2021	a summary of its full text
Vynatheya and colleagues 2022	a summary of its full text
Shevchenko 2015	a summary of its full text
Popova and Shevchenko 2013	a summary of its full text
Gallardo and colleagues 2021	a summary of its full text; its abstract on the publisher's page
Holman and Wiegert 1999	its abstract, its fit for the outer region (equation 3) and its passages on that region's islands
Karney 1983	its abstract
Laskar and colleagues 1992	its record; the method as later papers describe it
Meiss 1992	as Meiss's later review describes it (arXiv 1501.04364)
Mardling 2007	its abstract
Georgakarakos 2003	in full (arXiv 1408.5890)
Georgakarakos 2002	its abstract and its short-period equations
Quarles and colleagues 2018	a summary of its full text; its passages on resonances read in its text
Quarles and Lissauer 2016	a summary of its full text; its passages on circumbinary orbits
Doolin and Blundell 2011	its abstract and its passages on stability, in its text
van der Weele and colleagues 1986	its abstract
Érdi and colleagues 2007	its abstract
Franklin and Soper 2012	its abstract
Antoniadou and Libert 2018	its abstract
Henrard and Lemaître 1983	named by Mardling; not read
Mardling and Aarseth 2001	its formula as quoted; not read
the six non-twist papers	as Dullin and Worthington cite them
Chirikov 1979	as Mudryk and Wu cite it; its pendulum model as Shevchenko cites it (arXiv 1312.7099)
the doubling constants	records checked; not read
the integrator, SciPy, Hénon	standard sources; not read
Schulz 2026a and 2026b, the field guide and the Crossover Atlas	the author's own

Provenance

Each finding names the sections that carry it, and each number in those sections comes from a run whose tool, run files and ledger are deposited with the paper (Appendix F); each figure's caption names its own runs. The table below maps the claims of each section to their runs. A script checks it (triple_provenance_table v1.13, run 20261029_054653): every run it names has its files in the deposit, and in every row that carries numbers, the numbers and phrases it checks for that row are printed by that row's own runs and stated in the paper as printed. A row may also give figures the paper rounds or leaves out, such as a ledger's unrounded value; those are not checked against the paper. A second script looked for every number in the paper with three or more significant digits in the runs' own ledgers, leaving out the table's own (triple_provenance_scan v1.7, run 20261029_054709): 1,438 of 1,446 are printed by a ledger. A number found there is a lead to read, not a proof, since a ledger can print the same digits for another quantity; the table's row-by-row checks are the proof for the numbers they check, among them the percentages, which it checks against the fractions their ledgers print, such as 0.9606 for 96.1%. Three such coincidences are known: the

0.000111 by which 7.345’s slow line misses half its distance from a fifth is 0.000771 less 0.000660, arithmetic, though a ledger prints the same digits for another quantity in Appendix C; the circular start’s 5,000 of the pair’s periods, arithmetic on its laps, is printed by a ledger of 28 October as a count of passes; and 6,000 passes counted as 40,000 of the pair’s periods is printed by the long run’s looks as the end of the span they search for near returns. Of the other eight, three are printed only by run logs: two rounded bounds (1,050, above the ledger’s 1,043, and 2,300, above the ledger’s 2,293) and the 4,500 at the low end of the circular start’s 4,500 to 5,000. Five are printed by no ledger: one is passes counted in the pair’s periods (30,000 passes as 200,000), two are percentages whose fractions only the table checks (91.3% and 97.4%, for 0.9126 and 0.9737), and two are rates less a fifth (0.000320 and 0.000481). One more of the 1,438 is not a result either: 4,200, which is 600 passes counted in the pair’s periods and which a ledger happens to print as a pass number. Where a number was first computed by a one-off script, it is computed again by a deposited one (triple_paper_numbers and triple_crossmachine_check). A third script set every sentence and paragraph of the paper that carries numbers against the runs whose ledgers print them and the runs the table names (triple_source_check v1.2, run 20261030_020754; against the table before these rows, run 20261019_214932). Its leads, read by hand against the ledgers, gave nine runs whose numbers the paper states but no row named: probes 26 and 27, five checks and looks on the heavy star, the look that read the grip about each run’s own centre, and the drawing of Figure 1. Each now has its row. It cannot trace a claim made only of small whole numbers, which nearly every ledger prints.

where in the paper: Front page (Figure 1)

what it rests on: the door map’s cross for the twin test agrees with the scans’ own twin rule at 144 of 146 births, on the two scans whose series were kept (69 of 70 and 75 of 76)

runs (probe or check, tool, run): the door map’s drawing, triple_paper_figure3b v1.0, run 20261014_165101

where in the paper: Section 3

what it rests on: the registered predictions counted by script from the register: 490, of which 236 came out as written, 211 did not and 43 could not be read; 92 redesigns of earlier misses, 35 of which came out as written; a cold check of 40 drawn at random, and of the 46 added from 26 to 28 October

runs (probe or check, tool, run): the count, triple_ledger_count v1.1, run 20261028_210006

where in the paper: Section 3

what it rests on: the author’s marked pictures set side by side as the eye figure, each cropped to one panel and scaled, with no mark changed

runs (probe or check, tool, run): the eye figure, triple_paper_figure_eye v1.0, run 20261025_033626

where in the paper: Section 3

what it rests on: zooming out: the bend a smooth surface of two angles over 600 passes at 7.200 and 7.010 (0.1126, 0.1309) and not at 6.810 (0.2225 against 0.2; T1); over 6,000 passes at 7.200, 0.0603 (Z1), and over 30,000, 0.0251 (LZ1); the 600-pass page’s dots linked 5 passes apart, 242 links against 66.3 shuffled (RW); the twin of 7.200 with it over 600 passes; the author’s marked bend pictures above the zoom-out run in the zoom-out figure

runs (probe or check, tool, run): T1, triple_heavy_torus_look, run 20261009_071003; the zoom-out, Z1, triple_heavy_zoomout_run v1.0, run 20261009_080853; RW, triple_rf_reads_check, run 20261010_182543; probe 35, triple_heavy_edge v1.0, run 20261008_131041; the zoom figure, triple_paper_figure_zoom v1.0, run 20261025_180150; LZ1, triple_heavy_zoomout_long v1.0, run 20261026_074708

where in the paper: Section 3

what it rests on: the zoom-out at the tight tolerance (relative 1e-11): its readings over 6,000 passes a rotation of two angles, the angle about the circle’s centre and psi, returning at 1, 3, 6, 9, 19, 28, 47 and 701 passes, at each rung within 10% of the rotation (LG3’); the slow line at 722 passes, the 3:2 combination of the two rates

runs (probe or check, tool, run): DR2’s run, triple_heavy_zoomout_tight v1.0, run 20261026_085300; the ladder look, triple_zoomout_ladder_look v1.5, run 20261026_091321

where in the paper: Section 3

what it rests on: the arithmetic with the heavy star: over 30,000 passes at the kit’s tolerance the pair’s mean eccentricity falls from 0.04899 to 0.04883 and the two angles’ rates drift steadily, the 3:2 beat shortening from 700 to 450 passes; a hundred times tighter the rates change 200 to 300 times less and the beat holds at 722 (DR2)

runs (probe or check, tool, run): the long run, triple_heavy_zoomout_long v1.0, run 20261026_074708; its blocks, triple_zoomout_ladder_look v1.5, run 20261026_091321; DR2, triple_heavy_zoomout_tight v1.0, run 20261026_085300

where in the paper: Section 3

what it rests on: the arithmetic’s alarm at a second birth: at 7.300 at the kit’s tolerance over 12,000 passes the beat (3 x the angle about the centre’s rate - 2 x psi’s) shortens from 603 to 281 passes in steps keeping pace with the pair’s mean eccentricity (ratios within 25% of their mean; AL1), 7.2 times faster than 7.200’s (AL2); at the tight tolerance it holds at 746 passes, changing by 0.0019 of the kit’s step (AL3); the steps’ spread 0.18% against the eccentricity’s 26%

runs (probe or check, tool, run): case B, triple_heavy_beat_runs v1.0, run 20261027_041758; case C, triple_heavy_beat_runs v1.0, run 20261027_043455; the look, triple_beat_alarm_look v1.3, run 20261027_150905

where in the paper: Section 3

what it rests on: the zoom-out at the tight tolerance over 30,000 passes: on its two-angle rotation out to the rotation's near returns 4,328 and 13,732 passes, at 0.94 and 1.57 times the rotation's closeness (LD1a); two rules building each long rung from earlier ones missed (LD1b, LD1b'), fitting 21 and 25 of 200 random pairs of rates respectively; the halves' likeness swinging every 736 passes against 7.300's tight beat of 746 (UN2')

runs (probe or check, tool, run): case A, triple_heavy_beat_runs v1.0, run 20261027_041756; case C, triple_heavy_beat_runs v1.0, run 20261027_043455; the look, triple_beat_alarm_look v1.3, run 20261027_150905

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where in the paper: Section 3

what it rests on: the surprises' second round: the tight run's rates still drift steadily, psi's by 2.8e-6 over 24,000 passes, and at lag 13,732 the pairs centred on the run's middle give 1.00 of the rotation's distance (TD1); the slow approach after 4,328 passes is ordinary, 79 of 200 random pairs of rates coming no closer (RW1); the eccentricity's steps' spread mostly the leftover of its fast swing every 3.17 passes, its swing at the beat 2.6e-6, 3.5% when smoothed (WB1, WB1')

runs (probe or check, tool, run): case A, triple_heavy_beat_runs v1.0, run 20261027_041756; case B, triple_heavy_beat_runs v1.0, run 20261027_041758; the look, triple_surprise_look v1.1, run 20261028_001911

---

where in the paper: Section 3

what it rests on: the third round: the eccentricity's fast lines at 7.200 at the angle about the centre's period and at psi + 2 x that angle (4.704 and 9.470 passes; ML1), peaking as the hand points away from the pole at both births (CF1); the loosening at lag 13,732 within 15% of a steady creep's prediction in every 2,000-pass bin (TD2); at relative 1e-10 the creep 25.4 times the tight run's (TS1 of 28 October)

runs (probe or check, tool, run): case A, triple_heavy_beat_runs v1.0, run 20261027_041756; case B, triple_heavy_beat_runs v1.0, run 20261027_041758; case D, triple_heavy_beat_runs v1.1, run 20261028_044339; the look, triple_surprise_look v1.2, run 20261028_045131

---

where in the paper: Section 3

what it rests on: the fourth round: loosened threefold from the tight run, the creep 5.38 times the tight run's, a power of 1.53 (CR1); over the tight run's 30,000 passes a straight ramp, its bend 0.7 of its standard error (SE1 of 28 October, missed); ten times tighter, over 60,000 passes, 106.1 times less than the tight run's (LR1 of 28 October, missed); the creep moving the two rates in the twin's proportion, -0.490 against the creep's -0.544 (TW1 of 28 October)

runs (probe or check, tool, run): case G, triple_heavy_beat_runs v1.2, run 20261028_061710; case A, triple_heavy_beat_runs v1.0, run 20261027_041756; case E, triple_heavy_beat_runs v1.3, run 20261028_102627; case F, triple_heavy_beat_runs v1.3, run 20261028_102629; the look, triple_surprise_look v1.3, run 20261028_062708; the look, triple_surprise_look v1.3, run 20261028_104854; the look, triple_surprise_look v1.4, run 20261028_105236

---

where in the paper: Section 3

what it rests on: LR1's redesign (LR1' of 28 October): three times tighter than the tight run, relative 3e-12 over 30,000 passes in 6 segments, the creep 6.7 times less than the tight run's, a power of 1.58 (bar 1.02e-11 to 1.87e-11 for psi's rate's slope; +1.7518e-11)

runs (probe or check, tool, run): case H, triple_heavy_beat_runs v1.4, run 20261028_113830; the look, triple_surprise_look v1.4, run 20261028_114806

---

where in the paper: Section 3

what it rests on: the energy's drift (EN1 and EN1' of 28 October, both missed): over 6,000 passes at relative 1e-10, 1e-11 and 1e-12 its powers 1.46 and 1.92 over the two tenfold steps, where psi's creep has 1.41 and about 2.0; over case E's 60,000 passes, from its segments' saved states, the drift +2.9125e-14 a pass, rising where the looser tolerances' fall; the 7.300 twins (PS1): the angle's slope over psi's -0.492, psi plus twice the angle 0.016 of psi's change

runs (probe or check, tool, run): the energy runs, triple_energy_drift v1.0, run 20261028_140844; case E's energy, triple_energy_drift v1.1, run 20261028_143037; case I, triple_heavy_beat_runs v1.5, run 20261028_120829; case J, triple_heavy_beat_runs v1.5, run 20261028_120830; the look, triple_surprise_look v1.5, run 20261028_142911; the look, triple_surprise_look v1.6, run 20261028_143058

---

where in the paper: Section 3

what it rests on: the energy and the angular momentum over case H's 30,000 passes at relative 3e-12, read from its segments' saved states (FL1 and FL2 of 28 October): the energy's drift -2.6683e-13 a pass, the angular momentum's +7.0290e-14

runs (probe or check, tool, run): case H, triple_heavy_beat_runs v1.4, run 20261028_113830; the reading, triple_segment_invariants v1.0, run 20261028_170633

---

where in the paper: Section 3

what it rests on: the angular momentum's drift (EN2 of 28 October, missed): over 6,000 passes at relative 1e-10, 1e-11 and 1e-12 its slopes -6.6043e-12, +1.0983e-14 and +3.7374e-14 a pass, changing sign where psi's creep keeps its own; above island 7 (TC1 and PS2 of 28 October): over the first 600 passes of probe 35's 41 births from 7.100 to 7.300 whose twins stay together, the angle's rate falls 1.0627 a unit of birth and spans 0.21133, psi plus twice the angle 0.01125

runs (probe or check, tool, run): the momentum runs, triple_momentum_drift v1.0, run 20261028_154913; probe 35, triple_heavy_edge v1.0, run 20261008_131041; the look, triple_surprise_look v1.7, run 20261028_160946

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where in the paper: Section 3 and Appendix A.1

what it rests on: the long run at relative 1e-12 over 60,000 passes in 12 segments: the rotation's near returns after 15,181, 15,882

and 31,063 passes, all pairs at 0.98, 0.99 and 0.93 of the rotation's closeness (LR2); its twin, 1e-10 further out in birth, 6.39e-6 turn away at pass 60,000, the gap's log-log slope 0.97 (LR3); 600 passes in two segments within 1.19e-9 turn of one in φ_N (F8); by case E's rates the rotation's near returns past 701 at 5,076 and 10,105 passes, not case A's 4,328 and 13,732 (RG1, missed)

runs (probe or check, tool, run): case E, triple_heavy_beat_runs v1.3, run 20261028_102627; case F, triple_heavy_beat_runs v1.3, run 20261028_102629; F8, triple_heavy_beat_runs v1.3, run 20261028_082621; the look, triple_surprise_look v1.3, run 20261028_104854; the look, triple_surprise_look v1.5, run 20261028_120909

---

where in the paper: Section 3

what it rests on: his first marks and the blind figure at 7.230: the lanes not seen by cells looking five each way; the dots triplets, one place every 201 passes (575 of 599); arrows along φ_N; rows 6 of 7 and 6 of 8, one short of the bar; the walls' step 0.778 of its usual size, 2,000 random sets never below 0.856

runs (probe or check, tool, run): the lattice look, triple_heavy_lattice_look v1.0, run 20261009_085502; the blind score, triple_heavy_blind_score, run 20261010_010034; the row fit, triple_heavy_row_fit v1.0, run 20261010_012847

---

where in the paper: Section 3

what it rests on: the maypole's clockwork (C24a): over the first 120 passes the median look steps 0.2154 turn about the circle's own centre against the author's 5/24 = 0.2083, and 0.2579 shuffled

runs (probe or check, tool, run): C24a, triple_maypole_m1, run 20261012_033532

---

where in the paper: Section 3

what it rests on: the dead zones on runs he never drew on: 3 of 484 runs in the lower zone (Z1 of 12 October); 1,462 of 1,475 rows agree on the upper (Z2)

runs (probe or check, tool, run): Z1, Z2, triple_dead_zone_check v1.0, run 20261012_062504

---

where in the paper: Section 3

what it rests on: the hidden pattern at the centre: his trace's 37 highs, 53.5 passes apart, between the centres' beats of 57.2 and 46.5 passes (BC0); two notes beating the null at 44 of 51 runs (BC2′)

runs (probe or check, tool, run): BC0 to BC2, triple_centre_chord v1.0, run 20261015_133950; BC1′, BC2′, triple_centre_chord v1.1, run 20261015_134034

---

where in the paper: Section 3

what it rests on: the squeeze-and-tilt reads at the pair's eccentricity 0.25: symmetry, the pinch and the waist that moves 0.02 in birth, all three borne out at both outline scales (SQ1 to SQ3)

runs (probe or check, tool, run): SQ1 to SQ3, triple_strobe_views v1.0, run 20261014_072942

---

where in the paper: Section 3

what it rests on: the author's line 'the ratio stays fixed, the rates don't': steady at three slices of eight over 600 passes (RF1); over 2,400 passes at the star's 0.20 to 0.24 every birth within 1% of the median while the rates spread 2.7 to 3.2 times as much (SM1)

runs (probe or check, tool, run): RF1, triple_ratio_atlas v1.0, run 20261015_140219; SM1, triple_slice_middles, run 20261015_151629

---

where in the paper: Section 3

what it rests on: the chains of smaller islands: 121 locked runs at simple fractions against 34.5 at decoys (E6a)

runs (probe or check, tool, run): E6, triple_address_look v1.1, run 20261015_230449

---

where in the paper: Section 3

what it rests on: relation against shape among the register's 74 picture-reads, sorted blind: 19 of 30 (0.633) against 11 of 28 (0.393), +0.240 against the bar of 0.2 (RC1); and the Rider in Two Waves session's three blind reads, all borne out on the data

runs (probe or check, tool, run): RC1, triple_read_count v1.0, run 20261017_040300; the Rider in Two Waves session's reads, triple_rf_reads_check, run 20261010_182543

---

where in the paper: Section 5

what it rests on: the held runs, and the lap ratio pinned at seven plus twice the apse turn (7.0185 against 7.0196; 8.0219 against 8.0226)

runs (probe or check, tool, run): probe 22, triple_door2, run 20261006_190322

---

where in the paper: Section 5

what it rests on: the island's centre moves in birth from 7.155 to 7.445 as the pair's eccentricity goes from 0.05 to 0.20, the period kept falling from 7.022 to 7.010

runs (probe or check, tool, run): probe 34, triple_binary_scan v1.1, run 20261008_070206; its look, triple_binary_scan_look, run 20261008_193959

---

where in the paper: Appendix C

what it rests on: Mardling's pendulum: the small-swing rocking rate is half the island's half-width, within 1.1% at the pair's eccentricity 0.10, 3% at 0.05 and 8% at the star's 0.24

runs (probe or check, tool, run): the check, triple_mardling_check, run 20261008_133322; probe 34's look, triple_binary_scan_look,

run 20261008_193959; Y3′′, triple_y3_p41, run 20261013_201222; LP1's numbers, triple_lp1_numbers_look, run 20261017_025855

---

where in the paper: Appendix B

what it rests on: Mardling's rule at the star's eccentricities 0.12 and 0.16 read with the second reader: within 2.3% and 3.5%, after the fact, a pair of twins that part in the last passes counted chaotic (the first reader, blind there, 149% and 51% off)

runs (probe or check, tool, run): the look, triple_y3_v2_look, run 20261016_185433; LP1's numbers, triple_lp1_numbers_look, run 20261017_025855; Y3′′, triple_y3_p41, run 20261013_201222; reader v2, triple_rocking_reader_v2 look, run 20261013_230139

---

where in the paper: Section 6

what it rests on: the lopsided chaotic layer at islands 7 and 8, and the edge's pendulum law (7.57 against 7.29; 7.37 against 7.83; 13.6 against 10.6), read step by step (1.23, 1.24 and 1.76 times)

runs (probe or check, tool, run): probe 23, triple_door2_edge_law, run 20261006_201311; probe 24, triple_door2_centres, run 20261006_204230; the local slopes, triple_edge_law_local, run 20261013_082012

---

where in the paper: Section 6

what it rests on: the layer grows with the pair's eccentricity (0.030, 0.085, 0.120 in birth, the late parter at 7.275 counted) while the held island stays 0.26 to 0.31 wide

runs (probe or check, tool, run): probe 34, triple_binary_scan v1.1, run 20261008_070206; its look, triple_binary_scan_look, run 20261008_193959; LP1c, triple_lp1c v1.0, run 20261017_173539

---

where in the paper: Section 6

what it rests on: turning the star's eccentricity instead: the chaos below, or even at 0.16, and never shrinking (0.020, 0.125, 0.250 in birth, the late parters counted); 135 of 139 escapes from below at 0.28

runs (probe or check, tool, run): probe 41, triple_star_ecc_scan, run 20261013_103221; its escapes counted by side, triple_paper_numbers v1.2, run 20261015_173032; LP1's numbers, triple_lp1_numbers_look, run 20261017_025855

---

where in the paper: Section 6 and Appendix B

what it rests on: at the star's 0.28 the chaos below the island unbroken for 0.735 in birth, down to 6.435, held regular stars from 6.430 (57 of the 63 walked escape); counted whole, the rule -34% (LR1 registered within 10%: not borne out; SE1, the sea ends: borne out)

runs (probe or check, tool, run): LR1, triple_layer_reach, run 20261017_024024; SE1, triple_sea_end, run 20261017_030727; the rule, triple_sea_rule_look, run 20261017_031501

---

where in the paper: Appendix B and Appendix C

what it rests on: at the pair's 0.25: 24 of the 30 births below the island chaotic, the thread counted; the rule +13% (registered 12%)

runs (probe or check, tool, run): LP1's numbers, triple_lp1_numbers_look, run 20261017_025855; LP1, triple_late_parters, run 20261017_024415

---

where in the paper: Appendix C

what it rests on: the island at the pair's eccentricity 0.225 (7.325 to 7.540; layers 0.125 and 0.010; all 27 escapes from below)

runs (probe or check, tool, run): probe 40, triple_binary_scan v1.3, run 20261013_093351

---

where in the paper: Appendix C

what it rests on: the island at 0.25 over 600 passes (7.390 to 7.555, its thread at 7.420, regular at 7.419 and 7.421)

runs (probe or check, tool, run): probe 39, triple_binary_scan v1.2, run 20261012_213903; the thread's check, its file stamped by day only, triple_th1_check, run 20261013

---

where in the paper: Section 6

what it rests on: the door: escapes on the low-birth side, 163 of 166; the chaos below turns its stars the far side's way, 17 of 18

runs (probe or check, tool, run): the recount, triple_escape_recount v1.2, run 20261016_182117; the backwards look, triple_backwards_look, run 20261015_032141

---

where in the paper: Appendix C

what it rests on: the long runs at 0.25: below the island eleven of twelve leave after 46 to 1,186 passes; above it, one leaves at pass 5,904

runs (probe or check, tool, run): probe 40, triple_binary_scan v1.3, run 20261013_093351

---

where in the paper: Section 6 and Appendix B

what it rests on: 30,000 passes above the 0.25 island: two of fourteen leave, at passes 5,904 and 14,278; the 0.20 island's lower half empties from its rim inward (11,977, 28,569)

runs (probe or check, tool, run): HK3 and HL1′, triple_long_pair, run 20261014_235048

---

where in the paper: Appendix C

what it rests on: held for a while: the four stars that read as held at 600 passes lose their hold after passes 651 to 922; five rim births, 1,602 to 2,615

runs (probe or check, tool, run): probe 40, triple_binary_scan v1.3, run 20261013_093351; recomputed, triple_paper_numbers v1.2,

run 20261015_173032; HK1, triple_hold_check v1.0, run 20261013_180851

---

where in the paper: Appendix C

what it rests on: the chaos above the 0.25 island hands back every star over 30,000 passes; the two escapes 478 and 504 passes after a held stretch

runs (probe or check, tool, run): HL2, triple_saved_reads v1.1, run 20261017_031204; HL1′, triple_long_pair, run 20261014_235048

---

where in the paper: Appendix C

what it rests on: held for a while: the first stay of 1 to 923 passes (medians 90 below and 255 above); ten of eleven escapes below within 273 passes of it; rank correlations 0.85 and 0.50; held again for whole stretches later

runs (probe or check, tool, run): probe 40, triple_binary_scan v1.3, run 20261013_093351; HK1, triple_hold_check v1.0, run 20261013_180851; recomputed, triple_paper_numbers v1.2, run 20261015_173032

---

where in the paper: Appendix C

what it rests on: over 6,000 passes the islands at 0.15 and 0.20 last (7.165 left at pass 3,917; 7.270 and 7.280 at 2,881 and 2,036; twins 1e-3 apart between passes 518 and 5,456)

runs (probe or check, tool, run): probe 43, triple_long_hold, run 20261013_233146

---

where in the paper: Section 6

what it rests on: over 6,000 passes the island at 0.25 dissolves (escapes after 442 to 5,292 passes; the rooms' median holds of 4,122, 2,018, 6,000 and 1,345 passes; each pair of twins' first pass 1e-3 apart)

runs (probe or check, tool, run): probe 42, triple_hold_scan, run 20261013_152519; its twins, triple_twin_growth_look, run 20261014_014526; recomputed, triple_paper_numbers v1.2, run 20261015_173032

---

where in the paper: Appendix C

what it rests on: on a grid five times finer the third thread is a band (7.446 to 7.456); leaking stops above 7.471; the narrowest swings at 7.524 to 7.531

runs (probe or check, tool, run): probe 44, triple_pinch_grid, run 20261014_090328; its scoring, triple_pinch_score v1.1, run 20261014_225036

---

where in the paper: Section 6

what it rests on: the door at the star's eccentricities 0.24 to 0.34 (21 of 177, 114 of 355 and 231 of 476 from above)

runs (probe or check, tool, run): probe 30's atlas, triple_rocking_atlas, run 20261007_004206; the sides, triple_escape_sides, run 20261013_083705

---

where in the paper: Section 6

what it rests on: two of the stability rules bracket the island (0.93 to 0.97 and 1.07 to 1.15 of the critical values; the escapes at 1.063 to 1.071)

runs (probe or check, tool, run): the look, triple_mardling_programme_look, run 20261009_013508

---

where in the paper: Section 6

what it rests on: Holman and Wiegert's fit for test particles around our pair: the inner unstable region ends at period ratios 4.03 to 5.36 at the pair's eccentricities 0.05 to 0.25, and the innermost whole-number resonance beyond it is the 5:1 or the 6:1

runs (probe or check, tool, run): the edge, triple_hw_edge, run 20261017_234929

---

where in the paper: Section 6

what it rests on: started on a circular orbit, with the pair's eccentricity at 0.20 and the star's mass as before, the star neither escapes nor is pumped across the 7:1 over about 4,500 to 5,000 of the pair's periods: none of 82 runs escapes, the largest eccentricity 0.053; started at 0.20, 11 of 91 escape within 600 laps (7.175 to 7.255)

runs (probe or check, tool, run): CS1, triple_cs1 v1.0, run 20261018_201130; read by, triple_cs1 v1.1 --score, run 20261018_203454; probe 34, triple_binary_scan v1.1, run 20261008_070206

---

where in the paper: Section 6

what it rests on: run on for 10^5 of the pair's periods, the 41 circular starts all stay and none is pumped, the largest eccentricity again 0.0527 (CS2 NO GAP)

runs (probe or check, tool, run): CS2 job 1, triple_cs2 v1.0, run 20261018_205711; job 2, triple_cs2 v1.0, run 20261018_221712; job 3, triple_cs2 v1.0, run 20261018_231408; job 4, triple_cs2 v1.0, run 20261019_001051; read together, triple_cs2 v1.0 --score, run 20261019_010756

---

where in the paper: Section 7

what it rests on: a lock rests about one axis (the arcs' centres within 1.1 degrees), and the centre's angle is two notes

runs (probe or check, tool, run): probe 22, triple_door2, run 20261006_190322; probe 24, triple_door2_centres, run 20261006_204230; Figure 10, triple_paper_figure4, run 20261015_075527; the chord, triple_centre_chord v1.0 and v1.1, run 20261015_133950; run 20261015_134034

---

where in the paper: Section 8

what it rests on: the rocking quickens steeply with the star's eccentricity: the quickest swing 11.0, 6.8, 4.5 and 3.0 passes at 0.20,

0.24, 0.28 and 0.32, a log-log slope of -2.77

runs (probe or check, tool, run): probe 27, triple_door4b_scope v1.0, run 20261007_054157

---

where in the paper: Section 8

what it rests on: the slow rocking at the star's eccentricities 0.12 to 0.16, read again; the middle's ratio to its beat slides (2.006, 2.233, 2.486 at 0.16, 0.165 and 0.17)

runs (probe or check, tool, run): the floor, triple_reader_floor_look, run 20261013_222629; reader v2, triple_rocking_reader_v2 look, run 20261013_230139; FW1, triple_fw1, run 20261014_095319; FW2, triple_fw2_laptop, run 20261014_200040; FW3, triple_long_pair, run 20261014_235048

---

where in the paper: Appendix C

what it rests on: the ratio across the atlas: a plateau at three of eight slices (0.22, 0.24, 0.34); the medians 4.78, 8.00 and 14.8 at 0.20, 0.22 and 0.24

runs (probe or check, tool, run): the look, triple_ratio_atlas, run 20261015_140219

---

where in the paper: Appendix C

what it rests on: the slice middles for 2,400 passes: the ratio within 1% at 0.20, 0.22, 0.24 (4.84, 7.98, 14.76) while the rates spread 2.7 to 3.2 times as much; 0.22 not locked at 8 (2 of 7 within 0.2%); plateaus at 0.26, 0.30, 0.34 (37.8, 44.4, 27.6)

runs (probe or check, tool, run): the slice middles, triple_slice_middles, run 20261015_151629; the rescore, triple_slice_middles --score, run 20261016_032715

---

where in the paper: Section 8

what it rests on: a real lock at a third (fourteen starting angles, 0.333311 to 0.333339); across the atlas, 31 runs at the third turn their ellipses from 0.0050 backward to 0.0029 forward

runs (probe or check, tool, run): probe 29, triple_lock_window, run 20261007_071039; the look, triple_third_look, run 20261015_165735; the crossings, triple_paper_numbers v1.2, run 20261015_173032

---

where in the paper: Section 8

what it rests on: the refuge is not for good: over 2,400 passes on start 0 the survivor at 7.170 escaped at pass 850, 7.240 at pass 881, 7.260 and 7.265 drifted off the third (0.333516, 0.332400), only 7.245 to 7.255 stayed

runs (probe or check, tool, run): the slice middles, triple_slice_middles, run 20261015_151629; the aftermath, triple_sm_aftermath, run 20261016_032840

---

where in the paper: Section 8

what it rests on: below the island at 0.32, over 2,400 passes, the three locked at a third born at 7.190 escaped (passes 693, 810, 862) and the three born at 7.195 to 7.205 on the angle -0.02 held at exactly a third

runs (probe or check, tool, run): SM6′, triple_offmirror v1.0, run 20261015_212604; rescored, triple_offmirror v1.0, run 20261016_161430

---

where in the paper: Section 8

what it rests on: most flat stretches are not locks (an atlas of 5,656 runs); re-run at births 0.001 apart, 11 of the 20 are real locks: six at the island's edge, four at a turn, one passing through

runs (probe or check, tool, run): probe 30, triple_rocking_atlas, run 20261007_004206; its look, triple_atlas_look, run 20261007_224331; probe 33, triple_crossing_sweep v1.0, run 20261007_211547

---

where in the paper: Appendix C

what it rests on: the real locks other than the third over 2,400 passes: the two longest locks' middles within 1.3e-7 (a fifth) and 8.1e-7 (a seventh), from 3.4e-6 and 3.0e-6 over 1,000 passes; their ends 2e-5 to 5e-3 off; no two neighbours at two ninths or an eleventh (LX1 not borne out)

runs (probe or check, tool, run): LX1, triple_followups v1.0, run 20261017_011407; RD1c, triple_reader_check v1.3, run 20261017_155035

---

where in the paper: Appendix C

what it rests on: over 6,000 passes each lock middle splits into a core and a shell (LX1′ not borne out): seven births at a fifth within 3.5e-9 and five at a seventh within 2e-8, the rest 9e-9 to 7e-4 off or escaped (7.452 at pass 3,077; 7.119 at 5,792), the shorter lock at a seventh unlocked; the eleventh's four births within 2e-7, locked

runs (probe or check, tool, run): LX1′, triple_followups2 v1.0, run 20261017_100655

---

where in the paper: Appendix C

what it rests on: over 12,000 passes (LX2 not borne out on all three): the fifth's core within 4.06e-9 (3.49e-9 over 6,000), HOLDING; the seventh's 7.121 loses its lock and drifts to 1.65e-7; 7.131 onto the fraction; 7.451 escapes at 10,538; the eleventh's 7.127 at 2.42e-7, the others within 3.2e-8; 7.4525 on the fifth

runs (probe or check, tool, run): LX2, triple_lx2 v1.0, run 20261017_200756

---

where in the paper: Appendix C

what it rests on: the locks at tolerances a hundred times tighter (LW6 XA MOVE, XB ARITHMETIC, XL FEWER, XE NOT HOLDING): the shells' median slide 1.18e-7 against 7.96e-7, 7.462 +1.62e-7; three of eight losses kept; 7.121 not locked; the fifth within 4.8e-9; the

eleventh's 7.127 at 4.05e-7, the others within 1.03e-7; over 12,000 (LX3, reported after LF8 failed) 7.121 escapes at 8,007, the eleventh within 2.3e-8; two machines part on 7.121 after pass 5,400

runs (probe or check, tool, run): LW6, triple_lw6 v1.0, run 20261018_001236; LX3, triple_lx3 v1.0, run 20261018_071538; LX3 read, triple_lx3 v1.0, run 20261018_173405; the look, triple_lx3_look v1.0, run 20261018_173657

---

where in the paper: Appendix C

what it rests on: the eleventh's line at the tight tolerance over 12,000 passes (LX4 E11W CONTIGUOUS): 7.128 to 7.135 within 1.2e-8 (7.135 within 7e-9); 7.125 a steady 1.53e-4 off; the anchor 7.129 exact

runs (probe or check, tool, run): LX4, triple_lx4 v1.0, run 20261018_173912

---

where in the paper: Appendix C

what it rests on: the seventh's core twins at the tight tolerance (C7 SLOWLY CHAOTIC, K5 MATCHES THE LOCKS): 7.121's twins part a thousandth of a turn at pass 5,515, the two other core births that lose their lock at 6,937 and 9,047, the two that stay locked never in 12,000 passes

runs (probe or check, tool, run): TW7, triple_tw7 v1.0, run 20261018_192644

---

where in the paper: Appendix C

what it rests on: 7.120's twins followed to 30,000 passes: they part a thousandth of a turn at pass 14,795, and the main run escapes at pass 29,323 while its twin stays held and locked (TW8 SLOWLY CHAOTIC)

runs (probe or check, tool, run): TW8 main, triple_tw8 v1.2, run 20261019_053814; TW8 twin, triple_tw8 v1.2, run 20261019_043245; read together, triple_tw8 v1.2 --read, run 20261019_062042

---

where in the paper: Section 8

what it rests on: locks at simple fractions over the whole atlas: 121 locked runs within 1e-4 of a fraction with denominator 2 to 12 against 34.5 at decoys, 31 of them the third, 90 against 34.0 without it; odd denominators 74 runs on 149 crossings against even 16 on 88

runs (probe or check, tool, run): probe 30, triple_rocking_atlas, run 20261007_004206; the look, triple_address_look v1.1, run 20261015_230449

---

where in the paper: Section 8 and Appendix C

what it rests on: odd chains off the mirror: 51 runs on 67 crossings against 10 on 40, 3.04 times per crossing (on the atlas's lines 2.73, which is 0.4966 over 0.1818, arithmetic); 61 at fractions against 16.0 at decoys; the fifth 23 on 8 crossings, the seventh 14 on 11; the count as first printed read 8 lines

runs (probe or check, tool, run): OM1, triple_offmirror v1.0, run 20261015_212604; read as registered, triple_offmirror v1.1, run 20261016_161718

---

where in the paper: Section 8

what it rests on: two dials: the low-birth flank rocks faster at every matched swing, by 0.013 to 0.076 cycles per pass

runs (probe or check, tool, run): the look, triple_two_action_look, run 20261008_063548

---

where in the paper: Section 8

what it rests on: Figure 12 at 0.24: the low flank's three widest swings (births 7.110 to 7.120) rock at a seventh, Figure 11's edge plateau

runs (probe or check, tool, run): probe 30, triple_rocking_atlas, run 20261007_004206; Figure 11, triple_paper_figure5, run 20261014_090645; Bear's marks, triple_bear_sheet_look, run 20261016_054242

---

where in the paper: Appendix C

what it rests on: one drag law on all eleven start lines with the band as a sixth term (98.3% to 99.3% pooled, at least 96.1% on every line; with the start, 97.1% to 98.0% and 82.8%); the law's coefficients not separately identified (grip3 added: 99.70% kept, the grip's coefficient moved 53%)

runs (probe or check, tool, run): batch 3, triple_batch3_reads v1.1, run 20261017_032534

---

where in the paper: Appendix C

what it rests on: the law with the band fitted on the atlas's seven lines and carried to the off-mirror four: 96.2% and 93.3% at 0.20 and 0.22, 80.3% and 70.5% at 0.24 and 0.26 (BL1 not borne out)

runs (probe or check, tool, run): batch 4, triple_batch4_reads v1.1, run 20261017_033852

---

where in the paper: Appendix C

what it rests on: the fin's two sides at 0.001 in birth on Figure 13's line: both closer to straight (FR1 not borne out); 7.358 held between unheld neighbours; the fastest turn 0.0172 at 7.356

runs (probe or check, tool, run): FR1, triple_fin_fine, run 20261017_032258

---

where in the paper: Appendix C

what it rests on: the fin on nine atlas lines: its tip at the hold's edge on all nine; back within 3% of free 0.025 to 0.045 beyond; more gently on four; a wake on six; a logarithmic point on one; at the atlas's spacing the fall outside looks logarithmic on eight and the climb inside straight on eight (seen after)

runs (probe or check, tool, run): FN1 to FN4, triple_fin_lines v1.2, run 20261017_031501; as scored, triple_fin_lines v1.1, run 20261017_030927

---

where in the paper: Section 9

what it rests on: the hold drags on the star's ellipse; the centre's ellipse turns backward from near the star's 0.28, the beat crossing zero at 7.251 and 7.333

runs (probe or check, tool, run): the grip, triple_apse_grip_look, run 20261008_065732; the harmonic, triple_drag_harmonic_look, run 20261008_220219; Figure 13, triple_paper_figure7, run 20261015_140349; the crossings, triple_paper_numbers v1.2, run 20261015_173032

---

where in the paper: Appendix C

what it rests on: the grip read about each run's own running centre: the grip spreads to -0.116, the line tightens to R^2 0.968, its zero-grip value still half again above the free rate

runs (probe or check, tool, run): the redesign look, triple_redesign_look v1.0, run 20261008_070736

---

where in the paper: Section 9

what it rests on: the slowest locked ellipse at each star eccentricity, forward 0.0084, 0.0067, 0.0045, 0.0022, then backward 0.0005, 0.0029, 0.0053 and 0.0078; and the locked runs' two branches about the line (26 of 27 below it, 15 of 17 above)

runs (probe or check, tool, run): probe 30, triple_rocking_atlas, run 20261007_004206; recomputed, triple_paper_numbers v1.2, run 20261015_173032

---

where in the paper: Section 9

what it rests on: on every starting angle at 0.28 the centre's ellipse is all but still (no more than 0.00045 of a turn per pass), against 0.0024 to 0.0029 forward at 0.26 and 0.0022 to 0.0027 backward at 0.30; the narrowest swing within 0.005 of a zero crossing on three of the six other lines, up to 0.024 from one on the rest

runs (probe or check, tool, run): probe 30, triple_rocking_atlas, run 20261007_004206; the look, triple_zero_centre_look v1.1, run 20261015_195901

---

where in the paper: Section 9

what it rests on: over 2,400 passes the centres' beats at 0.20 to 0.26 on a straight line within 2.3% of their mean size, zero at 0.286; between the centres at 0.28 (+0.0005) and 0.30 (-0.0054) the beat crosses zero at 0.282

runs (probe or check, tool, run): the slice middles, triple_slice_middles, run 20261015_151629; the aftermath, triple_sm_aftermath, run 20261016_032840

---

where in the paper: Appendix C

what it rests on: the zero pinned at 0.2811 (registered 0.2817); at 0.30 the turn's one sign change at 7.4284 (registered 7.4276); over 6,000 passes at 0.28's centre the beat at 7.330 and 7.335, and at 7.340 a slow line at 0.000997 near the rocking's distance from a fifth (0.001412 is 0.2 less 0.198588, the rocking at 7.335, arithmetic)

runs (probe or check, tool, run): SM2′, SM3′, SW1, triple_offmirror v1.0, run 20261015_212604; rescored, triple_offmirror v1.1, run 20261016_161718

---

where in the paper: Appendix C

what it rests on: the slow line beside a fifth tested on eight more births (SL5 MIXED, against the guess): only 7.335 and 7.340 near the rocking's distance from a fifth; at the new births the line at 0.000771 to 0.001356 while the distance runs 0.001320 to 0.008050

runs (probe or check, tool, run): SL5, triple_sl5 v1.0, run 20261018_203922

---

where in the paper: Appendix C

what it rests on: the slow line against the beat (SL6a NOT BORNE OUT; SL6b, printed BORNE OUT, withdrawn to no reading since its band lay below the reader's floor of 3/6,000): beside the beat's zero 3.61 and 2.76 times it, at 0.000972 and 0.000937, where the distance from a fifth is 0.000320 and 0.000481 (the rates 0.200320 and 0.200481 less a fifth, arithmetic); farther out 1.325 and 1.313 inside the band, 1.085 and 1.039 below it; across the middle 1.039 to 1.325 times the beat

runs (probe or check, tool, run): SL6, triple_sl6 v1.0, run 20261018_220340

---

where in the paper: Appendix C

what it rests on: over 24,000 passes at the kit's tolerance the test reads no reading: 7.340 against SW1's laptop run parts by 2.2e-5 by pass 6,000, past the 2e-6 set

runs (probe or check, tool, run): SL7, triple_sl7 v1.0, run 20261019_012918

---

where in the paper: Appendix C

what it rests on: at the tight tolerance over 24,000 passes the rates change by at most 1.04e-9 a pass, against 8.02e-8 to 2.77e-7 at the kit's; the centre's ellipse at 7.340 still (beat 0.000026); the line off the beat at 1.307 and 1.214 times it; beside the zero 0.001075 and 0.001108, 0.000103 and 0.000111 from the 6,000-pass values; the kit's first quarter 0.200320 against the tight 0.201703

runs (probe or check, tool, run): SL8, triple_sl8 v1.1, run 20261019_032214

---

where in the paper: Appendix C

what it rests on: beside the zero, 7.3425 in one clean run: the line 0.001050 over the first 6,000 passes and 0.001048 over all 24,000 (SL9 BORNE OUT)

runs (probe or check, tool, run): SL9, triple_sl9 v1.0, run 20261019_043449

---

where in the paper: Section 9

what it rests on: the two branches on all 28 start lines at the star's eccentricities 0.20 to 0.26: 936 of 957 locked runs born below the centre under their line, 604 of 668 born above it over theirs

runs (probe or check, tool, run): probe 30, triple_rocking_atlas, run 20261007_004206; the look, triple_branch_look, run 20261015_203534

---

where in the paper: Appendix C

what it rests on: the branches are not the star's eccentricity alone: at equal grip the low flank carries the higher mean eccentricity on all 28 lines; with it and the swing's next harmonic added the line reaches R^2 0.996 (start 0, 0.26), and 808 of 957 below still fall under it, 581 of 668 above over it

runs (probe or check, tool, run): probe 30, triple_rocking_atlas, run 20261007_004206; the look, triple_ecc_dial_look v1.4, run 20261015_225444

---

where in the paper: Appendix C

what it rests on: where the island is widest the rocking rate carries the branches: with it added, 99 of 192 below the centre under the line and 63 of 119 above it over at 0.20, against 141 and 95 on the same runs without it; at 0.26 about seven in ten stay on their side

runs (probe or check, tool, run): probe 30, triple_rocking_atlas, run 20261007_004206; the look, triple_ecc_dial_look v1.4, run 20261015_225444

---

where in the paper: Appendix C

what it rests on: the branches over 2,400 passes: 57 of 60 below the centre under their line, 58 of 60 above it over; the line with grip, the next harmonic and the rate, fitted on the atlas and applied to the 2,400-pass runs, explains 96.1% to 97.8%, misses 0.00014 to 0.00022 against the grip line's 0.00064 to 0.00074

runs (probe or check, tool, run): the slice middles, triple_slice_middles, run 20261015_151629; E3, triple_sm_branches, run 20261016_032728; E5, triple_sm_predict, run 20261016_032726; its coefficients, triple_sm_predict --freeze, run 20261015_230236

---

where in the paper: Appendix C

what it rests on: the drag law across the atlas's six other start lines: unchanged 95.9% to 97.2% explained, 98.5% to 99.2% at start 0.01, 87.2% to 91.3% at 0.05; with the mean eccentricity, 97.4% to 98.6%

runs (probe or check, tool, run): probe 30, triple_rocking_atlas, run 20261007_004206; E7's report, triple_om_predict v1.1, run 20261016_042956; its five-term fit, triple_om_predict --freeze5, run 20261016_042946

---

where in the paper: Appendix C

what it rests on: farther off the mirror the drag law does not carry: without the eccentricity 6.6% to 35.0%, with it 44.3% to 71.3%; line by line from 99.4% to 99.7% on its own line to 12.2% to 33.7% at 0.12; the miss at 0.24 and 0.26; 745 of 876 runs inside the fitted grips; the miss and the band

runs (probe or check, tool, run): E7 and E7c, triple_om_predict v1.1, run 20261016_161446; the miss, line by line, triple_om_miss_look v1.0, run 20261016_161828

---

where in the paper: Section 9

what it rests on: the drag's size from Mardling's coefficient: 0.74 to 0.90 of the measured slope at the starting eccentricity; 0.945 to 1.039 at the held runs' mean eccentricity, star e 0.20 to 0.26; 0.822 at 0.28

runs (probe or check, tool, run): D1, triple_drag_size v1.0, run 20261016_192342; D1′, triple_drag_ebar v1.0, run 20261016_194224

---

where in the paper: Section 9 and Appendix B

what it rests on: the gap at 0.28 priced: the next member of the harmonic's family at most 0.7% of the slope, 3.9% of the gap; the next order 2.7% to 2.9% on every line; with each run's eccentricity beside its grip (correlation 0.61; spread 0.2531 to 0.3164), 1.012 at 0.28 and 1.191 at 0.26, outside 0.85 to 1.18

runs (probe or check, tool, run): D1″, triple_harmonic_recipes v1.2, run 20261016_204919; D1‴, triple_drag_ecc_fit v1.0, run 20261016_205125; D1′, triple_drag_ebar v1.0, run 20261016_194224

---

where in the paper: Section 9 and Appendix B

what it rests on: each run's drag at its own eccentricity: 0.867 to 1.007 of the measured drag on all five lines, the scatter 0.27 to 0.82 of one slope a line's; the fifth try

runs (probe or check, tool, run): D1⁗, triple_drag_own_ecc v1.0, run 20261016_212538

---

where in the paper: Appendix B

what it rests on: the drag's size with a longer record: at 0.28's middle 0.975 (0.766 to 1.340) over 600 passes and 0.695 (0.621 to 0.789) over 2,400, its scatter falling to 0.326 (LB1‴ borne out); at 0.20 to 0.26 0.71 to 0.77 over 600 and 0.70 to 0.78 over 2,400 (DS1 not borne out), against 0.92 to 1.01 across the island; correlations -0.92 to -0.95 on the middles, -0.21 to -0.86 across

runs (probe or check, tool, run): LB1‴, triple_closeout v1.0, run 20261016_191854; LB's scale, triple_closeout_reads v1.0, run 20261017_061550; DS1, triple_followups v1.0, run 20261017_011407; D1⁗, triple_drag_own_ecc v1.0, run 20261016_212538

---

where in the paper: Appendix B

what it rests on: the drag's size over 2,400 passes on the runs across the island (DS2 not borne out): 0.82 to 0.87 on the runs held

2,400 passes, 0.80 to 0.87 on the same runs over 600; the rim's drop-outs 0.97 to 1.33 over 600; the outer halves above the inner on four lines, 0.76 to 0.91 against 0.66 to 0.85, 0.20 the other way (DS3 outer higher); the halves' correlations -0.91 to -0.94 at 0.26 and 0.28

runs (probe or check, tool, run): DS2, triple_followups2 v1.0, run 20261017_100655; DS3, triple_ds3 v1.0, run 20261017_210732

---

where in the paper: Appendix B

what it rests on: the starts at 0.348 followed for 1,600 passes all hold (bands 0.1658 to 0.1845), where all nine at 0.350 escaped

runs (probe or check, tool, run): NB4, triple_nb4_look v1.0, run 20261016_213908; NB1, triple_newborn_look, run 20261013_200621

---

where in the paper: Appendix B

what it rests on: the kick's trend across island 7: against the pair's swing 0.35, its mean 0.78, the star's mean eccentricity 0.86 (KW1 not borne out); predicted 0.01689 to 0.00821, measured 0.01140 to 0.01867

runs (probe or check, tool, run): batch 4, triple_batch4_reads v1.1, run 20261017_033852; KS1, triple_kick_size, run 20261016_214904

---

where in the paper: Appendix B

what it rests on: the conserved quantity at three more births: rising strictly with the starting angle at 7.150 and 7.300 on all seven lines (KM1 borne out), and at 7.350 on its three held lines

runs (probe or check, tool, run): KM1, triple_k_starts v1.1, run 20261017_035730

---

where in the paper: Appendix B

what it rests on: the free stars against the pair's secular pull: 0.89 of it beyond 0.10 (218 runs); the excess near the rim against one over the distance 0.75; the zero-grip lead against one over the beat 0.70

runs (probe or check, tool, run): batch 3, triple_batch3_reads v1.1, run 20261017_032534; the levels, triple_drag_size, run 20261016_192342

---

where in the paper: Appendix B

what it rests on: set B's flat trade made at once: 82% to 99% of it in the first 50 passes, 75% at 7.130 and 7.135

runs (probe or check, tool, run): batch 3, triple_batch3_reads v1.1, run 20261017_032534

---

where in the paper: Appendix B

what it rests on: the room stretch by stretch: 7.477 at 0.136 then 0.058; every stretch's median 0.093 to 0.105

runs (probe or check, tool, run): batch 3, triple_batch3_reads v1.1, run 20261017_032534

---

where in the paper: Appendix B

what it rests on: the room's hum over 2,400 passes: 23 births held that long rock 0.055 to 0.138, median 0.107 (HM1′ registered above 0.1117: not borne out)

runs (probe or check, tool, run): HM1′, triple_saved_reads v1.1, run 20261017_031204

---

where in the paper: Appendix B

what it rests on: the leftover at 0.28: not the beat's zero (0.7 standard errors), not the rim's chaos (four of the 76 held runs chaotic), not the locks (locked runs fit about five times better)

runs (probe or check, tool, run): LB1, triple_leftover_look v1.0, run 20261016_214034; LB1′, triple_leftover_chaos v1.0, run 20261016_214250; LB1″, triple_leftover_locks v1.0, run 20261016_214403

---

where in the paper: Appendix B

what it rests on: the kick's size from her coefficient through Lagrange's equation for the pair: medians 1.029 and 1.199; island 8's kick 0.247 of island 7's against 0.216; 1.350 to 0.598 as the pair's mean eccentricity falls from 0.0757 to 0.0341

runs (probe or check, tool, run): KS1, triple_kick_size v1.0, run 20261016_214904; the pump check, triple_heavy_pump_check, run 20261009_021434

---

where in the paper: Appendix B

what it rests on: the multiplet: both nearest members inside the holding island on all nine slices; the beat 0.001 to 0.485 of the rocking; the stronger member 0.0051 to 0.0814 of the holding term

runs (probe or check, tool, run): MW1, triple_multiplet_widths v1.0, run 20261016_215735

---

where in the paper: Appendix B

what it rests on: the second action inside one slice: 318 to 506 runs; 1.65 to 5.20 times the scatter with the root-mean-square swing, 1.95 to 4.93 with the band

runs (probe or check, tool, run): SA1, SA1′, triple_second_action v1.1, run 20261016_220144

---

where in the paper: Appendix B

what it rests on: the second action labelled by the conserved quantity: at one birth it differs from start line to start line and follows the mean eccentricity (correlation -0.9997); its drift within a run 0.239 and 0.245 of that spread

runs (probe or check, tool, run): KD1, triple_k_drift v1.1, run 20261017_011850

where in the paper: Appendix B

what it rests on: the conserved quantity's drift over 2,400 passes swings rather than creeps: one sign in 77% of runs at 0.28 and 52% to 68% at 0.20 to 0.26, where 80% was registered (KD3 and KD3′ not borne out); at three of the four lines smaller over the record than over 600 passes

runs (probe or check, tool, run): KD3, triple_closeout_reads v1.0, run 20261017_061550; KD3′, triple_followups v1.0, run 20261017_011407

---

where in the paper: Appendix B

what it rests on: odd and even chains weighed one at a time: every odd chain fuller than its even neighbours (1.10 against 0.477, 0.673 against 0.284, 0.475 against 0.155, 0.395 against 0.0744); a factor 0.783 a step in denominator, 3.16 for odd

runs (probe or check, tool, run): OC1, triple_chain_weigh v1.0, run 20261017_011954; the address look, triple_address_look v1.1, run 20261015_230449; OM1 read as registered, triple_offmirror v1.1, run 20261016_161718

---

where in the paper: Section 10

what it rests on: one doubling, then escape: the centre's rocking 0.310 to 0.497 from 0.320 to 0.350; at 0.348, 0.4964

runs (probe or check, tool, run): probe 31, triple_centre_follow, run 20261007_151231; the checks, triple_paper_checks, run 20261014_085115; Figure 14, triple_paper_figure8, run 20261014_090053

---

where in the paper: Section 11

what it rests on: a heavy third star: the islands below the whole number (the pair turning 7.9252 times per orbit at island 8's centre)

runs (probe or check, tool, run): probe 25, triple_door2_heavy, run 20261006_213610; probe 35, triple_heavy_edge, run 20261008_131041; its look, triple_heavy_edge_look, run 20261009_015816

---

where in the paper: Section 11

what it rests on: the heavy star's held runs read at closest approach: eight at births 6.85 to 7.05 and 7.95 to 8.05 (bands as small as 0.034 and 0.066 turn); a ninth at 7.10 turning 7.0944 times per orbit, its pair's eccentricity down to 0.011; the pair turning 6.9332 and 7.9245 times per orbit at the islands' centres

runs (probe or check, tool, run): probe 26, triple_door2_count v1.0, run 20261006_221108; its look, triple_heavy_section_look v1.0, run 20261008_152734

---

where in the paper: Section 11

what it rests on: the offset read on the pair's mean ellipse over 600 passes: at island 7's centre (7.010) the pair's apse turns 0.03604 a turn per orbit and the star's 0.00395, so $7 + 2 \times (0.00395 - 0.03604) = 6.93581$ against 6.93570 counted; at island 8's centre (8.015) 7.92516 against 7.92522

runs (probe or check, tool, run): the apse check, triple_heavy_apse_check v1.0, run 20261009_025258

---

where in the paper: Appendix C

what it rests on: read on the pair's mean ellipse the held runs above seven are the bend (drifts of 31.29 to 78.23 turns)

runs (probe or check, tool, run): the check, triple_heavy_mean_hold_check, run 20261009_025920

---

where in the paper: Appendix A.2 and Appendix C

what it rests on: the heavy star's readings re-run with the extra readings: twelve of probe 35's births reproduce its closest-approach readings within 7.3e-8 in eccentricity and 1.5e-5 turn; at 6.810 the star's own eccentricity at its farthest point 0.1787 to 0.2255

runs (probe or check, tool, run): the sampling check, triple_heavy_sampling_check v1.0, run 20261009_020101

---

where in the paper: Appendix C

what it rests on: where the star and the pair trade: the pair's eccentricity averaged over its own orbit moves by 0.0144 and 0.0072 at the islands' centres, 0.0072 to 0.0188 far outside, 0.046 to 0.118 at 6.810; 0.0422 at 7.050, 0.0329 at 7.920 and 0.0325 at 8.075; the kick follows $\sin 2\pi\varphi_N$ (median correlations 0.957 over 14 runs and 0.728 over 8, one sign in all 22); island 8's kick 0.216 of island 7's (0.235 on its four clean runs) against the harmonic's 0.268 to 0.297 at the star's 0.18 to 0.20, noticed after the run

runs (probe or check, tool, run): the mean-e check, triple_heavy_mean_e_check v1.0, run 20261009_020713; the pump check, triple_heavy_pump_check, run 20261009_021434; the ratio look, triple_heavy_pump_ratio_look v1.0, run 20261009_022809

---

where in the paper: Section 11

what it rests on: started on the pair's mean ellipse: the heavy and light stars at the same pair, and island 7's lower rim (Figures 15 and 16)

runs (probe or check, tool, run): probe 36, triple_heavy_mean_start v1.2, run 20261011_153059; probe 37, triple_held_chaos_long, run 20261012_020012; probe 38, triple_edge_twins_long, run 20261012_175845; Figure 15, triple_paper_figure10, run 20261015_074751; Figure 16, triple_paper_figure10b, run 20261015_075254; probe 38's scoring, triple_edge_twins_score, run 20261013_025848

---

where in the paper: Section 11

what it rests on: Figure 15d's birth followed for 6,000 passes: its first 600 repeat it, and the bend's neighbour ratio falls to 0.0603 from 0.1126

runs (probe or check, tool, run): the zoom-out, triple_heavy_zoomout_run, run 20261009_080853

---

where in the paper: Section 11

what it rests on: the bend checked against Georgakarakos's short-period formula for an eccentric outer orbit (his 2003 equations 6 to 15): the pair's short-period vector 1.115 to 1.117 of his prediction (correlations 0.982 to 0.993), the bend at closest approach 1.076 to 1.288 (0.966 to 0.975); in his own regime 1.015 (GK1 HOLDS)

runs (probe or check, tool, run): GK1, triple_gk1 v1.0, run 20261018_193951

---

where in the paper: Appendix B

what it rests on: the one-harmonic model: the pendulum's sizes, the lopsided island, the faster low flank

runs (probe or check, tool, run): the model, triple_sfm, run 20261015_082428; its flanks, triple_sfm_flanks, run 20261015_082744

---

where in the paper: Appendix B

what it rests on: the newborn's rocking before escape (0.4875; the newborn of 0.350 followed for 1,600 passes)

runs (probe or check, tool, run): NB1, triple_newborn_look, run 20261013_200621; NB3, triple_nb3_look, run 20261014_054907; the checks, triple_paper_checks, run 20261014_085115; probe 32, triple_beat_lock, run 20261007_224303

---

where in the paper: Appendix B

what it rests on: why the heavy islands are ragged where they trade; the kick's size; whether a lock's width follows its fraction; why the ellipse turns faster than free near the rim

runs (probe or check, tool, run): the trade's edges, triple_heavy_trade_edges, run 20261013_045529; the jump, triple_heavy_trade_jump, run 20261013_060207; the pump, triple_heavy_pump_check, run 20261009_021434; LW1, triple_lock_width, run 20261013_194815; the rim at 0.25, triple_binary_scan_look25, run 20261013_062911

---

where in the paper: Appendix B

what it rests on: the close-out line's four smooth crossings: no lock at 1e-5 apart (under 2.9e-6 to 6.5e-6) or at 1e-6 apart (under 4.8e-7, 8.9e-7, 4.5e-6, 1.7e-6); the seventh's flattening (slope 0.011 against 0.24 to 0.55; within 2.5e-5 over 0.0036 in birth, the span arithmetic on its printed births); a tenth crossed at 4.5 per unit birth against 0.29, an eleventh three times; the reader's pull near them at most 6e-7; at a tenth nothing flat on births 1e-7 apart (under 1.8e-6); slopes beyond a tenth and an eleventh a quarter of those before; the drop through a tenth 2.79 and 7.25 times steeper over 2,400 passes than over 1,200 (the 8e-5 resolution is arithmetic: 2/(10 × 2,400))

runs (probe or check, tool, run): LW2, triple_closeout v1.0, run 20261016_191854; the 1/7 crossing, triple_closeout_reads v1.0, run 20261017_061550; LW3, triple_followups v1.0, run 20261017_011407; RD1, triple_reader_check v1.3, run 20261017_155035; LW4, triple_lw4 v1.0, run 20261017_170131; TS1, triple_ts1 v1.0, run 20261017_172324; TS2, triple_ts2 v1.0, run 20261017_172328

---

where in the paper: Appendix B

what it rests on: the passage through a tenth over 12,000 passes (LW5): the eighth STRAIGHT (1.000 of its slope), the tenth NO READING; every unlocked rate falls steadily, 6.61e-5 a window of 2,400 passes on the eighth's line (2.8e-8 a pass), 2.6e-8 a pass on the tenth's (FA0)

runs (probe or check, tool, run): LW5, triple_lw5 v1.2, run 20261017_140817; FA0, triple_fa0 v1.0, run 20261018_031414

---

where in the paper: Appendix A.1 and Appendix B

what it rests on: tolerances a hundred times tighter (DR1): the eighth's fall shrinks to 0.0093 of itself; the pair's eccentricity falls 5.1e-11 a pass against 1.13e-8; near the tenth the tight rate jumps by 5e-5 between windows as φ_N's swing narrows

runs (probe or check, tool, run): DR1, triple_dr1 v1.0, run 20261018_031349; the look, triple_dr1_look v1.0, run 20261018_031943

---

where in the paper: Appendix B

what it rests on: the tenth at tolerances a hundred times tighter (LW6 TEN-G GAP, TEN-C NO CHAOS): crossing between 7.3925998 and 7.3926098; none nearer than 1.42e-5; the step +1.424e-5 to -1.958e-5; no pair parts; the slow swing at 1/(10d) (FA2 SLOW SWING, 11 of 13; within 1.2% from 800 to 2,300 passes), from 2e-6 to 7e-5

runs (probe or check, tool, run): LW6, triple_lw6 v1.0, run 20261018_001236; FA2, triple_fa2 v1.0, run 20261018_173550

---

where in the paper: Appendix A.1

what it rests on: the integrator's own fall of the pair's eccentricity over 600 passes (6.8e-6 at 0.10, 1.3e-5 at 0.20, 1.1e-5 at 0.25); the same within 0.3% with the star's mass 1e-12; 2.3% of itself with tolerances a hundred times tighter

runs (probe or check, tool, run): LS1′, triple_light_pair_drift, run 20261017_025444; the look, triple_pair_drift_look, run 20261017_025649

---

where in the paper: Appendix A.4

what it rests on: twins that part in the last passes: five of the light star's pairs over 600 passes (7.420, 7.275, 7.265, 7.055, 7.175), 7.275 widening the lower layer at 0.20 from 0.105 to 0.120; one over 6,000 (7.360); two of the heavy star's at the stored precision, none among the heavy edge scan's 243 pairs

runs (probe or check, tool, run): LP1, triple_late_parters, run 20261017_024415; LP1c, triple_lp1c v1.0, run 20261017_173539

---

where in the paper: Appendix E

what it rests on: beside the light star a circular pair stays circular: its eccentricity below 2e-7 over 600 passes

runs (probe or check, tool, run): LS1, triple_light_pair v1.1, run 20261017_025304

---

where in the paper: Appendix A.5

what it rests on: the long runs of 28 October made again on the laptop (XC1 of 28 October): every verdict the same; 83 of the look's

lines the same but for time stamps in file names and 20 the same verdict with different digits; the three tests registered after the runs were sent not part of it; the energy's drift at 1e-10 -7.9009e-11 there against -7.9006e-11 here, at 1e-11 -2.7120e-12 against -2.7119e-12, at 1e-12 +3.2743e-14 against +3.2590e-14; the twins' gaps at pass 60,000 5.41e-06 against 6.39e-06, the angles' proportion -0.490 on both, and -0.492 at 7.300 on both

runs (probe or check, tool, run): the laptop's comparison, triple_crosscheck v1.0, run 20261028_132326; its run folder, triple_crosscheck v1.0, run 20261028_095855

---

where in the paper: Appendix A.1

what it rests on: the arithmetic's floor (tolerances a hundred times tighter)

runs (probe or check, tool, run): TT1, triple_tolerance_check, run 20261014_025405; its control, triple_tolerance_control, run 20261014_031147

---

where in the paper: Appendix A.1

what it rests on: the escapes recorded at the last pass: three of the 81 in the light star's runs of 6,000 passes

runs (probe or check, tool, run): recounted, triple_paper_numbers v1.2, run 20261015_173032

---

where in the paper: Appendix A.4

what it rests on: twins share their arithmetic: on regular orbits at the pair's eccentricity 0.10 they end 600 passes a median 1.5×10^{-6} turn apart; the twins' slope on planted gaps (1.00, 6.34)

runs (probe or check, tool, run): TW1, triple_paper_checks, run 20261014_085115; probe 43's selftest, triple_long_hold, run 20261013_233146

---

where in the paper: Appendix A.5

what it rests on: fidelity: probe 43's 67 main runs within 2×10^{-6} of their 600-pass scans; across machines, 4.8×10^{-7} and 1.7×10^{-6}

runs (probe or check, tool, run): probe 43, triple_long_hold, run 20261013_233146; run again here, triple_crossmachine_check, run 20261015_172849

---

where in the paper: Appendix A.3

what it rests on: the readers' accuracy on planted lines between bins: within 1e-8, and 9e-7 for a line below the first reader's floor (RA1)

runs (probe or check, tool, run): RA1, triple_saved_reads v1.1, run 20261017_031204

---

where in the paper: Appendix A.4

what it rests on: slow chaos under 1e-5 at pass 600: three of six such pairs part later (7.175, 7.185, 7.290 at 659, 2,389, 934); the three between the bars all part, by 1,354

runs (probe or check, tool, run): batch 3, triple_batch3_reads v1.1, run 20261017_032534; probe 43, triple_long_hold, run 20261013_233146

---

where in the paper: Appendix A.3

what it rests on: a second line a tenth as strong beside the rocking moves the second reader's rate by at most 1.9e-6

runs (probe or check, tool, run): batch 3, triple_batch3_reads v1.1, run 20261017_032534

---

where in the paper: Appendix A.4

what it rests on: twins' gaps by slice: two of ten at 0.10 between the bars, eight of nine at 0.20, at the rims; 7.300 at 8.6e-4 (TW2: not borne out)

runs (probe or check, tool, run): TW2, triple_saved_reads v1.1, run 20261017_031204

---

where in the paper: Appendix A.3

what it rests on: the readers' known answers (0.05016 below the old floor; 169 runs read by both)

runs (probe or check, tool, run): the floor, triple_reader_floor_look, run 20261013_222629; reader v2, triple_rocking_reader_v2 look, run 20261013_230139

---

where in the paper: Section 4 and Appendix A.6

what it rests on: the bench on known maps (8.721082; 4.669197; 3.259608 to 7.284707; 3.0001470)

runs (probe or check, tool, run): the cascade, triple_cascade_bench, run 20261007_051341; the humps, triple_hump_bench v1.1, run 20261007_181741; the loop, triple_circle_bench v1.3, run 20261007_200417

---

where in the paper: Section 4

what it rests on: the damping dial: fifteen dampings on one curve of the area kept per trip, passing a tenth at 0.49 (0.4889 to 0.4950)

runs (probe or check, tool, run): the crossover, triple_cascade_crossover, run 20261007_051639; the knob, triple_friction_knob v1.1, run 20261016_201627

---

where in the paper: Appendix A.6

what it rests on: whether our curve turns back: pooled, the ratios fall to 4.6639 and climb back to 4.6689; read against each level's own damped approximant (det 0.30), the departure passes the damped value by at most 0.0012 and comes back, under the 0.01 registered, so by that mark it does not turn back (VW1 MONOTONIC)

runs (probe or check, tool, run): VW1, triple_vw1 v1.0, run 20261018_193103; the knob, triple_friction_knob v1.1, run 20261016_201627

Tool and computational resource disclosure

Tools used. The work was done with an AI assistant, Claude (Anthropic), which wrote and ran the tools, drafted the prose and checked the citations, under the author's direction and checks. The integrations used Python with NumPy and SciPy (solve_ivp with DOP853); the figures were drawn with matplotlib, and the pages typeset with pandoc and wkhtmltopdf.

Responsibility. Responsibility for the correctness of the arguments and for the accuracy of the citations rests with the author. No automated system is an author of this work.

Verification performed. Every citation was checked, field by field, against its publisher's record, as set out above. The numbers in the sections were traced to the runs' own records by script, and the provenance table is checked by script against those runs and against this text. Each figure names the runs it is drawn from. The final file was checked with lint_deposit.py before deposit.

Attribution limits. Two gaps are known. Mardling's derivation of the eccentricity a companion induces in the pair, which her 2013 paper cites as submitted, could not be found in print (Appendix B). Van der Weele and colleagues' abstract does not give the size of their curve's turn back, so ours is not compared with it (Appendix A.6). How each cited work was read is in the table above.

Computational resources. The long runs were made on the author's laptop and the shorter ones in the assistant's workspace, each through a launcher that checks its tool's fingerprint before it runs.